Strong and Weak Ties

EM1421 - Social media and web analytics

{Slides courtesy of Prof. Fabiana Zollo}

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Strong and Weak Ties

- How does information flow through a social network?
- Can different nodes play structurally distinct roles in this process?
- How do structural considerations shape the evolution of the network over time?

Strong and Weak Ties

- In 1960s, sociologist Granovetter formulated the famous "strenght of weak ties" hypothesis
- He interviewed people who had recently changed job to learn how they discovered about that job opportunity
- He found that many people learned information through personal contacts
- What personal contacts?

Strong and Weak Ties

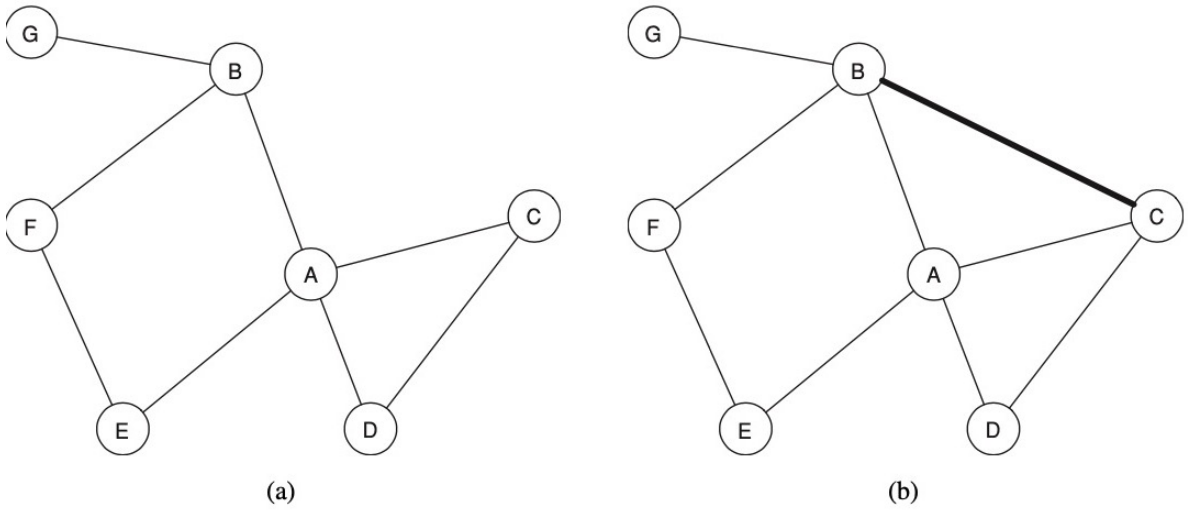
- Personal contacts were often described as **acquaintances** rather than close friends!
- Why? Granovetter proposed two different perspectives on distant friendship:
  - Structural: how these friendships span different portions of the full network
  - Interpersonal: considering the local consequences of strong or weak friendships
- This can help us thinking about the architecture of social networks more generally

Triadic Closure

- It is useful to consider how a network evolves over time
- What are the mechanisms by which nodes arrive and depart?
- What are the mechanisms by which edges form and vanish?

Triadic Closure

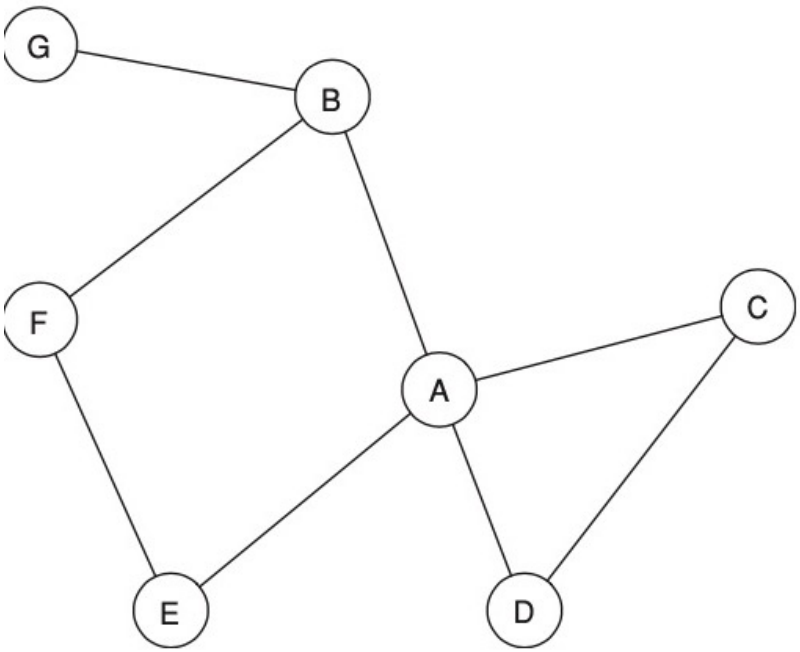
If two people in a social network have a friend in common, then there is an increased likelihood that they will become friends themselves at some point in the future.



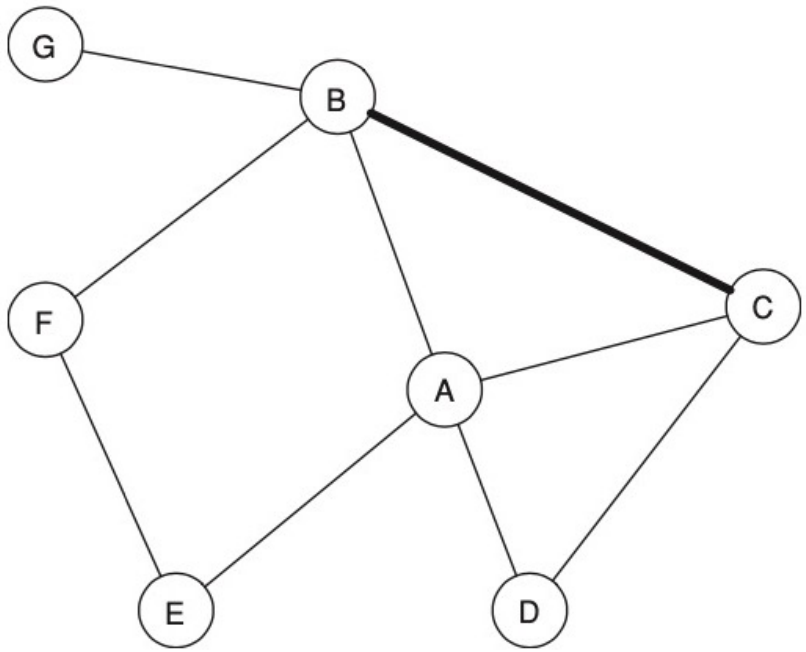
Anatole Rapoport. "Spread of information through a population with socio-structural bias: I. Assumption of transitivity." The bulletin of mathematical biophysics 15.4 (1953): 523-533.

Triadic Closure

- The formation of an edge between B and C produces a **triangle**
- The term *triadic closure* comes from the fact that the new edge has the effect of closing the third side of this triangle



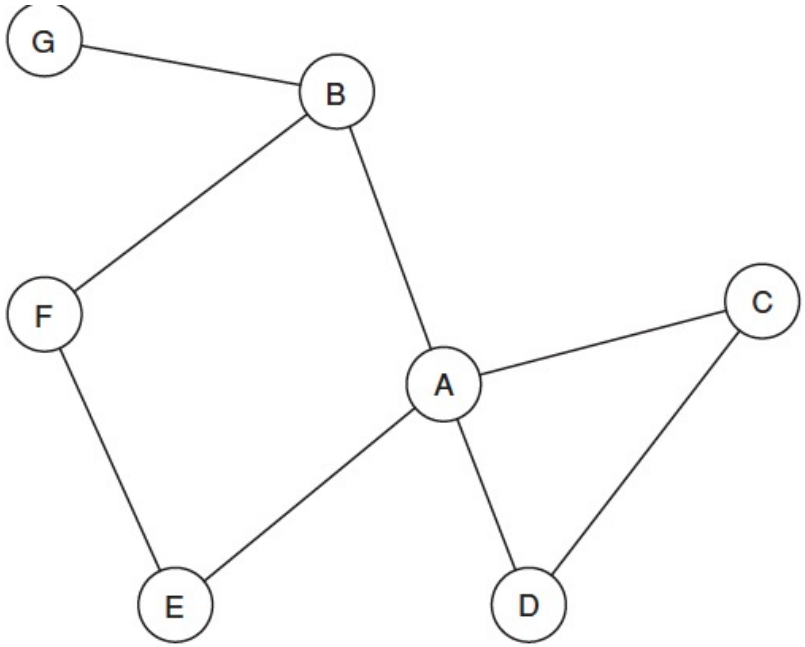
(a)



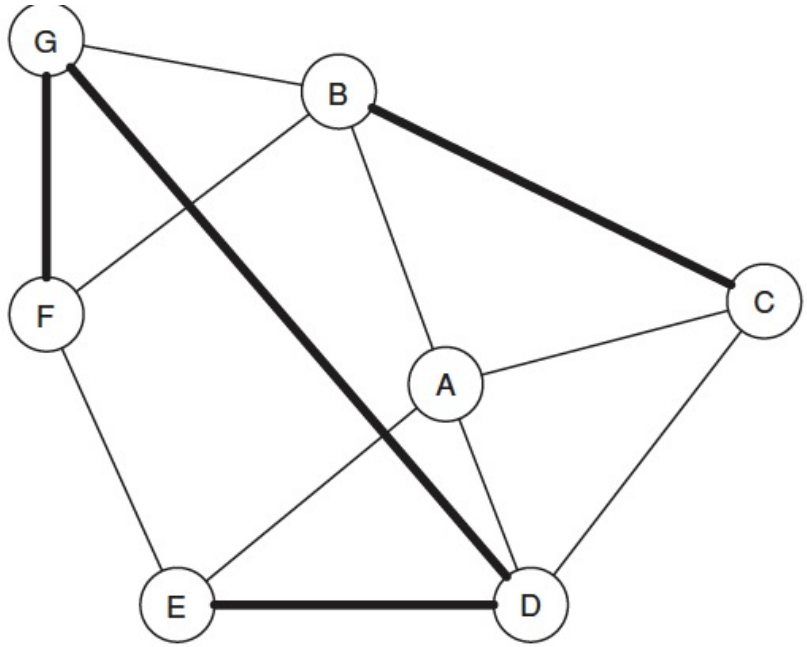
(b)

Triadic Closure

- Generally, we find a significant number of new edges formed through triadic closure over time



(a)



(b)

Triadic Closure

- The basic role of triadic closure in social networks has motivated the formulation of measures to capture its presence
- One of these measures is the **clustering coefficient**

Clustering Coefficient

- The clustering coefficient captures the degree to which the neighbors of a given node link to each other
- In other words, it's the fraction of the node's friends that are connected to each other by edges

Clustering Coefficient

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- In other words, it's the fraction of the node's friends that are connected to each other by edges
- For a node *i* with degree k_i the local clustering coefficient is defined as

$$C_i = \frac{2L_i}{k_i(k_i - 1)},$$

- L_i represents the number of links between the neighbors of node *i*

Clustering Coefficient

C_i is **between 0 and 1**:

- $C_i = 0$ if none of the neighbors of node *i* link to each other
- $C_i = 1$ if the neighbors of node *i* form a complete graph, i.e. they all link to each other
- The more strongly the process of triadic closure operate in the network, the higher the clustering coefficient will be

Clustering Coefficient

The local clustering coefficient measures the **local density** of links in a node's vicinity:

- C_i is the probability that two neighbors of a node link to each other
- $C = 0.5$ implies that there is a 50% chance that two neighbors of a node are linked

Average Clustering Coefficient

- The local clustering coefficient is a property of a single node
- The average clustering coefficient is a property of the entire graph, representing the average of C_i over all nodes

$$\langle C \rangle = \frac{1}{N} \sum_{i=1}^{N} C_i$$

Reasons for Triadic Closure

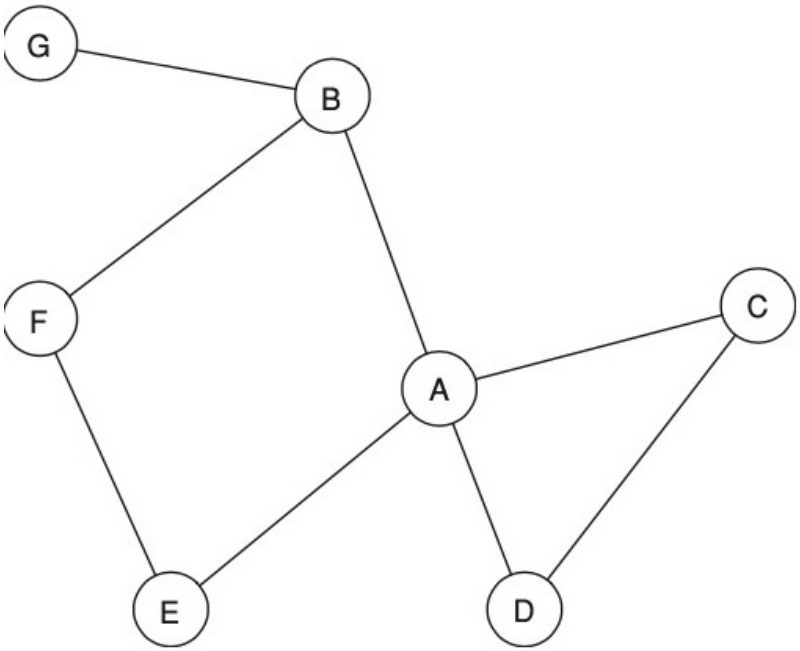
Why are B and C are more likely to become friends?



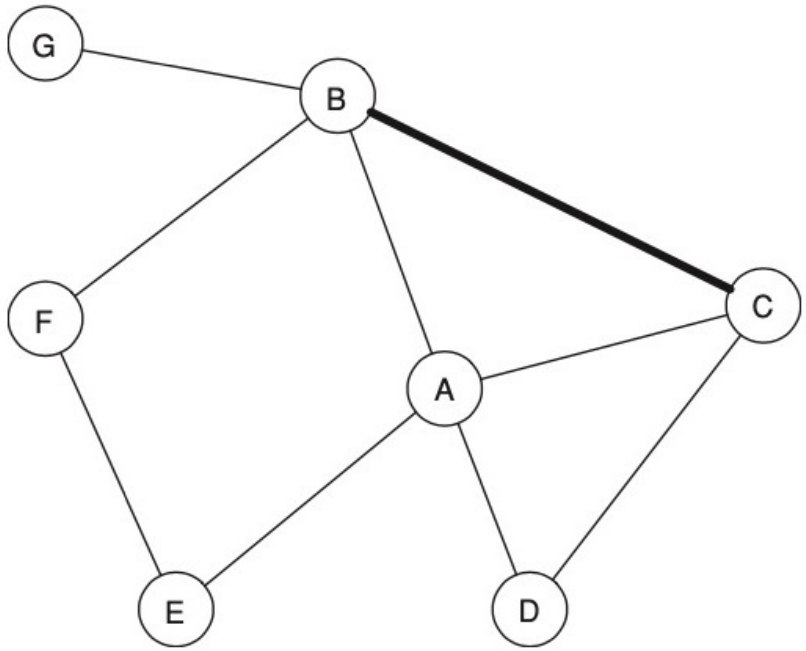
Reasons for Triadic Closure

B and C are more likely to become friends because:

1. Opportunity (increased)



(a)

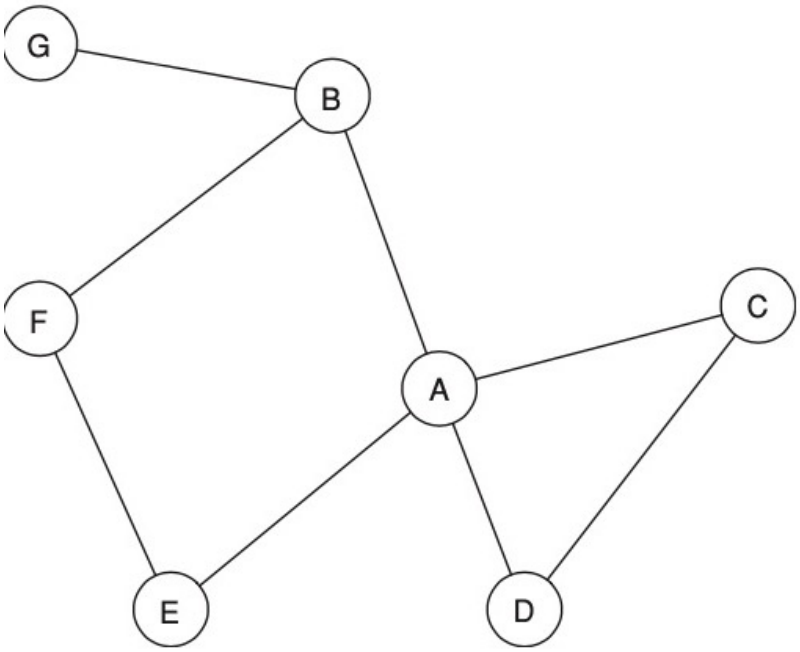


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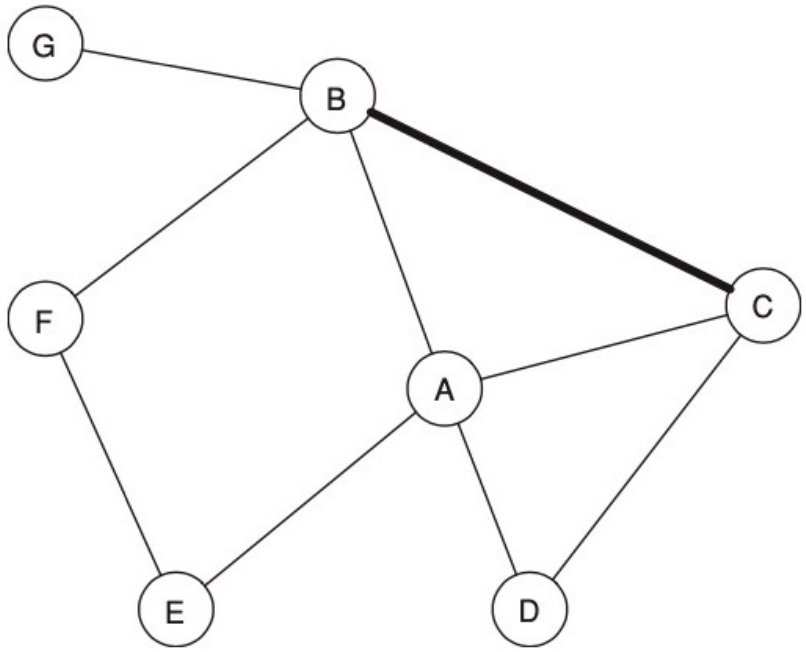
Reasons for Triadic Closure

B and C are more likely to become friends because:

1. Opportunity (increased)
2. Trusting (might lack in arbitrary pair of unconnected people)



(a)

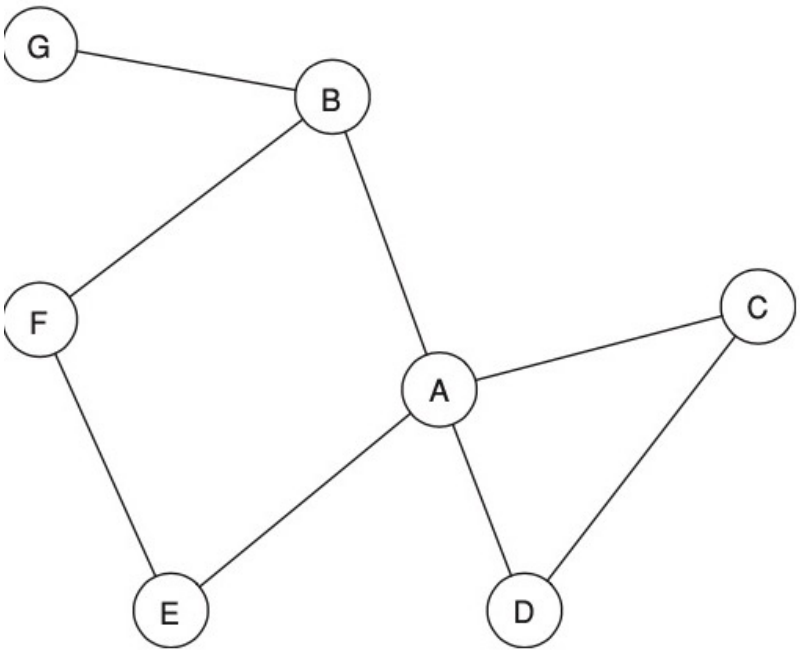


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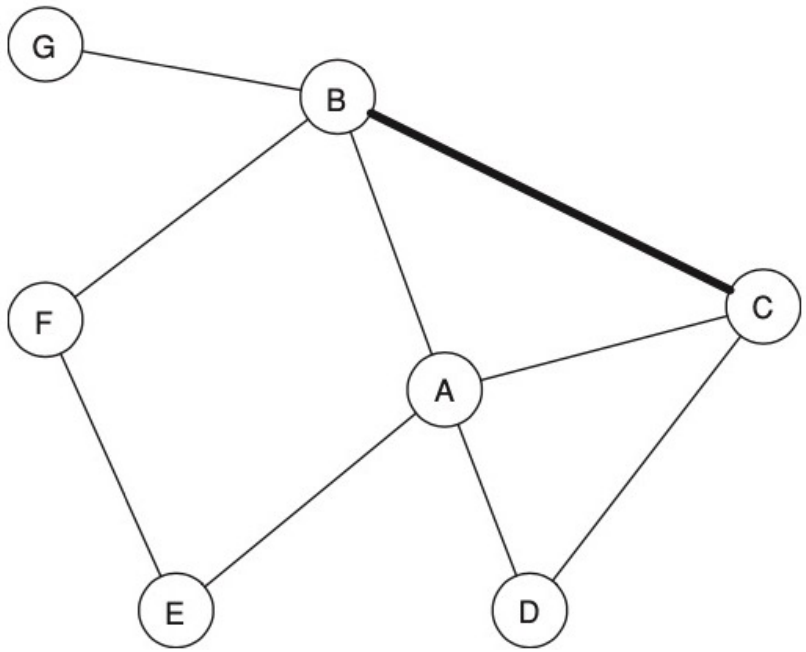
Reasons for Triadic Closure

B and C are more likely to become friends because:

1. Opportunity (increased)
2. Trusting (might lack in arbitrary pair of unconnected people)
3. Incentive



(a)



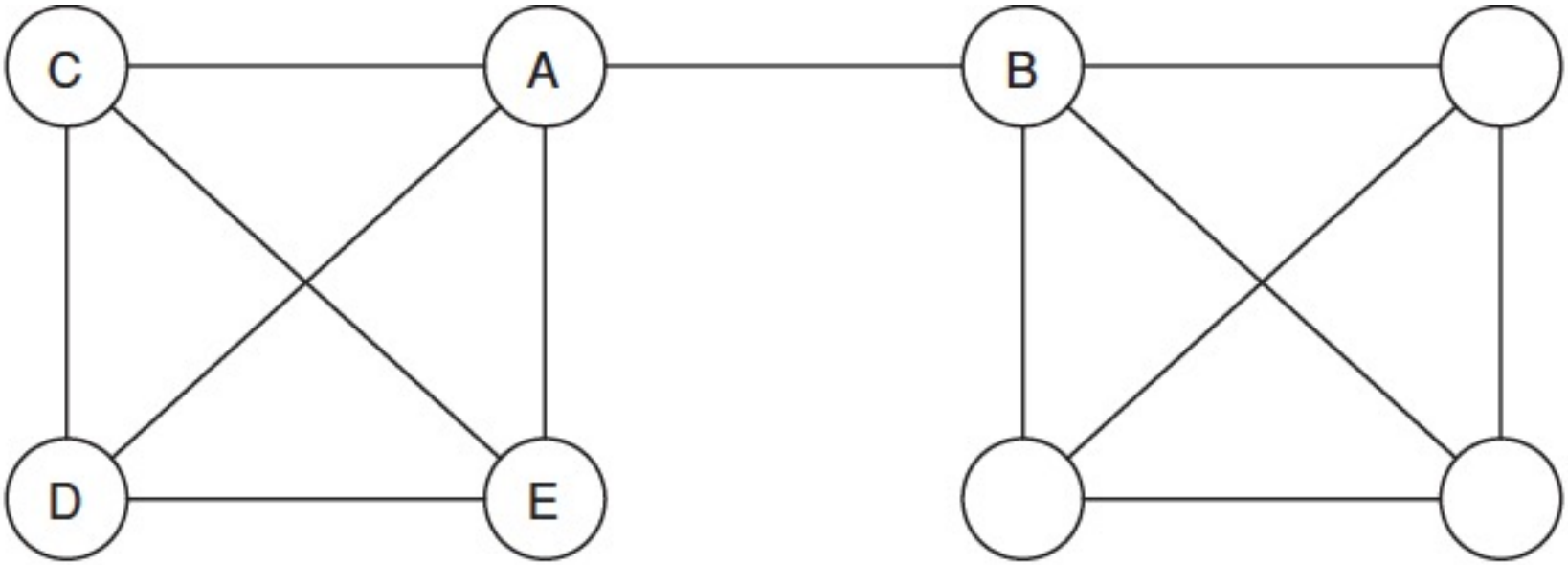
(b)

The Strength of Weak Ties

Let's see how triadic closure relates to Granovetter's interviews..

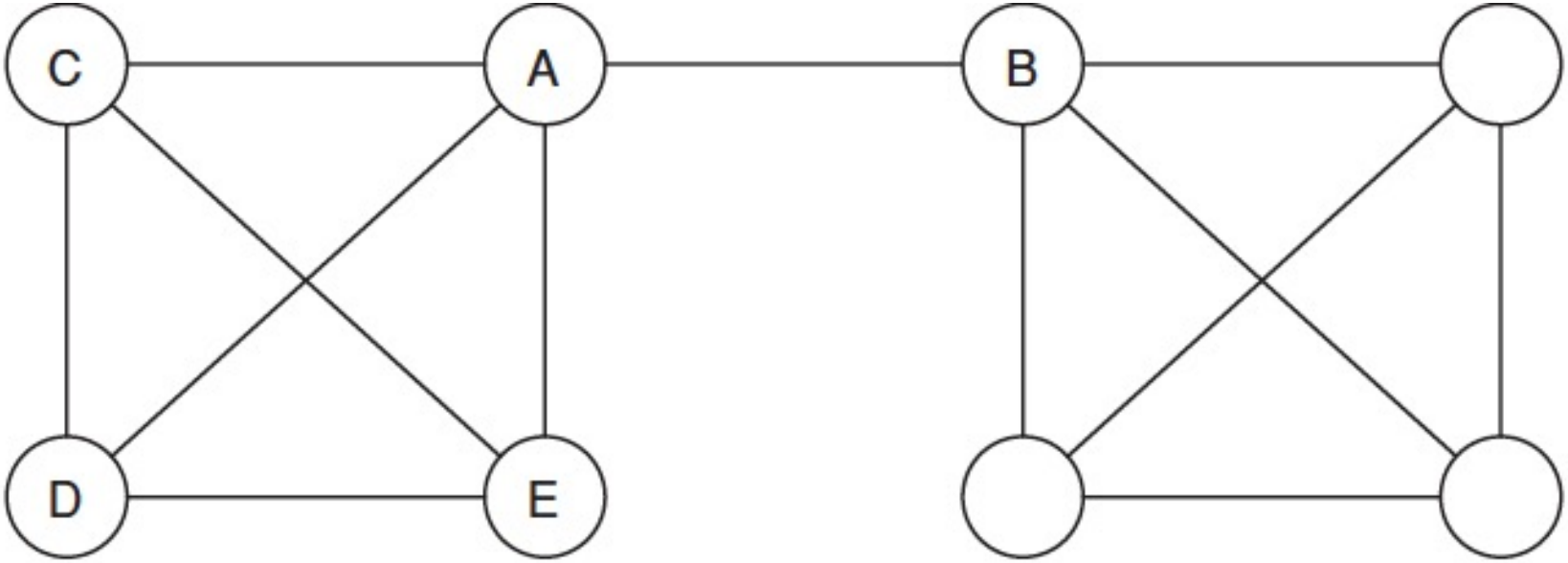
Bridges

- Let's assume that information about new jobs is relatively scarce
- If someone knows about a promising job opportunity, then they have information that you don't have access to
- Let's consider this simple network:



Bridges

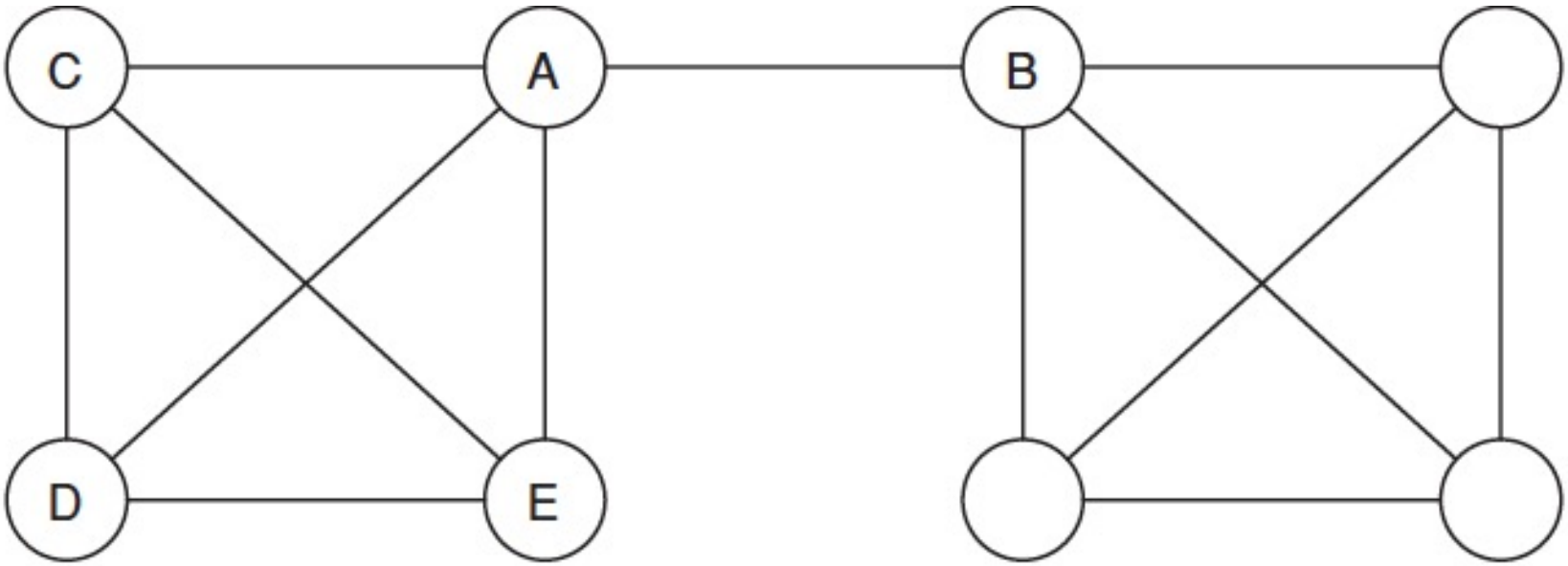
- A has four friends, but the link with B reaches a different part of the network
- We can assume that A,C,D,E tend to be exposed to similar opinions/information, differently from B



Bridges

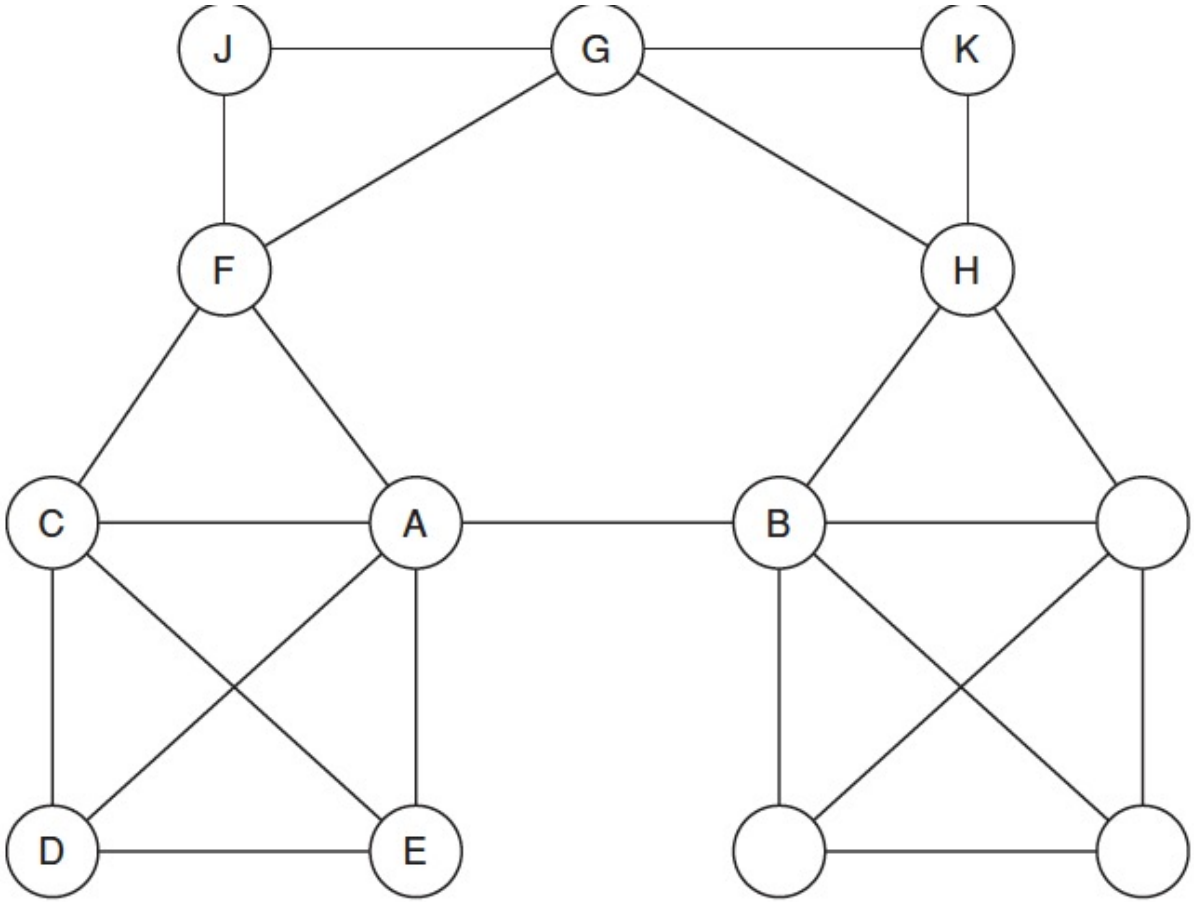
*An edge that joins two nodes A and B is called a **bridge** if deleting that edge would cause A and B to be in 2 different components*

- That edge is the only path between A to B



Bridges and Local Bridges

- But from our discussion on giant components and the small-world phenomenon, we also know that bridges are extremely rare in real social networks..
- In a larger social network we would probably have:



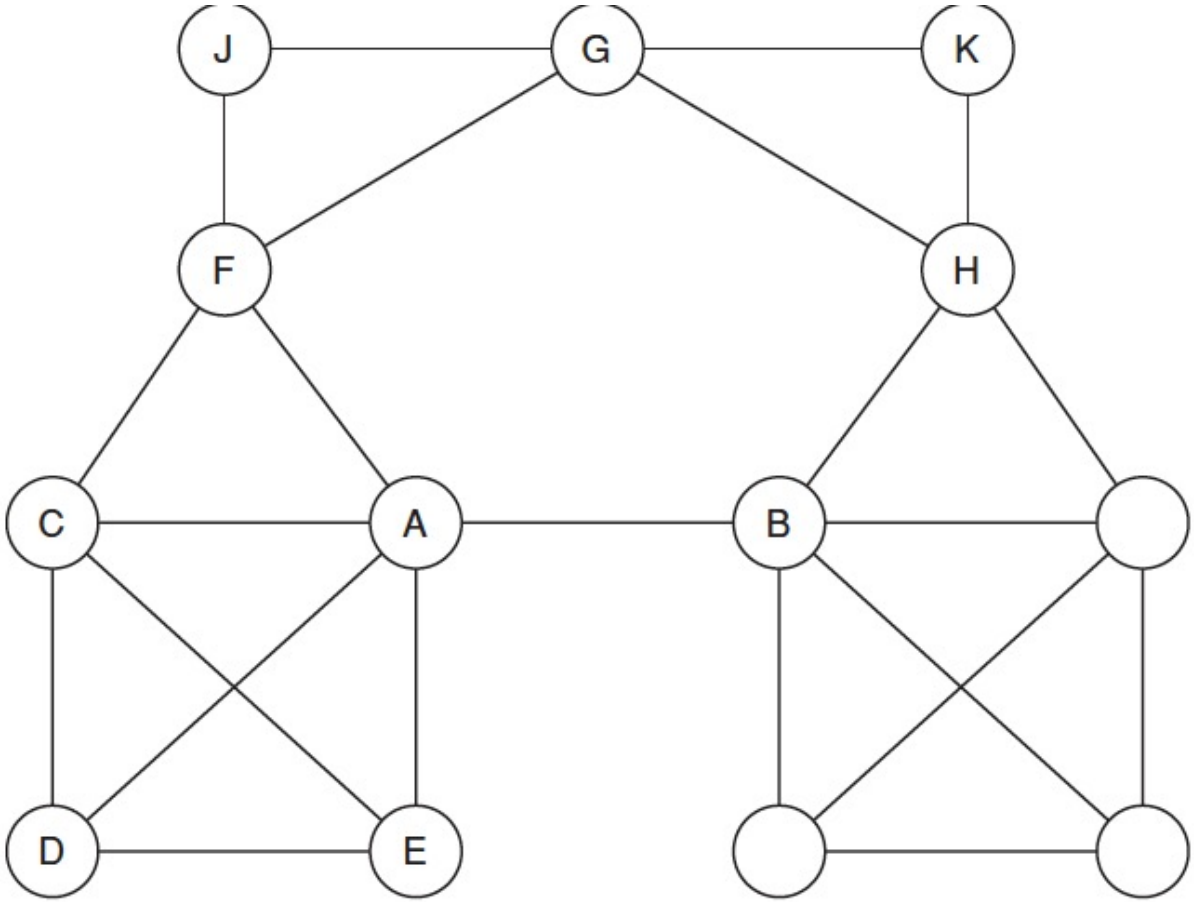
Local Bridges

*An edge joining two nodes A and B in a graph is a **local bridge** if its endpoints A and B have no friends in common*

- In other words, deleting the edge would increase the distance between A and B strictly > 2
  The span of a local bridge is the distance its endpoints would be from each other if the edge were deleted

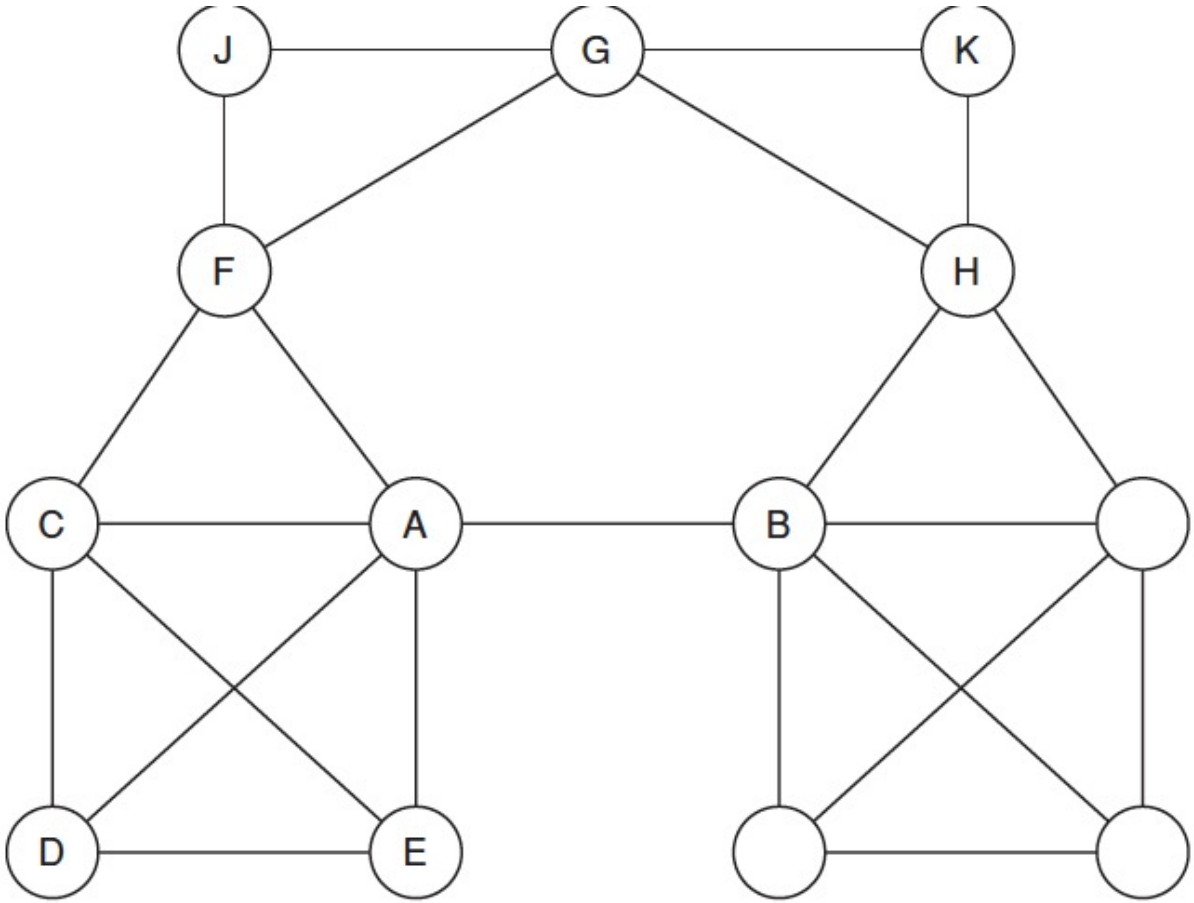
Local Bridges

- The A-B edge is a local bridge with span 4
- No other edge in the graph is a local bridge. Why?



Local Bridges

- The A-B edge is a local bridge with span 4
- No other edge in the graph is a local bridge. Why?
- Indeed, for each other edge, the endpoints would still be at distance of 2



Local Bridges

- Any connection with triadic closure?



Local Bridges

- The notion of triadic closure and local bridges are opposite!
- An edge is a local bridge precisely when it does not form the side of any triangle in the graph

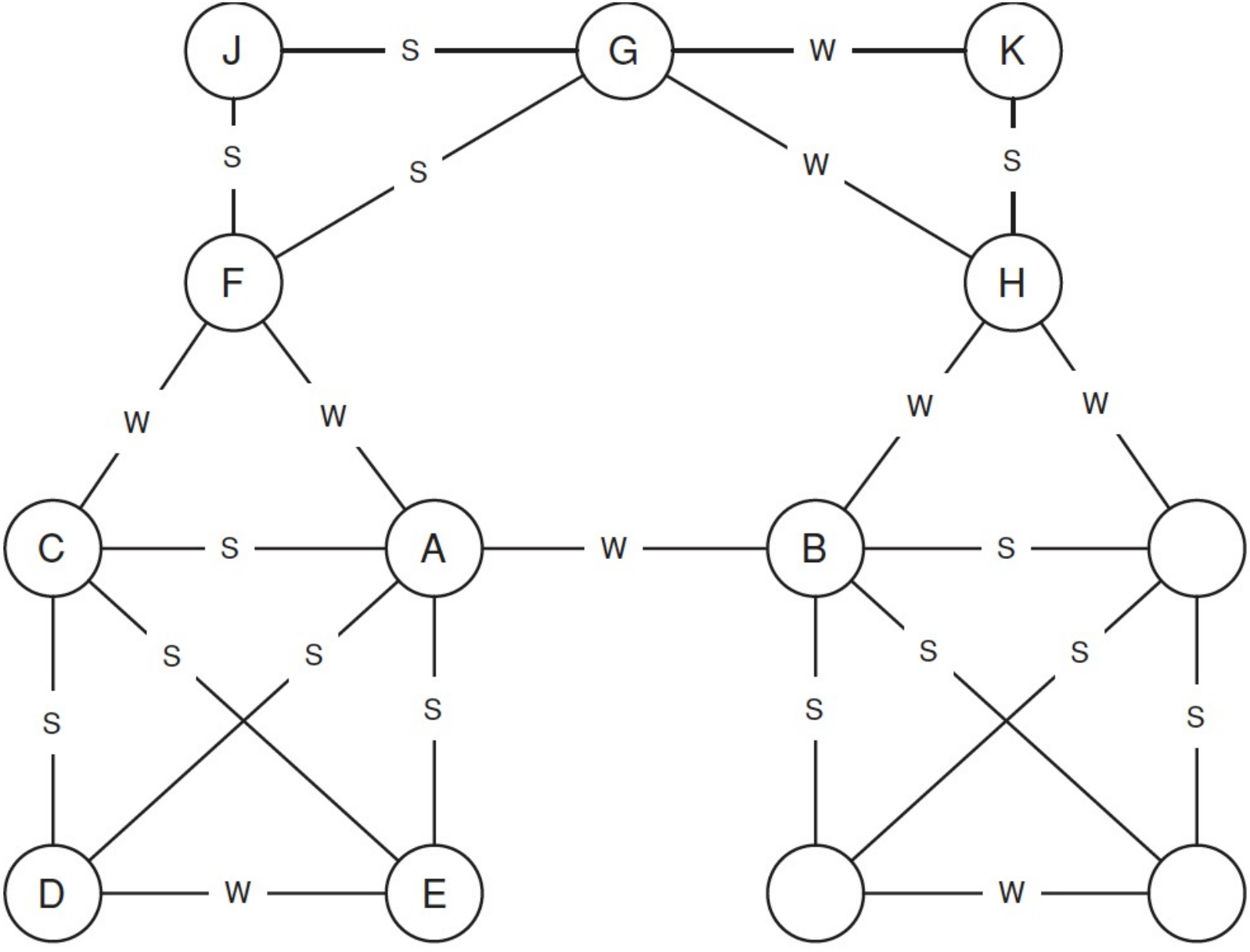
Local Bridges

- Going back to Granovetters..
- If a node like A is going to get truly new information, it might often come from a friends connected by a local bridge
- The closely-knit group to which A belongs is filled with people who roughly have the same information as A

The Strong Triadic Closure Property

- For the sake of simplicity, we can categorize the links in a social network as:
- **Strong** ties (stronger = friends)
- **Weak** ties (weaker = acquaintances)

The Strong Triadic Closure Property



The Strong Triadic Closure Property

If a node A has edges to nodes B and C, then the B-C edge is especially likely to form if A's edges to B and C are both strong ties

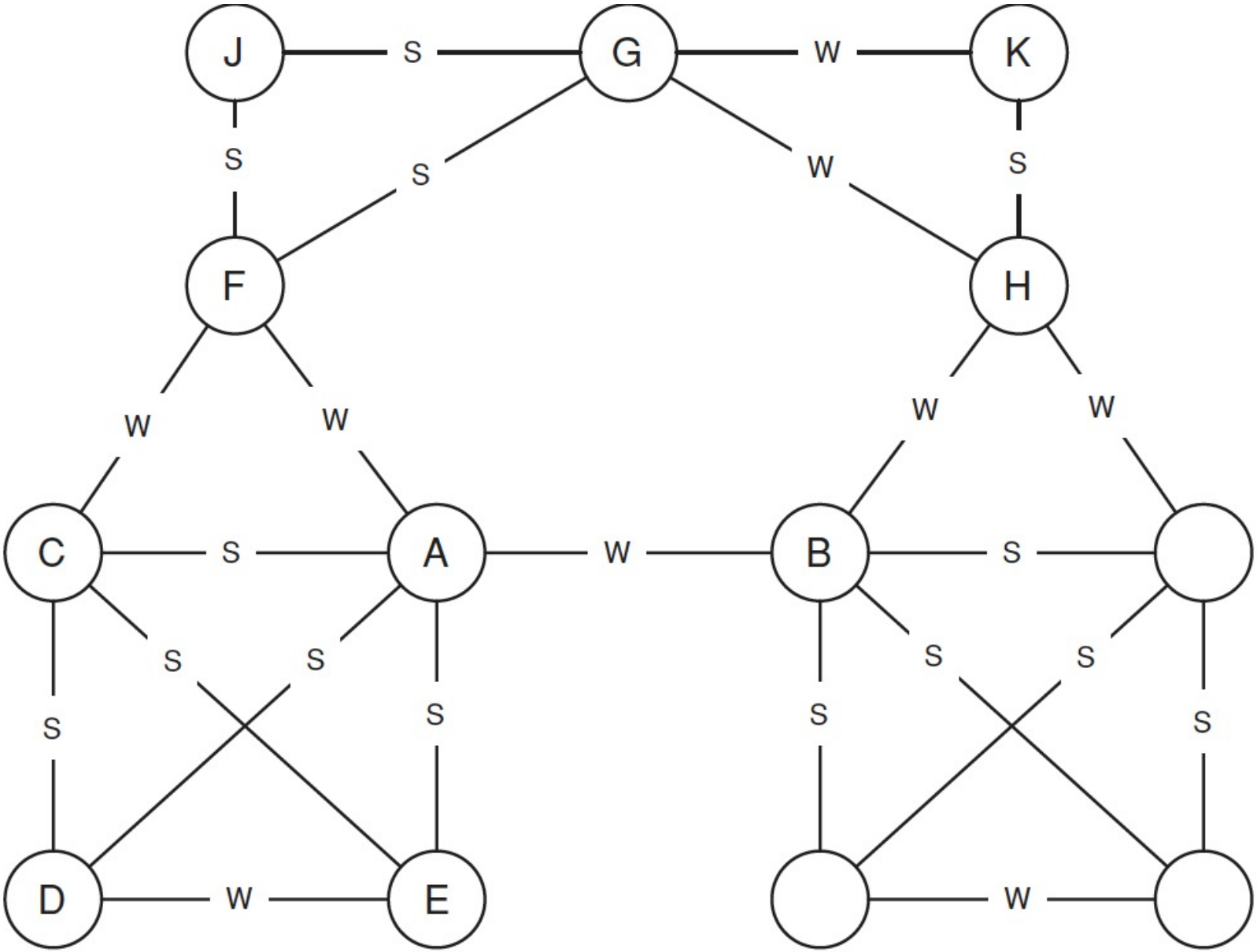
(Remember the reasons behind triadic closure? Opportunity, Trust, Incentive)

The Strong Triadic Closure Property

Granovetter suggested a more formal version of this assumption: *We say that a node A violates the Strong Triadic Closure (STC) property if it has strong ties to two other nodes B and C, and there is no edge at all (either a strong or weak tie) between B and C. We say that a node A satisfies the STC property if it does not violate it.*

The Strong Triadic Closure Property

- No node here violates the STC property



The Strong Triadic Closure Property

- The STC property is too extreme for us to expect it hold across all nodes of a large social network
- But it's useful for now to reason about the structural consequences of strong and weak ties

Local Bridges and Weak Ties

Claim: If a node A in a network satisfies the STC property and is involved in at least two strong ties, then any local bridge it is involved in must be a weak tie

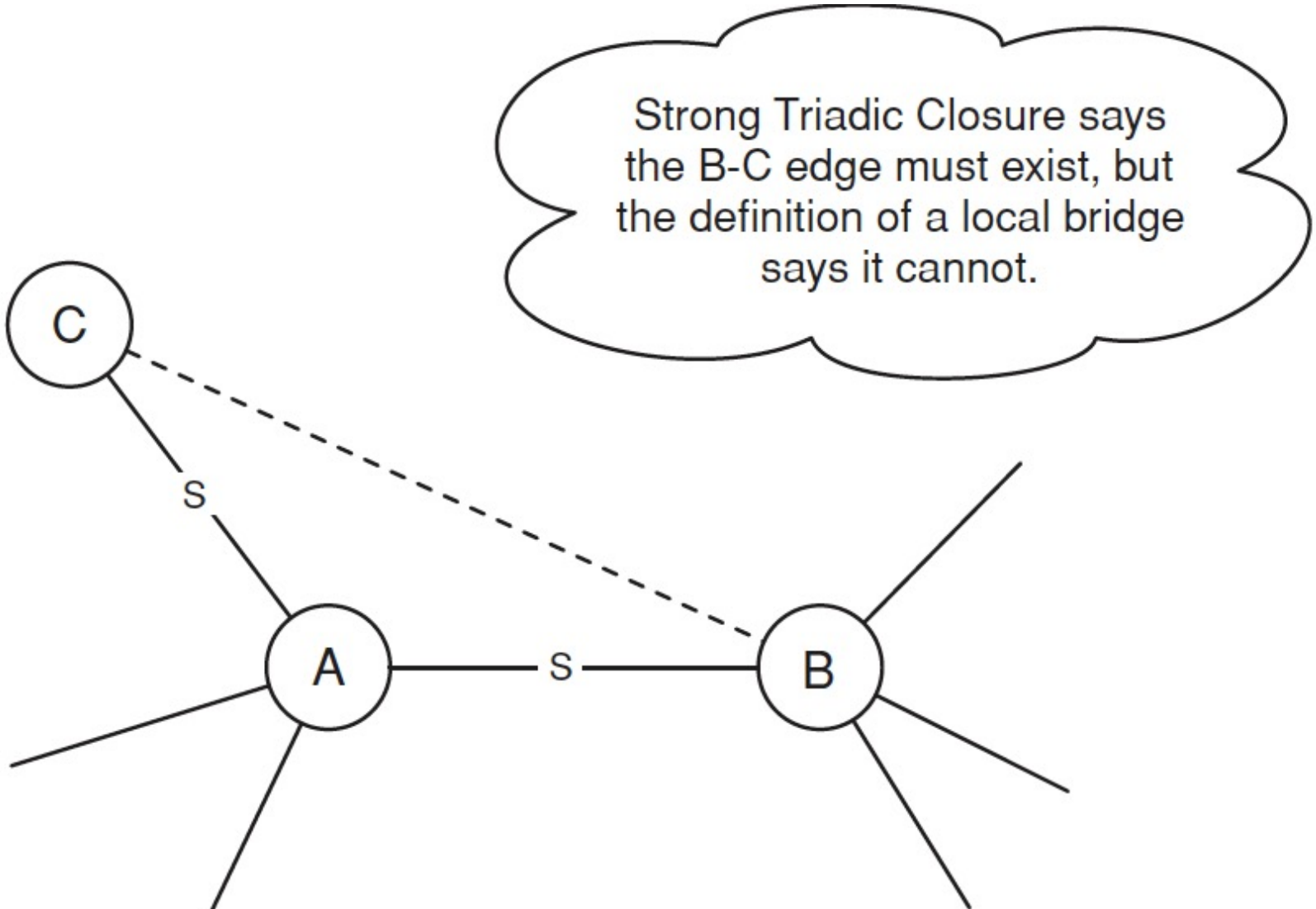
We can justify this claim as a mathematical statement.

Local Bridges and Weak Ties

- Let's take a network and consider a node A that satisfies the STC property and is involved in at least two strong ties
- Suppose A is involved in a local bridge (to a node B) that is a strong tie
- We want to argue that this is impossible

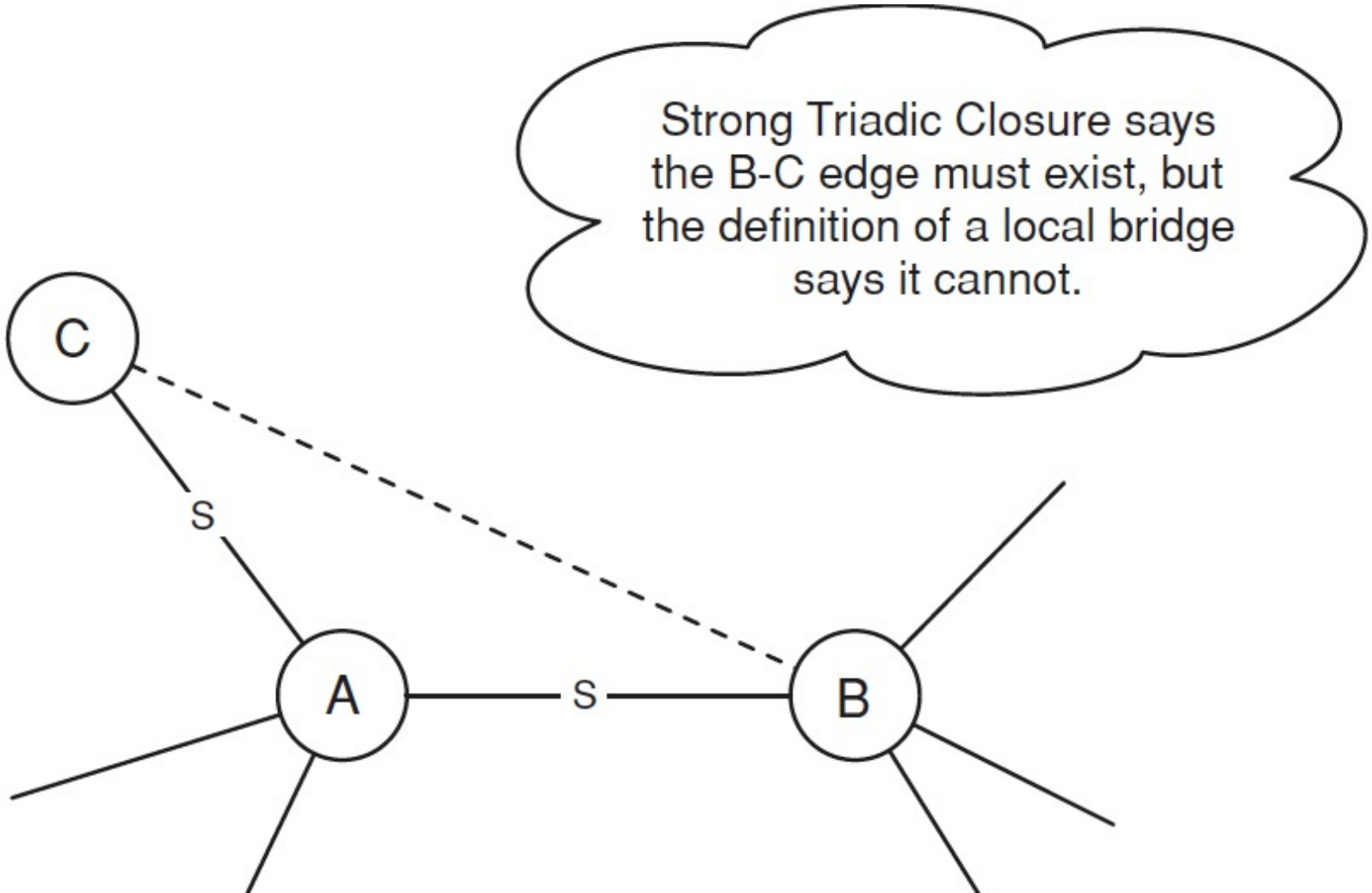
Local Bridges and Weak Ties

- A is involved in at least two strong ties
- The edge of B is only one of these ties $\rightarrow$ A must have a strong tie to some other node C



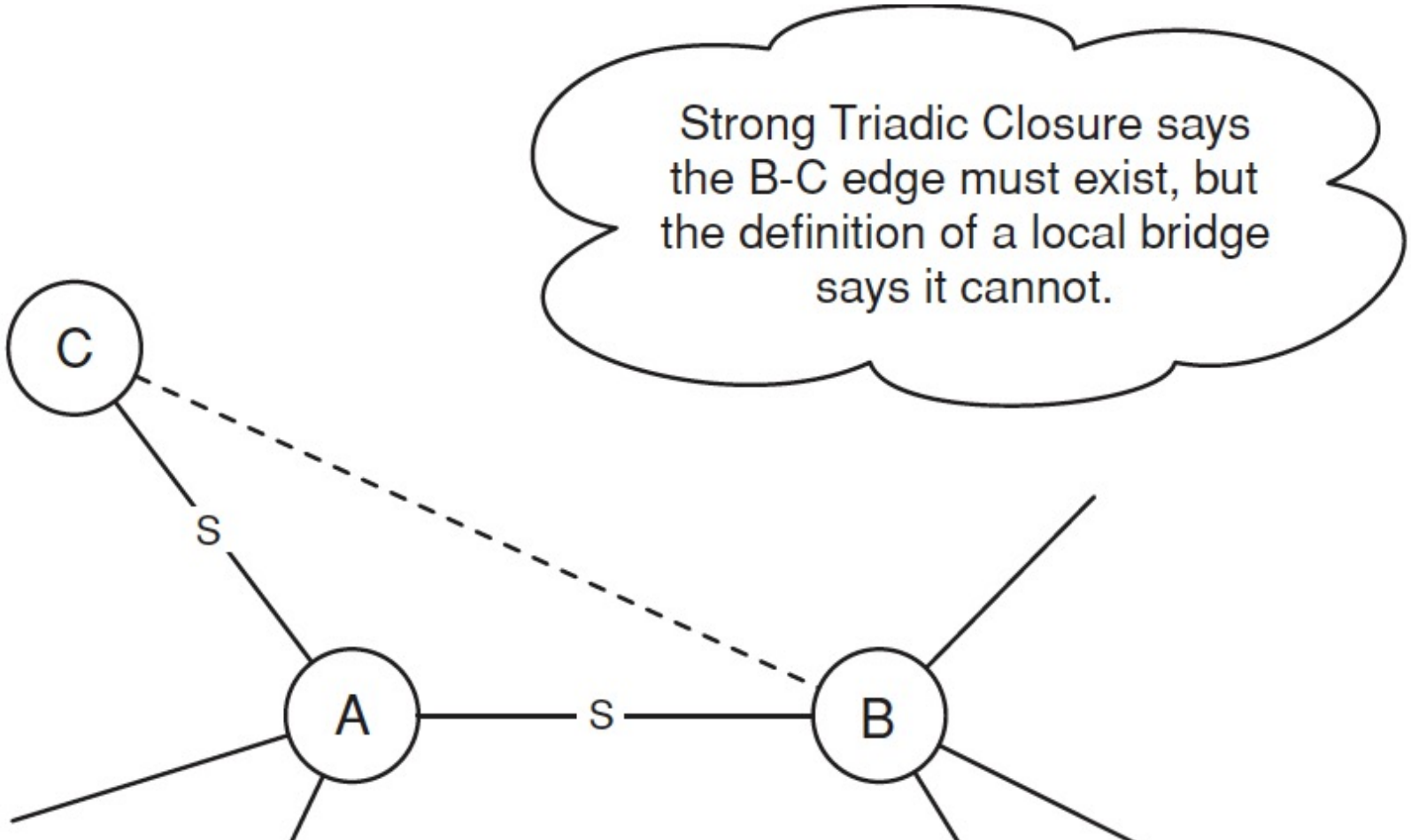
Local Bridges and Weak Ties

- Is there an edge connecting B and C?
- The edge from A to B is a local bridge (by assumption) $\rightarrow$ A and B must have no friends in common $\rightarrow$ B and C cannot be connected



Local Bridges and Weak Ties

- If B and C are not connected, then this contradicts the STC property (for which since A-B and A-C are both strong ties, the B-C edge must exist)
- We have shown that the existence of a local bridge that is a strong tie (our hypothesis) cannot hold



Local Bridges and Weak Ties

- Weak ties connect us to new sources of information and new opportunities
- Their local bridge property is directly related to their weakeness
- Weak ties have a dual role:
  - Weak connections
  - Valuable to enter hard-to-reach parts of the network
- This dual role is the strength of weak ties

Tie Strength and Network Structure in Large-Scale Data

- For many years Granovetter's predictions have remained untested on large social networks
- Difficult to find data that captured the strengths of edges in realistic settings
- With the advent of digital communications, this began to change but..

Tie Strength and Network Structure in Large-Scale Data

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- .. the definitions of weak ties and local bridge impose a sharp dichotomy:
  - An edge is either a strong tie or a weak tie
  - An edge is either a local bridge or it isn't

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- .. the definitions of weak ties and local bridge impose a sharp dichotomy:
  - An edge is either a strong tie or a weak tie
  - An edge is either a local bridge or it isn't
- We need **smoother gradations** when working with real data on a large scale

Structure and Tie Strengths in Mobile Communication Networks

- Onnela et al. studied the who-talks-to-whom network maintained by a cell phone provider
- Nodes are cell phone users, an edge joins two nodes if they called each other
- How can we distinguish between strong and weak ties?



Structure and Tie Strengths in Mobile Communication Networks

Tie Strength

- The strength of the edge can be defined as a numerical quantity, i.e.:
  - The total number of minutes spent on the phone by the two nodes
- We can sort all the edges by tie strength
- For each edge we can ask what percentile occupies

Structure and Tie Strengths in Mobile Communication Networks

Local Bridge

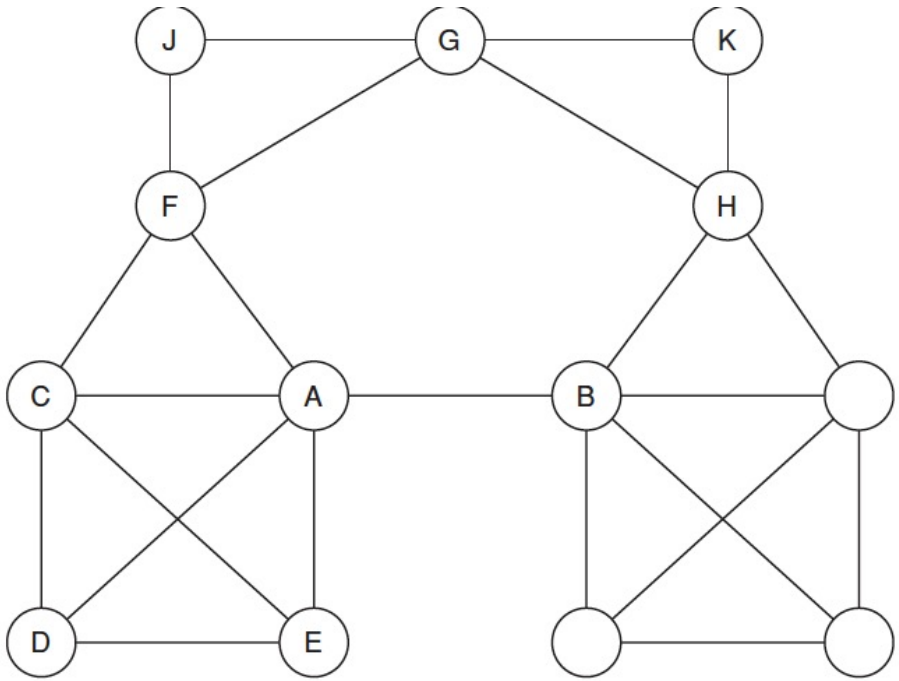
- Very few edges in the data constitute a local bridge
- We can relax the definition and consider **"almost" local bridges**
- Onnela et al. define the **neighborhood overlap** of an edge connecting A and B as:

$$\frac{\text{Number of nodes who are neighbors of both A and B}}{\text{Number of nodes who are neighbors of at least one of A or B}}$$

Structure and Tie Strengths in Mobile Communication Networks

Local Bridge

- Example: Edge A-F
- Nodes who are neighbors of both A and F: C (1)
- Nodes who are neighbors of at least one of A or F: B, C, D, E, G, J (6)
- Neighborhood overlap $= \frac{1}{6}$



Structure and Tie Strengths in Mobile Communication Networks

- This ratio is 0 when the numerator is equal to 0, that is..?

$$\frac{\text{Number of nodes who are neighbors of both A and B}}{\text{Number of nodes who are neighbors of at least one of A or B}}$$

Structure and Tie Strengths in Mobile Communication Networks

- This ratio is 0 when the numerator is equal to 0, that is..?
- When the edge is a local bridge!
- Edges with small neighborhood overlap can be considered as "almost" local bridges

$$\frac{\text{Number of nodes who are neighbors of both A and B}}{\text{Number of nodes who are neighbors of at least one of A or B}}$$

Structure and Tie Strengths in Mobile Communication Networks

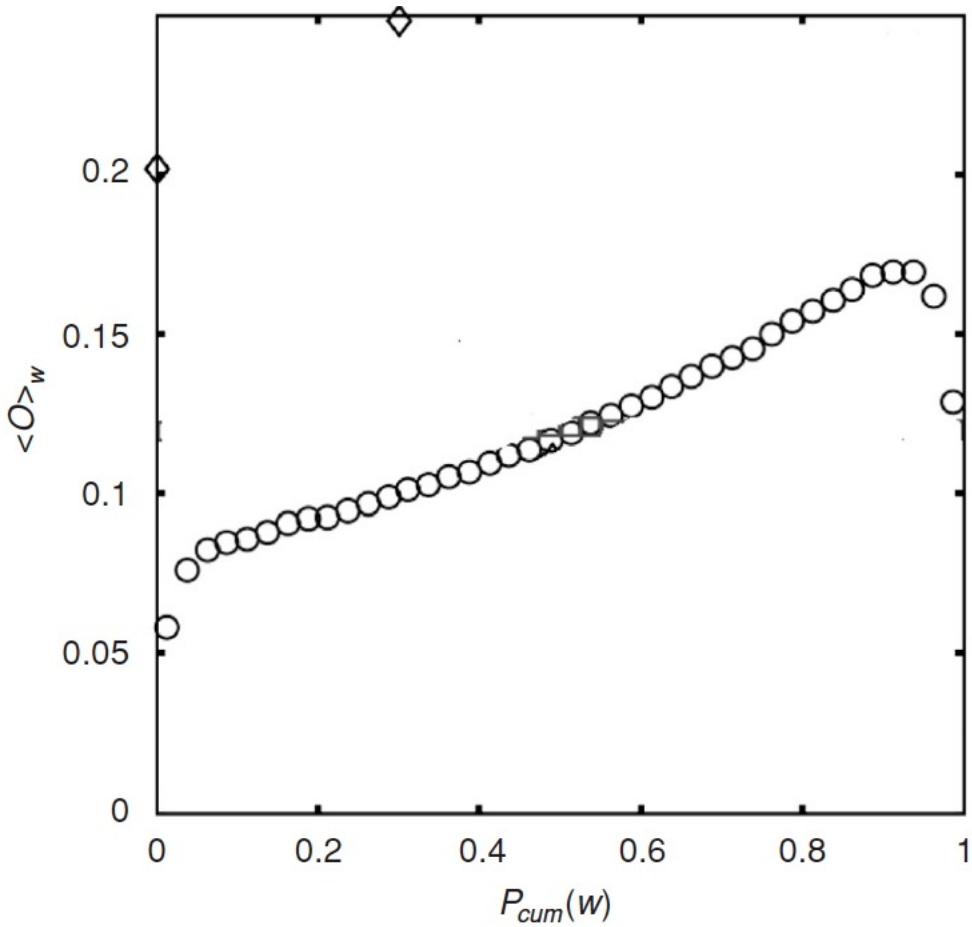
- Using these definitions, Onnela et al. formulated a series of questions based on Granovetter's predictions

1) Does the neighborhood overlap depends on its strength?

(The strength of weak ties predicts that the neighborhood overlap should grow as tie strength grows)

Structure and Tie Strengths in Mobile Communication Networks

- Neighborhood overlap of edges as a function of their percentile on the sorted order of all edges by tie strength (x-axis)
- The overlap increases with increasing tie strength



Structure and Tie Strengths in Mobile Communication Networks

2) Do weak ties serve to connect different tightly knit communities each containing a larger number of stronger ties?

Structure and Tie Strengths in Mobile Communication Networks

2) Do weak ties serve to connect different tightly knit communities each containing a larger number of stronger ties?

Onnela et al. addressed this question as follows:

- They first deleted edges from the network one at a time, in decreasing order of strength
- The giant component shrank steadily, reducing in size gradually
- Then, they did the same but starting from the weakest ties
- The giant component shrank more rapidly
- Its remnants broke apart abruptly once a critical number of weak ties had been removed

Structure and Tie Strengths in Mobile Communication Networks

- What can we learn from this?



Structure and Tie Strengths in Mobile Communication Networks

- What can we learn from this?
- Weak ties provide the more crucial connective structure for holding together different communities and keeping the giant component intact

Tie Strength on Facebook

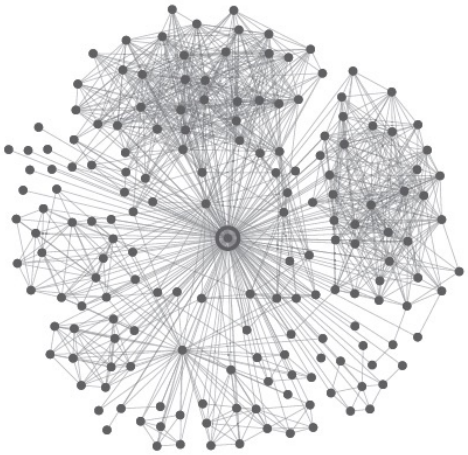
- In 2009, researchers at Facebook analyzed the friendship links reported in users' profiles
- To what extent is each link used for social interaction?
- In other words: What are the strong ties among a user's friends?

Tie Strength on Facebook

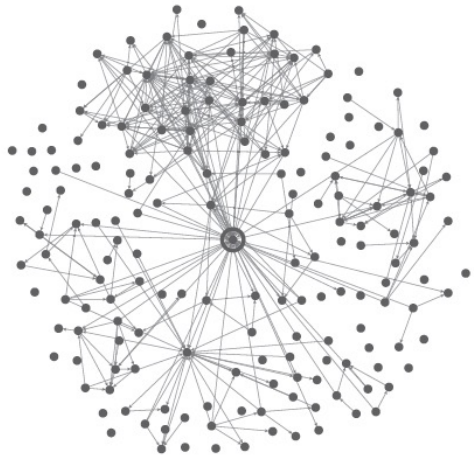
- They defined 3 categories of links:
  - **Mutual communication**: both users sent messages to each other
  - **One-way communication**: the user sent one or more messages to the other end (whether or not they were reciprocated)
  - **Maintained relationship**: the user followed information about the friend (whether or not a communication took place)
- The categories are not mutually exclusive

Tie Strength on Facebook

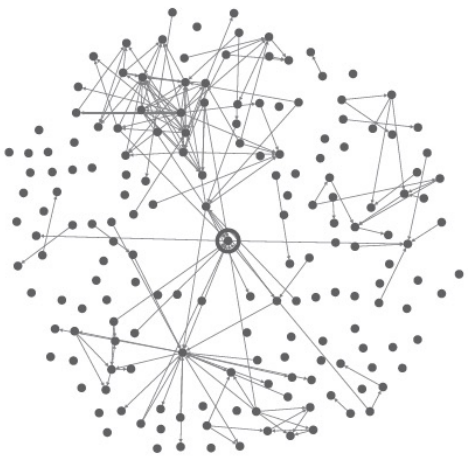
All Friends



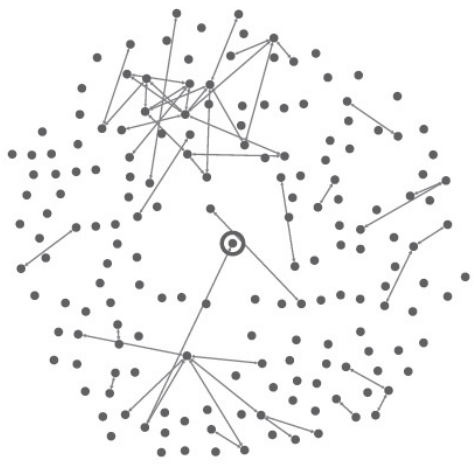
Maintained Relationships



One-way Communication

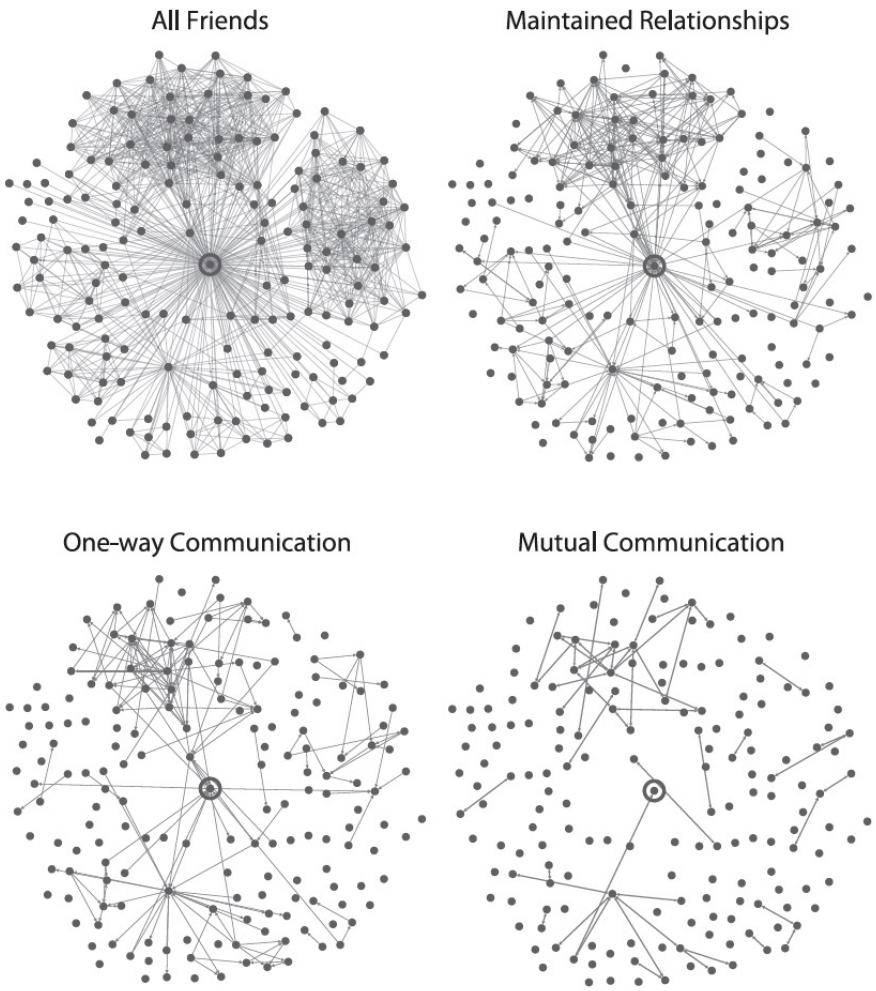


Mutual Communication

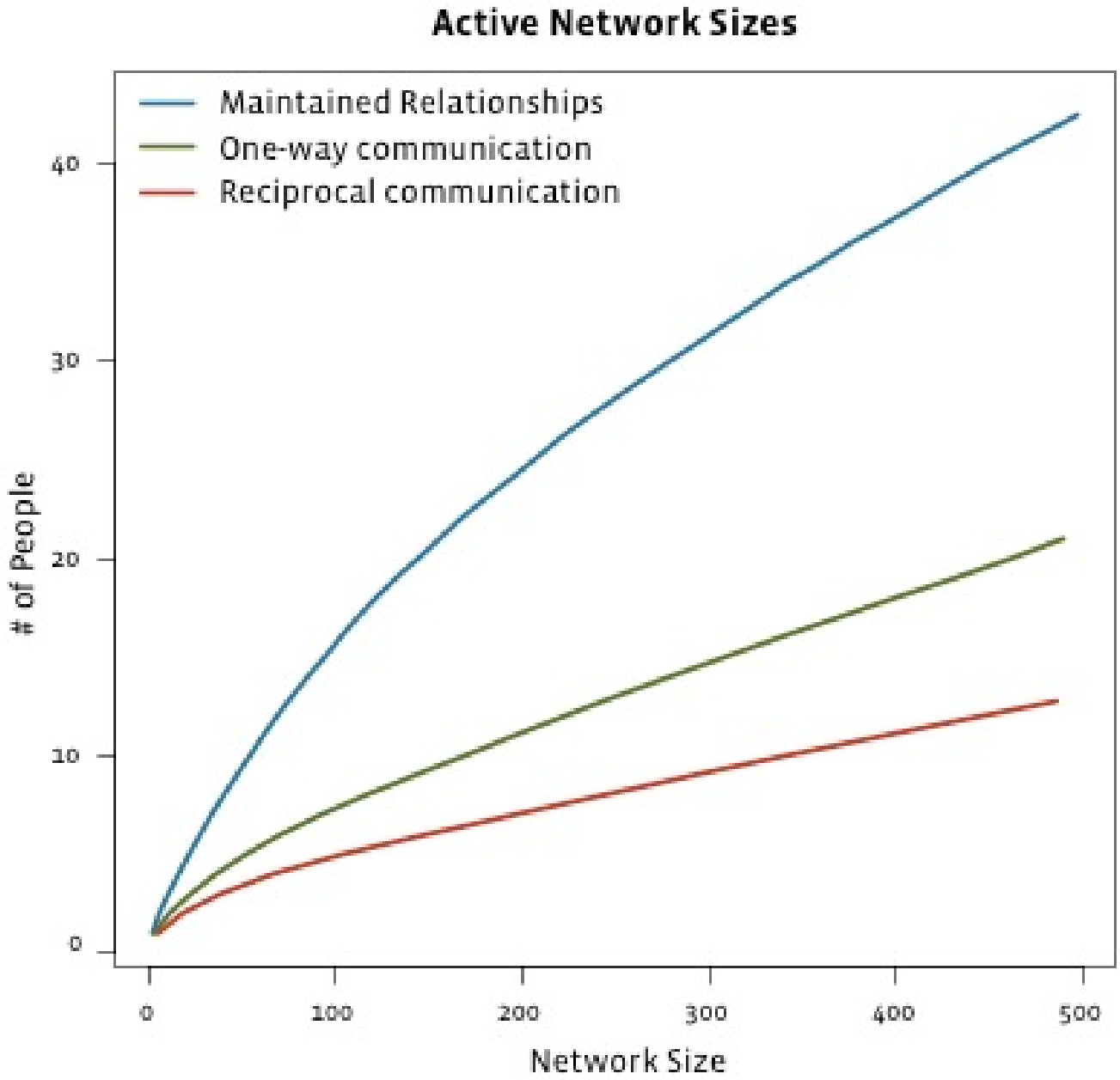


Tie Strength on Facebook

- 2 distinct regions with a large amount of triadic closure
- In communication and maintained relationships, many more links survive in the upper region



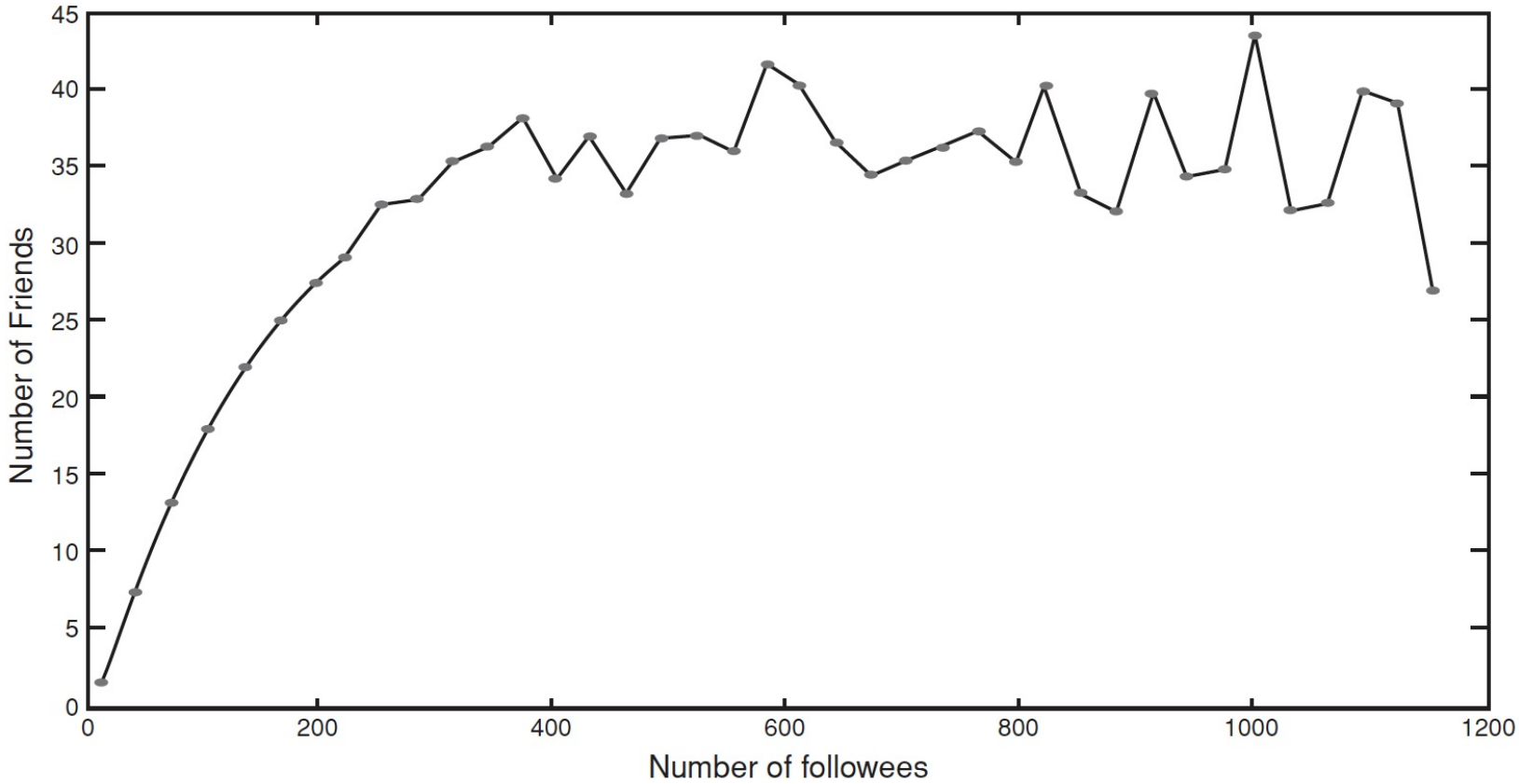
Tie Strength on Facebook



Tie Strength on Twitter

- Huberman, Romero, and Wu did something similar on Twitter
- For each user they considered the number of his/her followees
- They defined strong ties as the users to whom s/he had directed at least two messages

Tie Strength on Twitter



Closure, Structural Holes, and Social Capital



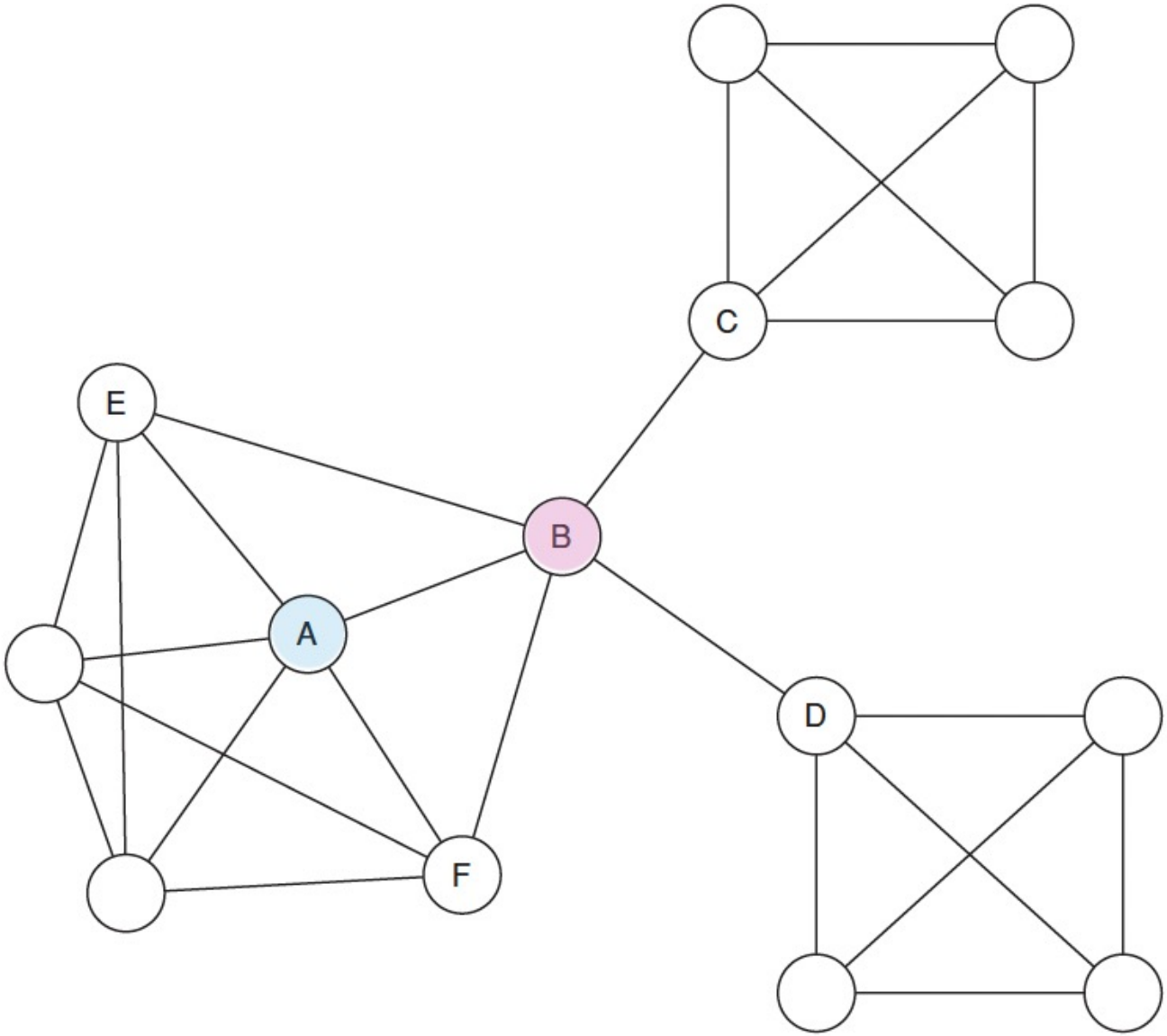
Introduction

- We have focused so far on the role played by different kind of edges in a SN
- What about **nodes**?

Introduction

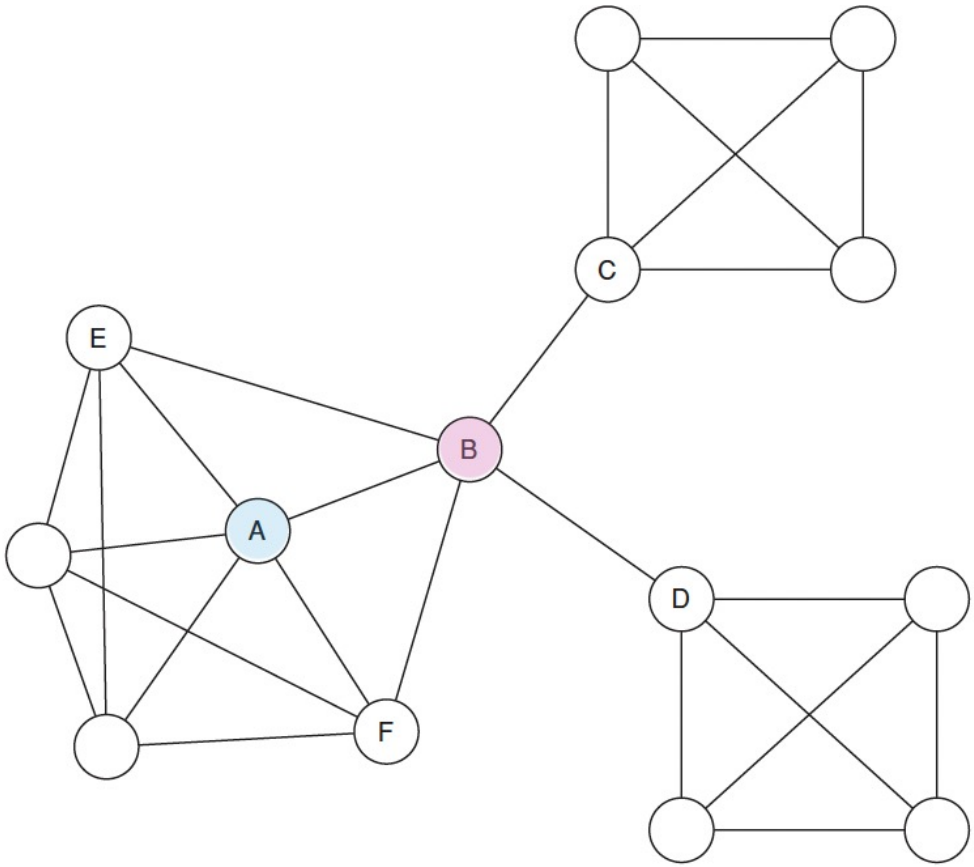
- We have focused so far on the role played by different kind of edges in a SN
- What about **nodes**?
- Not all nodes are equal

Introduction



Introduction

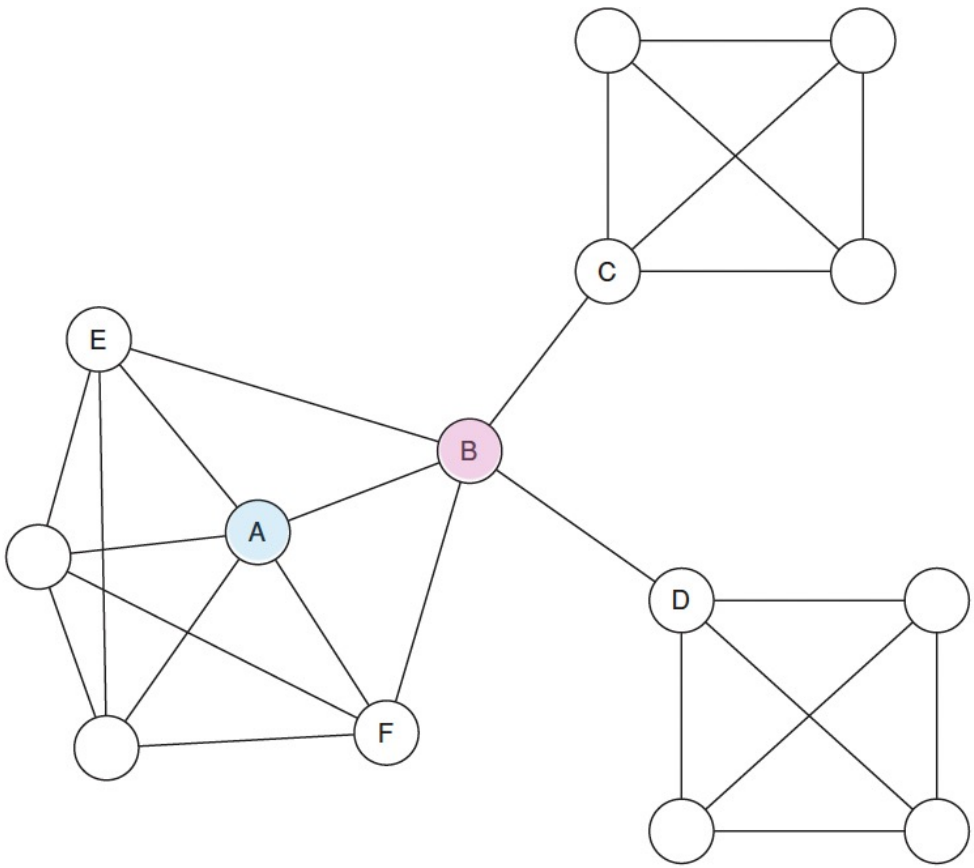
- A was subject to considerable triadic closure
- A has a high clustering coefficient



Embeddedness

*We define the **embeddedness** of an edge to be the number of common neighbors shared by two endpoints*

- Edge A-B has an embeddedness of 2



Embeddedness

Notice that:

1) embeddedness = the numerator in the ratio defining the **neighborhood overlap**:

$$\frac{\text{Number of nodes who are neighbors of both A and B}}{\text{Number of nodes who are neighbors of at least one of A or B}}$$

Embeddedness

Notice that:

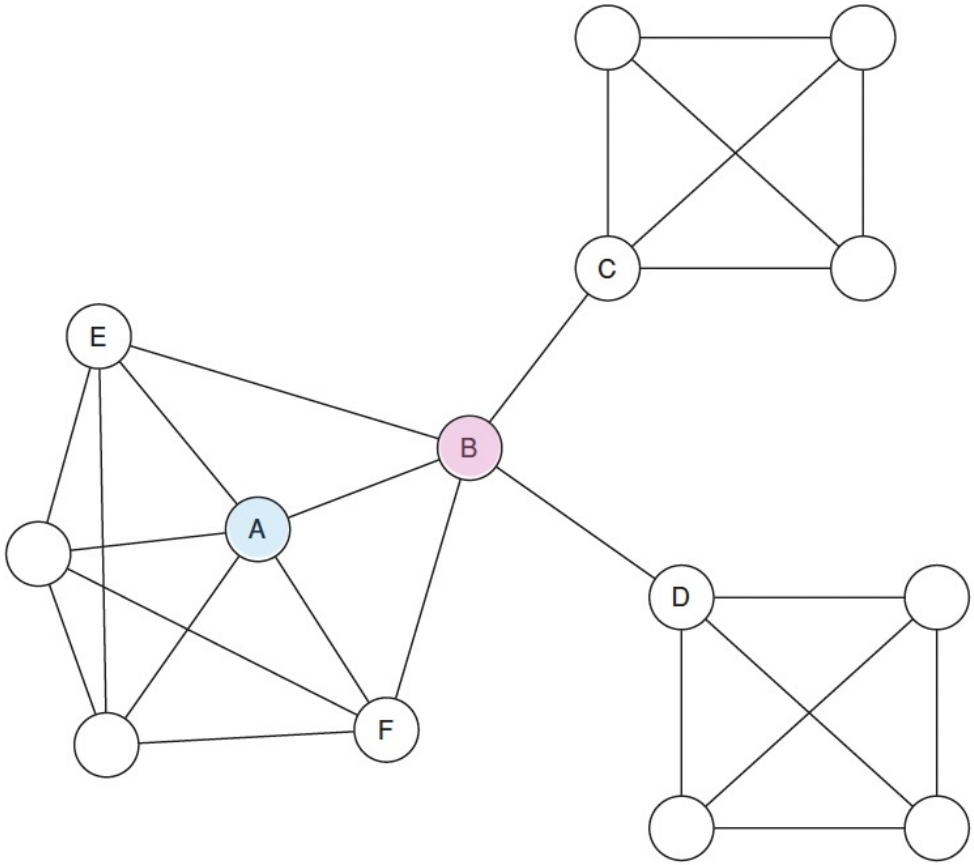
2) local bridges are edges with embeddedness $= 0$

$$\frac{\text{Number of nodes who are neighbors of both A and B}}{\text{Number of nodes who are neighbors of at least one of A or B}}$$

An edge joining two nodes A and B in a graph is a ***local bridge*** *if its endpoints A and B have no friends in common*

Embeddedness

- All the edges of A have significant embeddedness



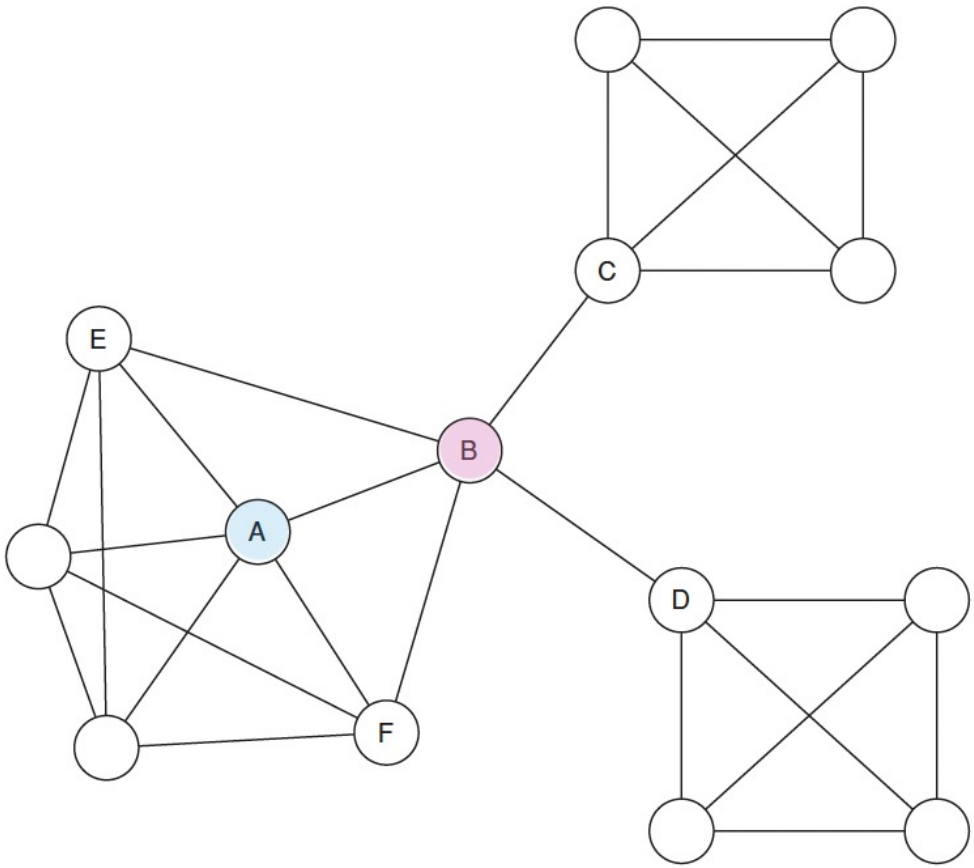
Embeddedness

- If two individuals are connected by an embedded edge $\rightarrow$ more trust and confidence in the transactions (social, economic, etc..) between them
- Why?



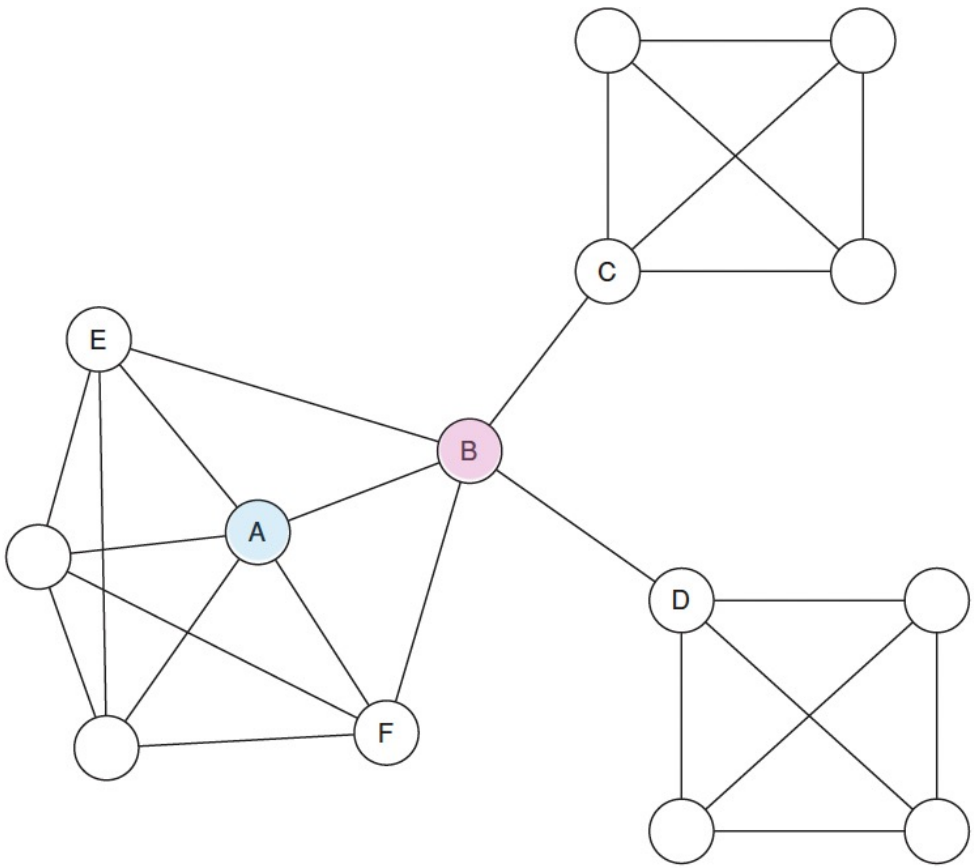
Embeddedness

- The presence of mutual friends makes the interactions between two people "public"
- Misbehavior can have reputational consequences
- A can trust his embedded interactions



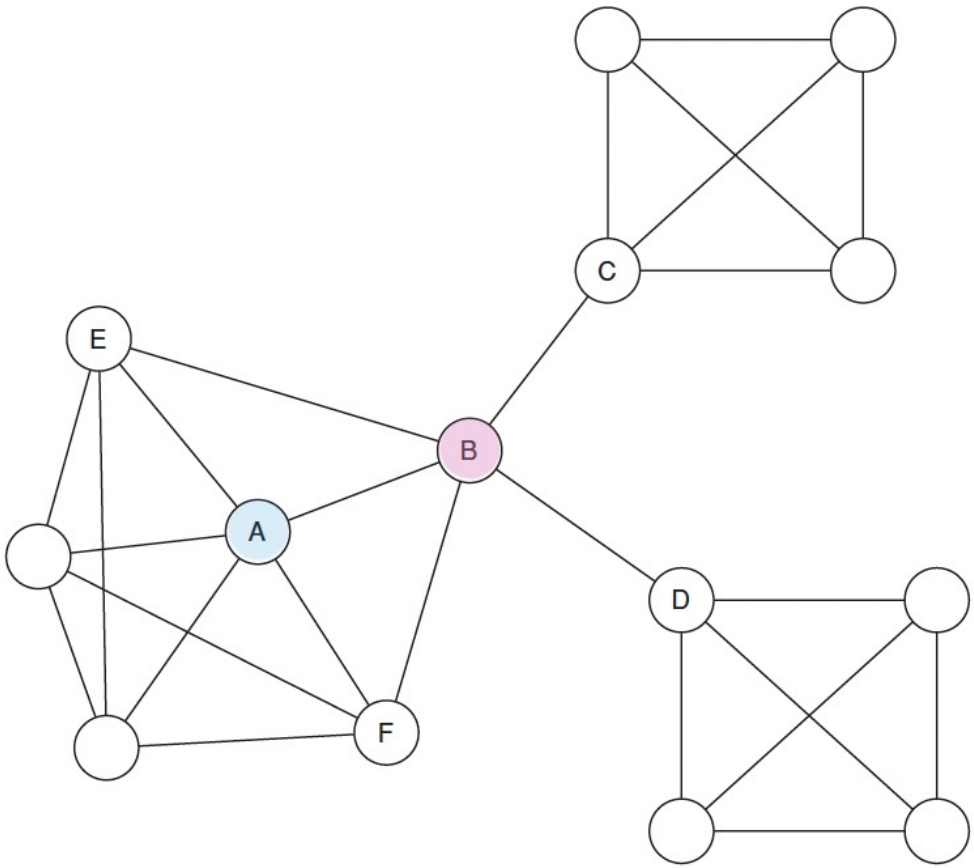
Embeddedness

- This threat does not exist for edges with embeddedness = 0
- The interactions of B with C and D are riskier
- B has more complicated constraints (norms and expectations from different groups)



Structural Holes

- So the node A has a series of advantages for its embedded position
- But the position of B at the end of multiple local bridges can also offer fundamental advantages Burt, 1996

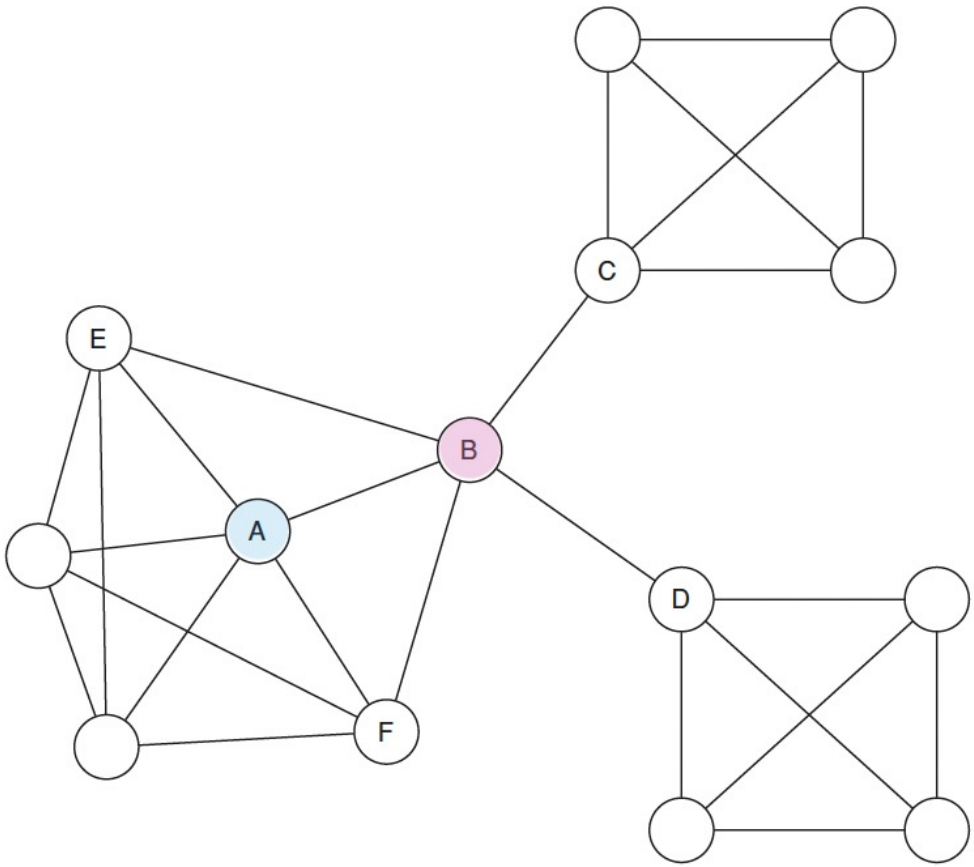


Structural Holes

- Example: social network in an organization
- Informal network of who knows whom, who talks to whom
- Empirical studies have shown that an individual's success correlates with his/er access to local bridges

Structural Holes

- Imagine the network represents interactions among managers in a company
- For Burt, node B spans a **structural hole** in the network
- B fills the empty space between two sets of nodes



Structural Holes

What are the advantages for B?



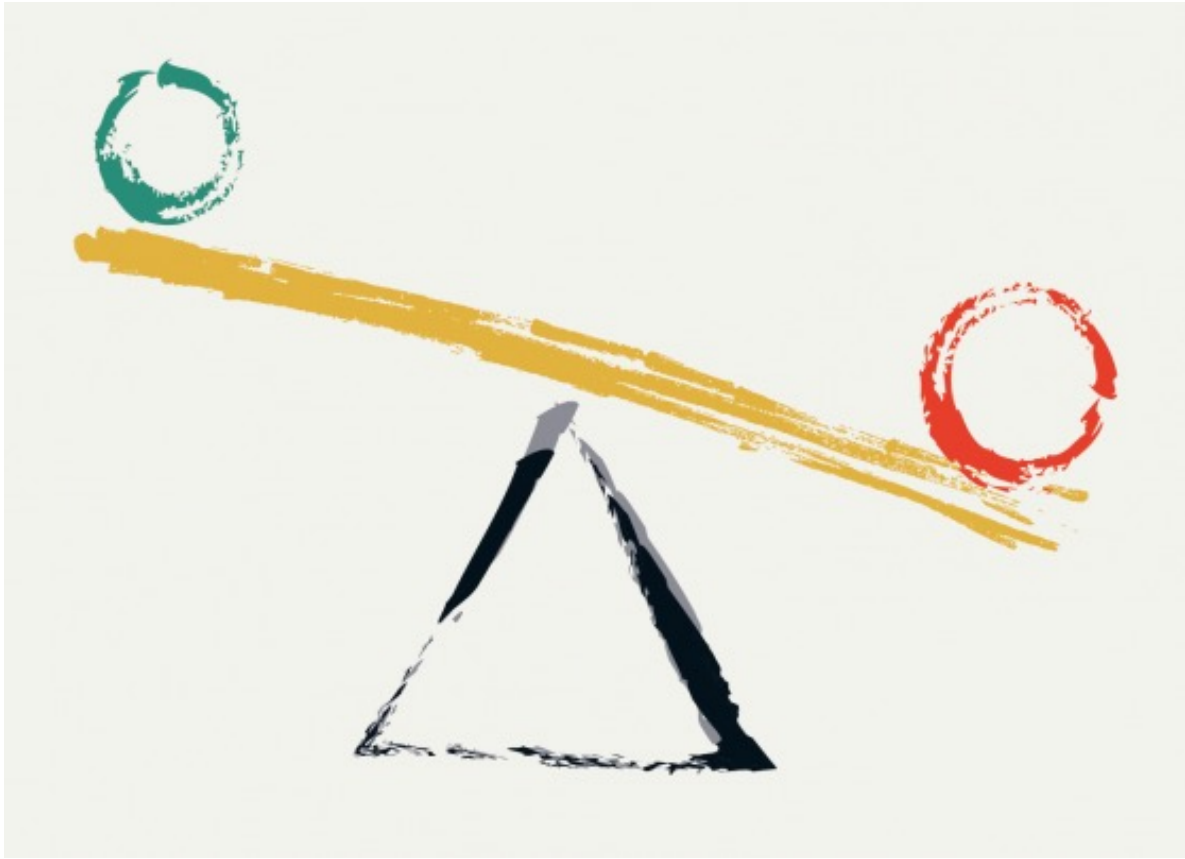
Structural Holes

What are the advantages for B?

- Access to information from multiple sources
- Opportunity for novel ideas by combining such information
- B is a *gatekeeper* $\rightarrow$ source of power
- The interests of B and the network are not aligned

Closure and Structural Holes: Trade-offs

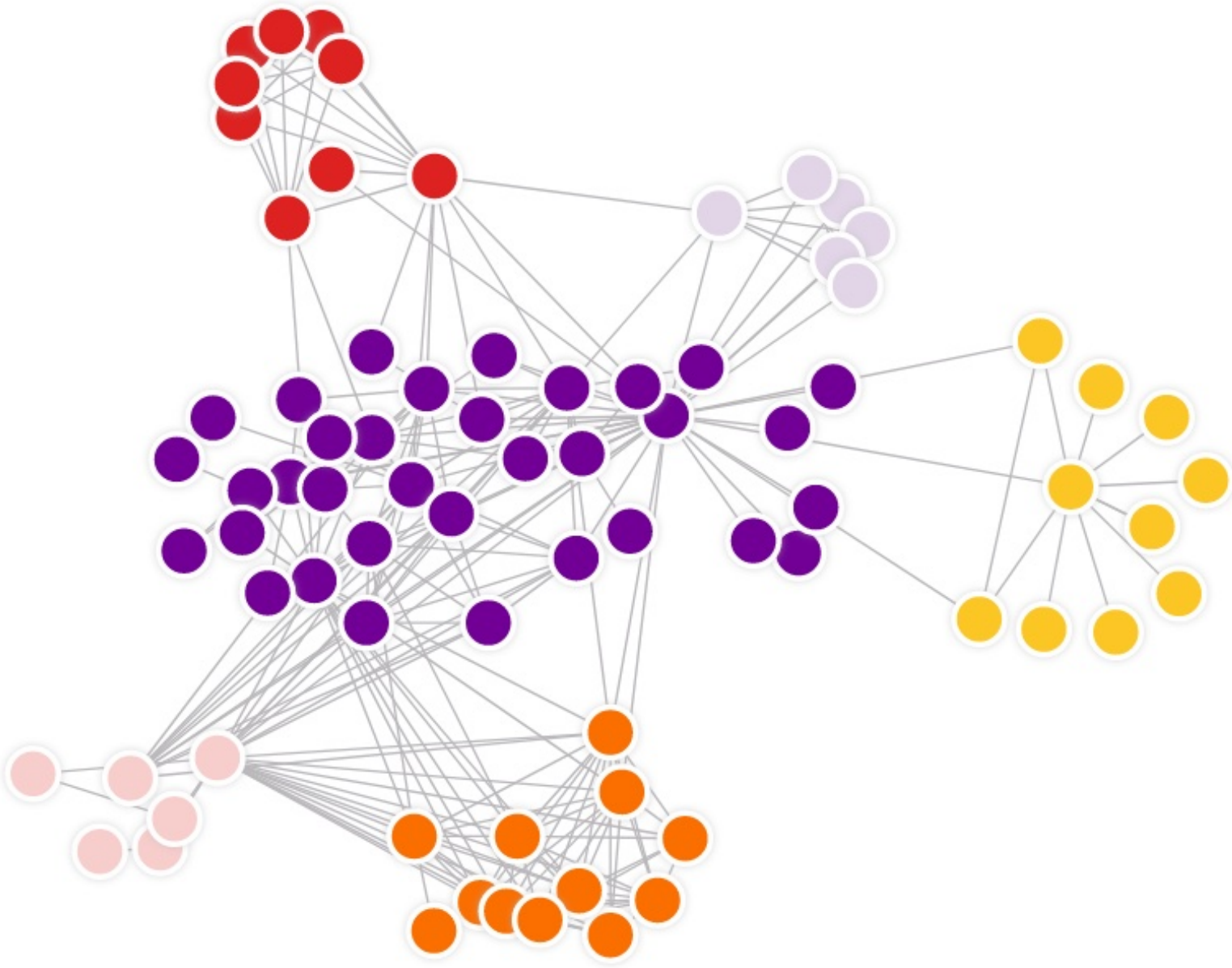
- B's interactions are less embedded within a group and less protected by the presence of mutual friends
- This position also provides B with access to valuable information



Social Capital

- All these concepts are related to the notion of **social capital**
- Social capital is difficult to define
- Borgatti, Jones and Everett summarized the definitions:
  - As a property of a group or of an individual
  - As a property intrinsic to the group
  - As a property based on the interactions of the group with the outside
- We can think of social structures as **facilitators** (for individuals ans/or groups)

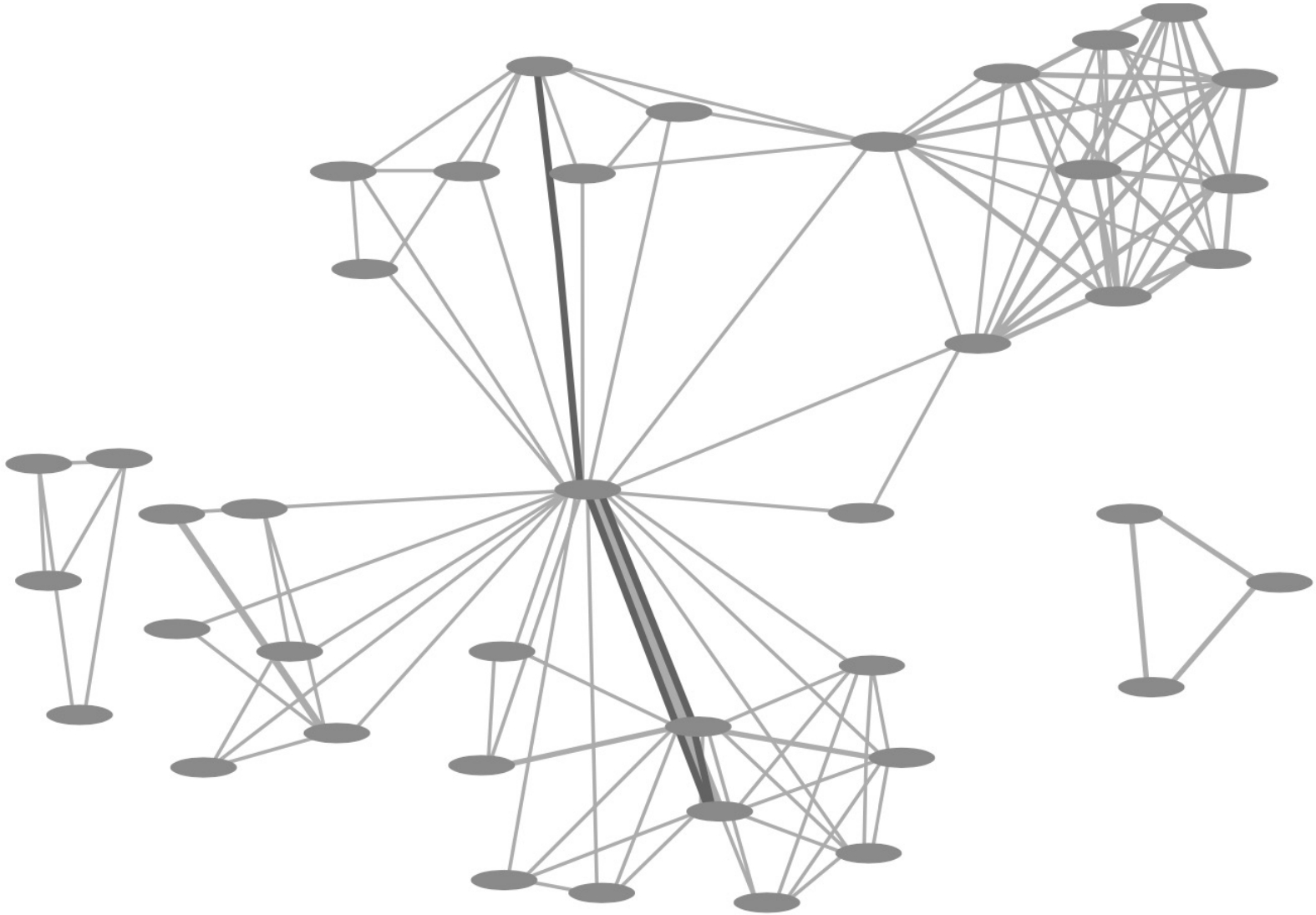
Betweenness Measures and Graph Partitioning



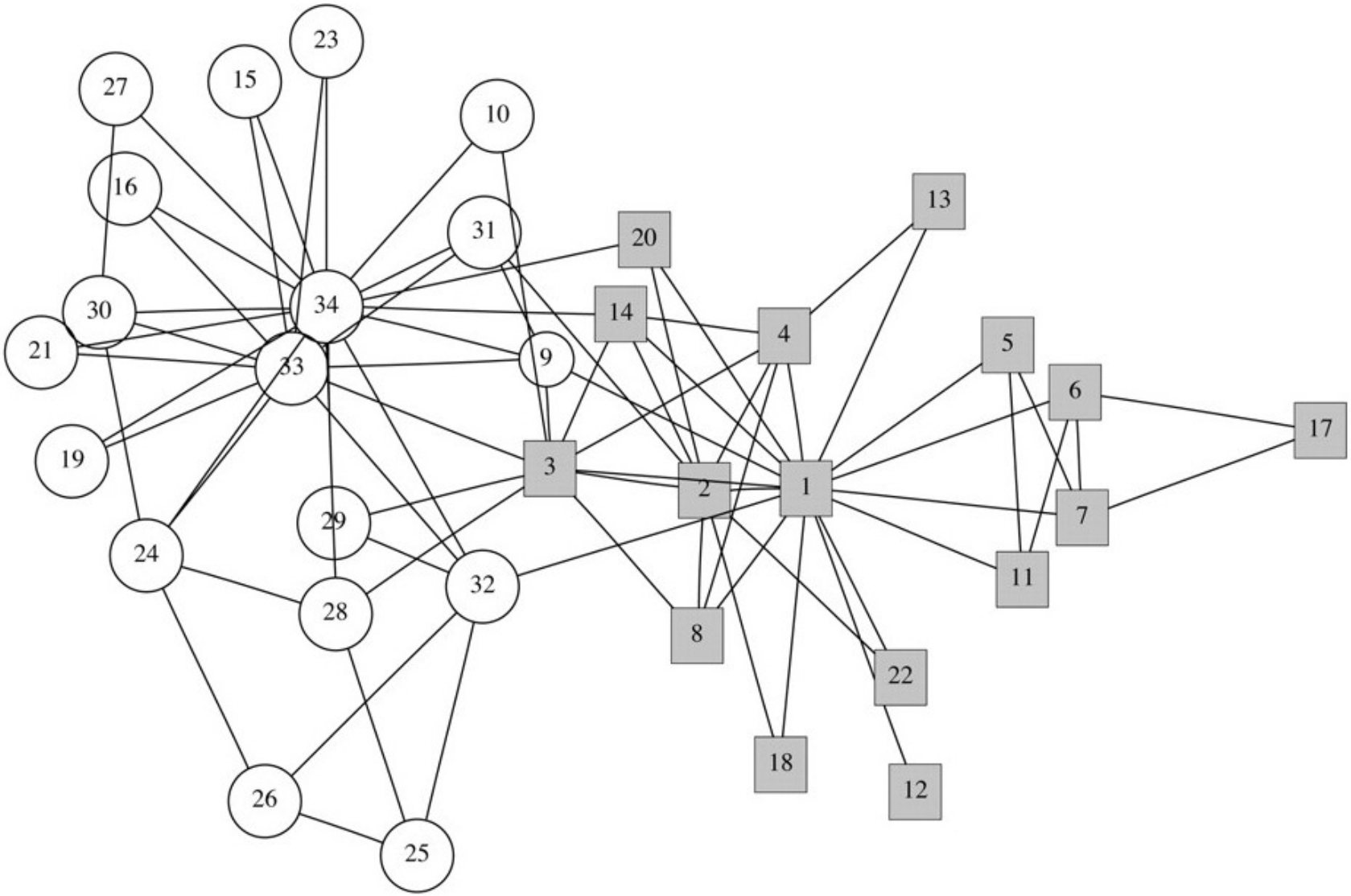
Introduction

- We will focus on methods for taking a network and breaking it down in a set of dense regions with sparse connections
- We refer to this problem as **graph partitioning**

Introduction



Introduction



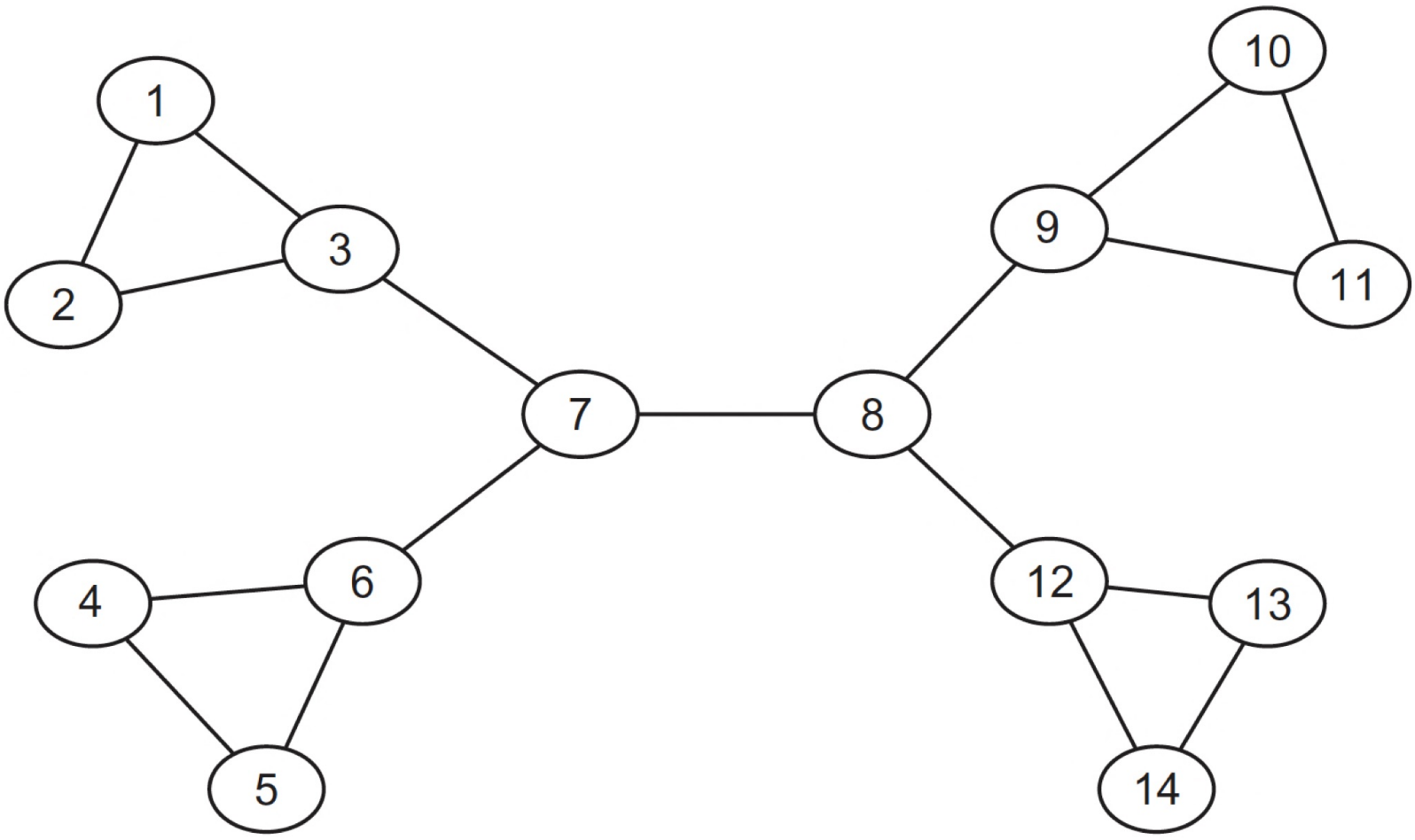
General Approaches to Graph Partitioning

Two main classes of methods:

- **Divisive**:
  - Identifying and removing the links between densely connected regions (top-down)
- **Agglomerative**:
  - Finding nodes that are likely to belong to the same region and merge them together (bottom-up)

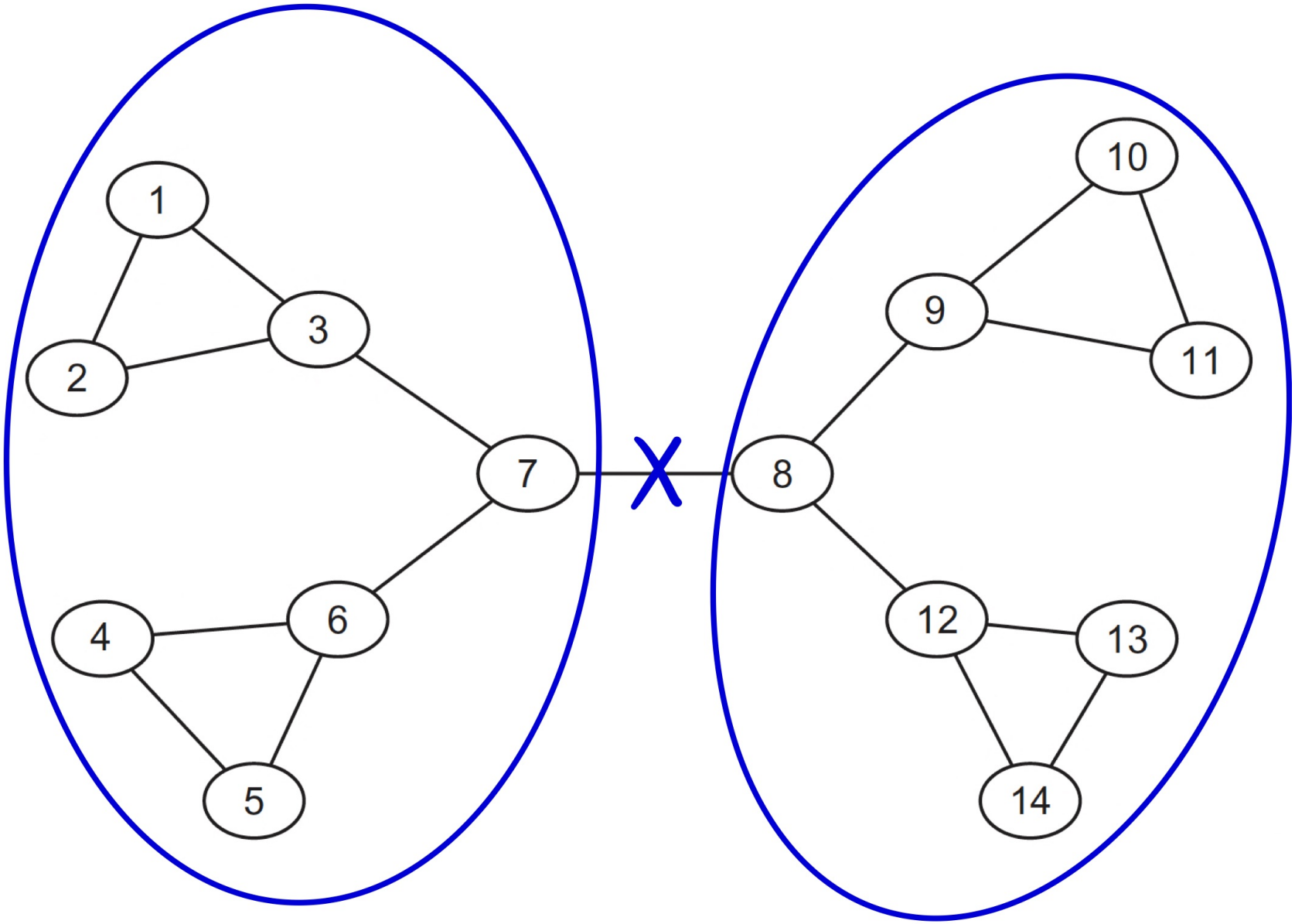
General Approaches to Graph Partitioning

Example



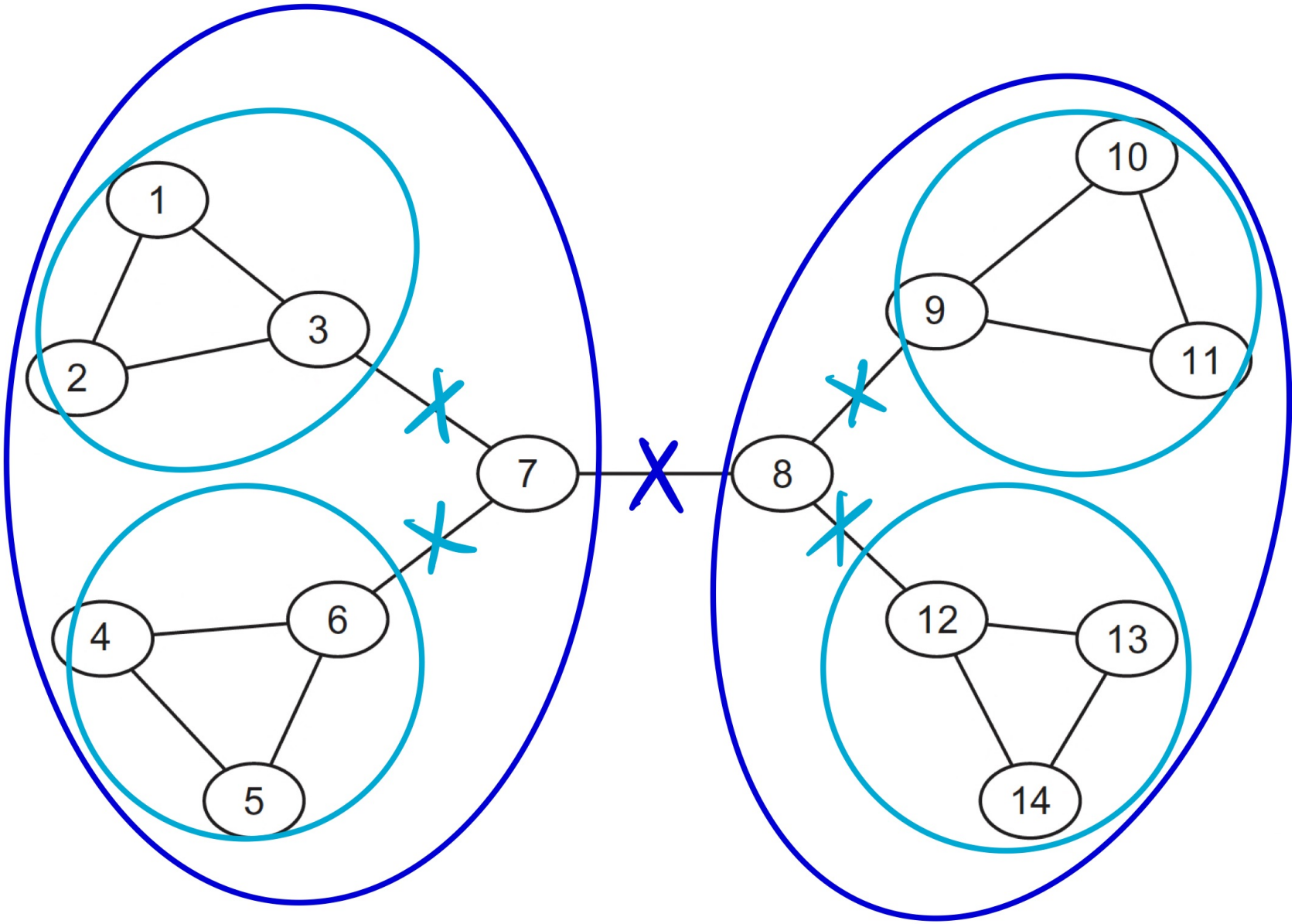
General Approaches to Graph Partitioning

Divisive methods



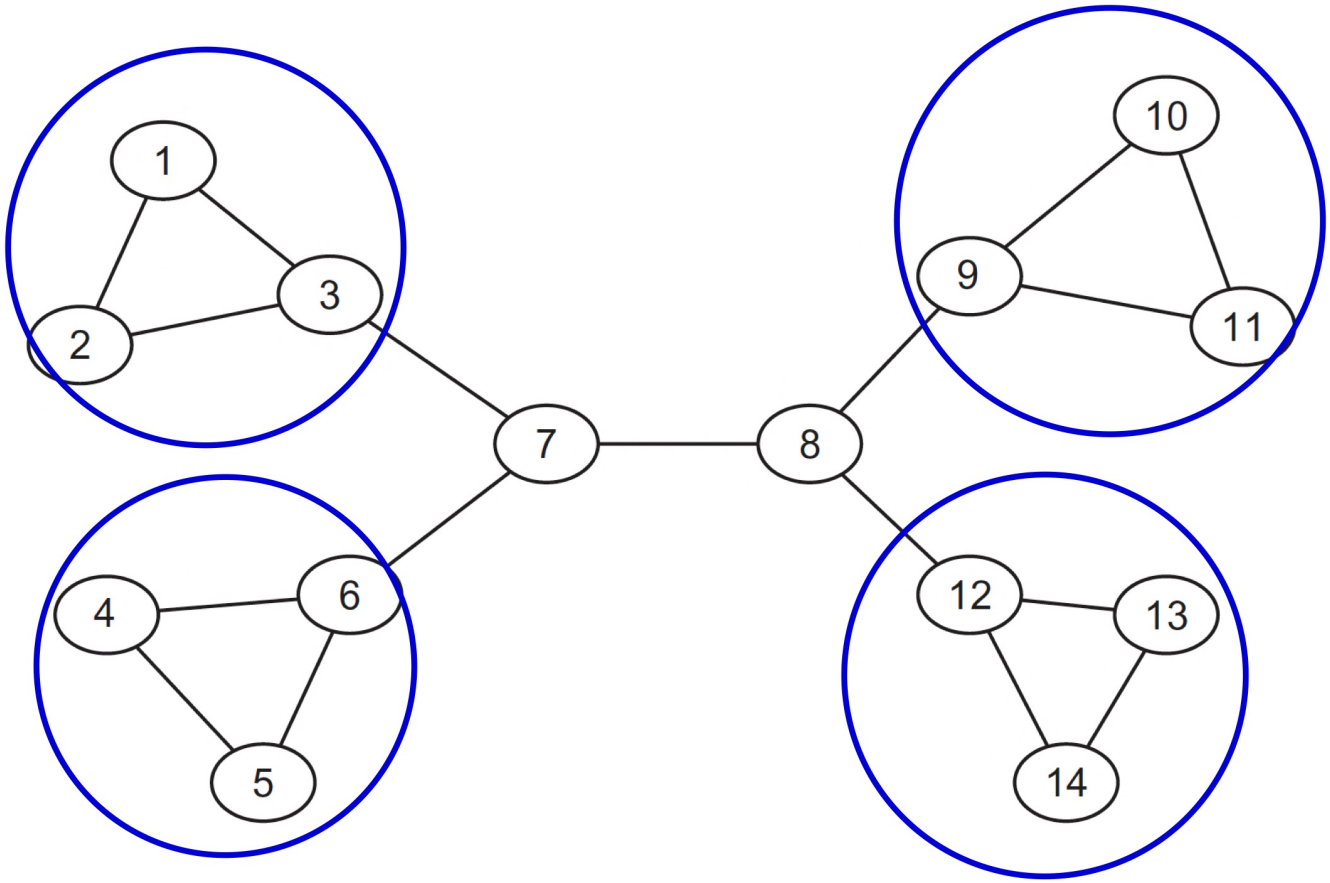
General Approaches to Graph Partitioning

Divisive methods



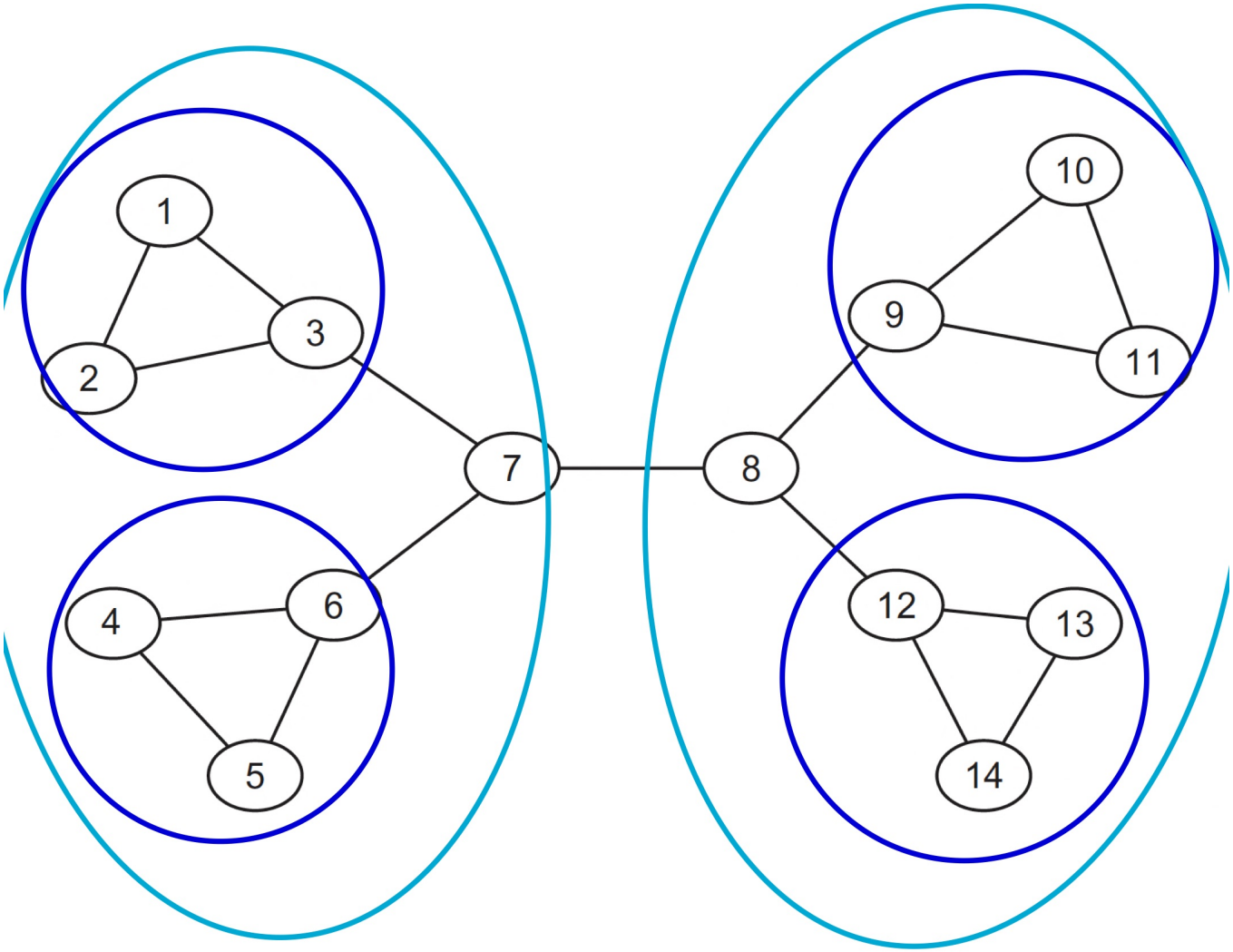
General Approaches to Graph Partitioning

Agglomerative methods



General Approaches to Graph Partitioning

Agglomerative methods

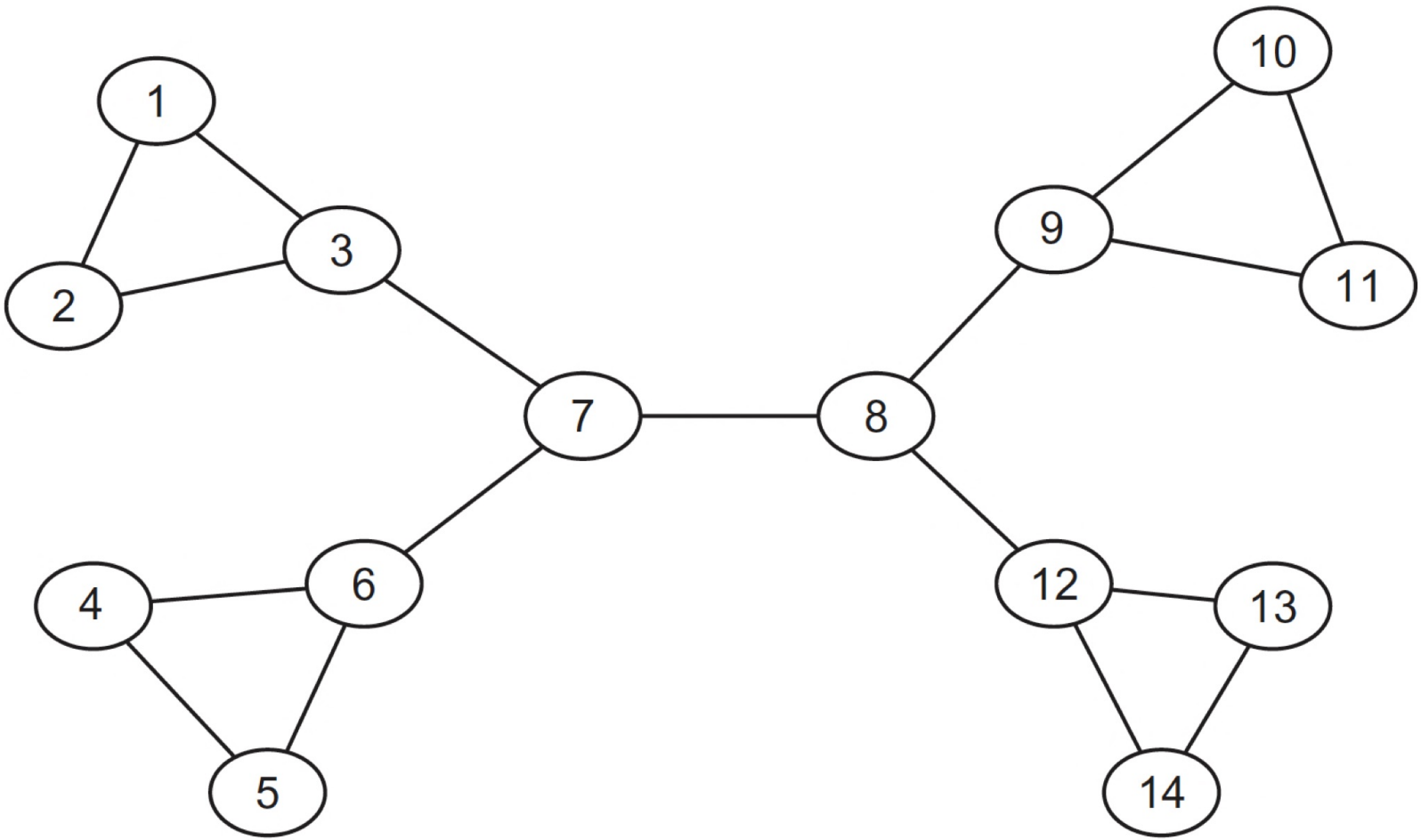


General Approaches to Graph Partitioning

- A wide range of different approaches and methods is in use
- Understanding what methods work best in specific settings is a subject of research
- Here we focus on a particular divisive method proposed by Girvan and Newman

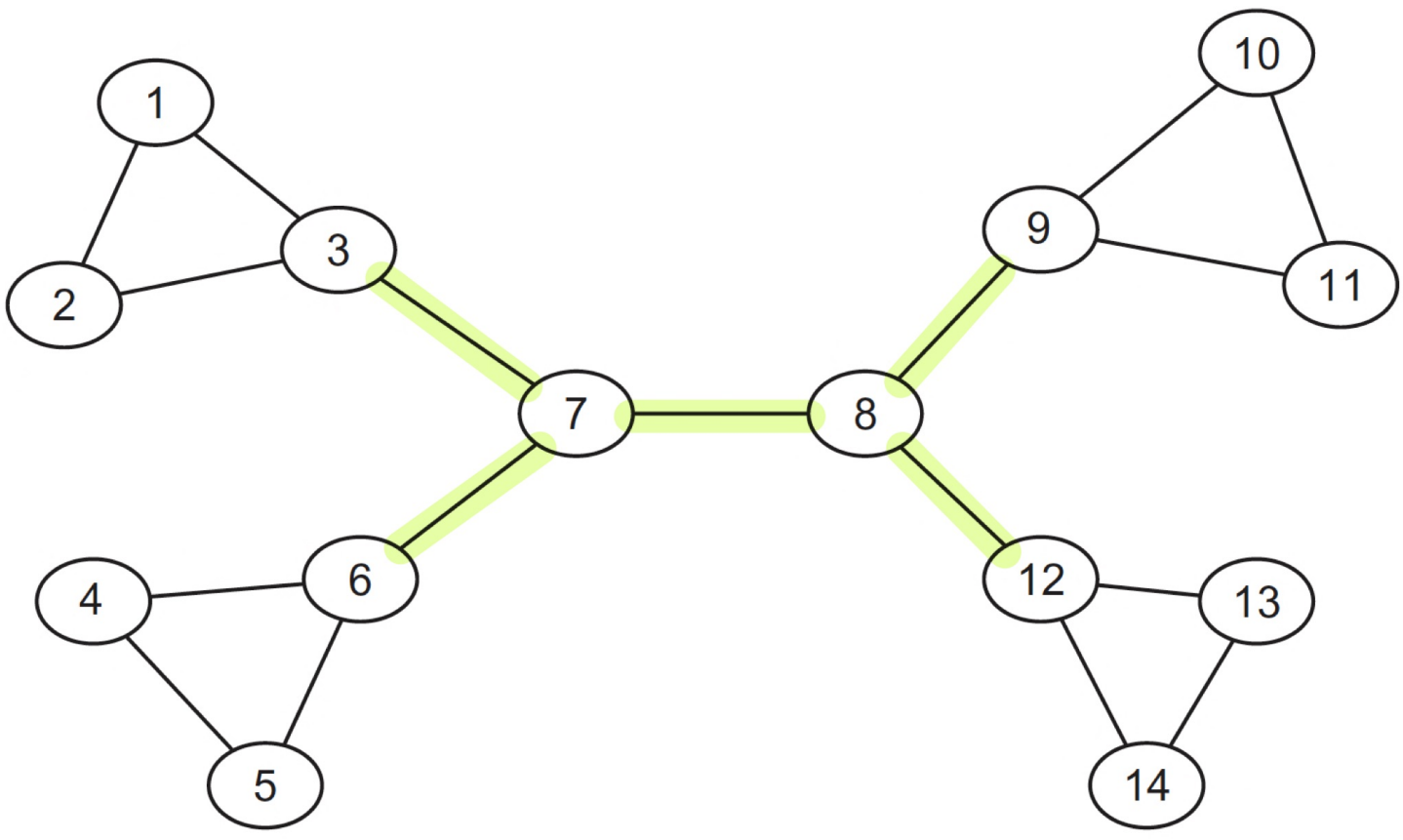
Betweenness

- How can we identify the edge(s) to be removed?



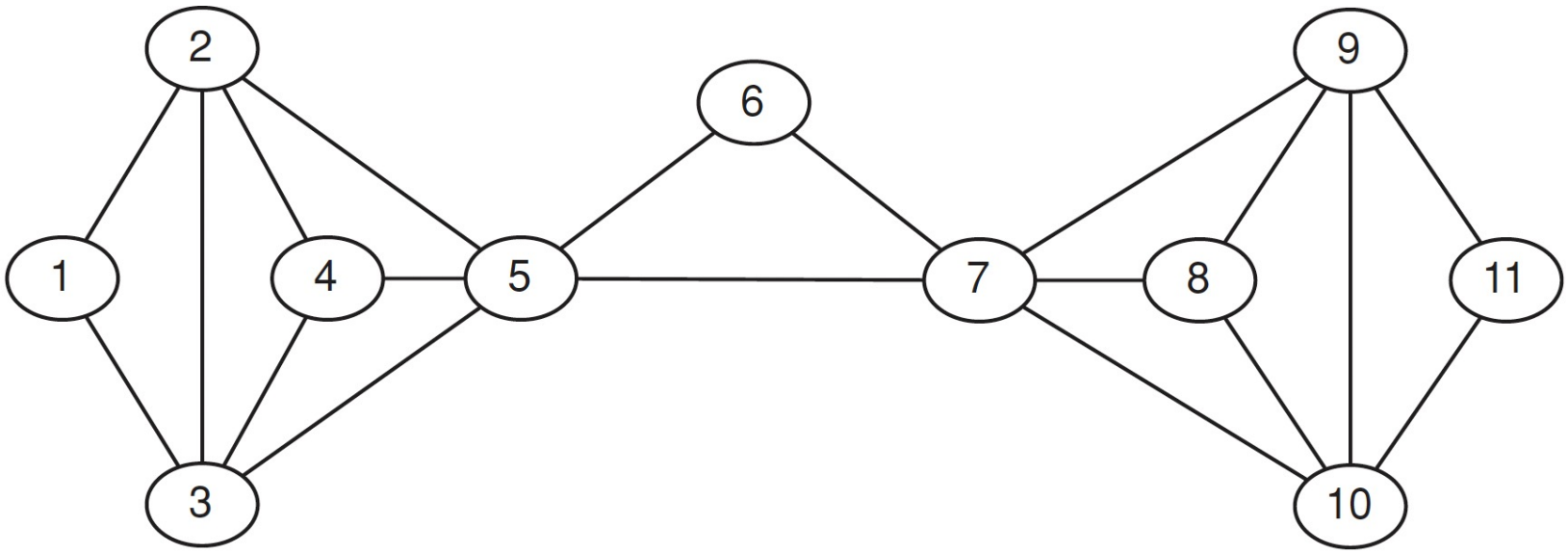
Betweenness

- Bridges and local bridges
- Problem #1:
  - There are several bridges – which to remove first?



Betweenness

- Bridges and local bridges
- Problem #2:
  - There is no local bridge



Betweenness

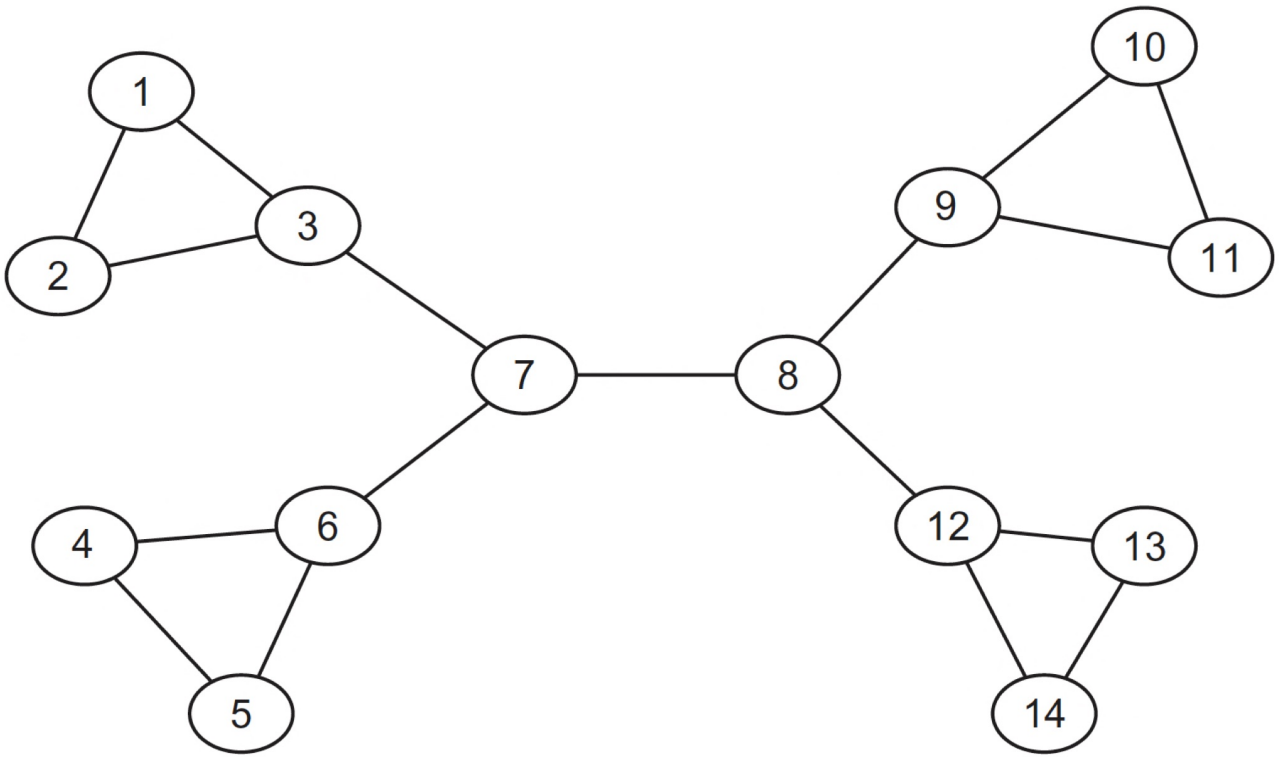
- However..
- Local bridges are important because they are **part of the shortest path** between pairs of nodes in **different regions** of the network
- Without a certain local bridge, paths might be longer

Betweenness

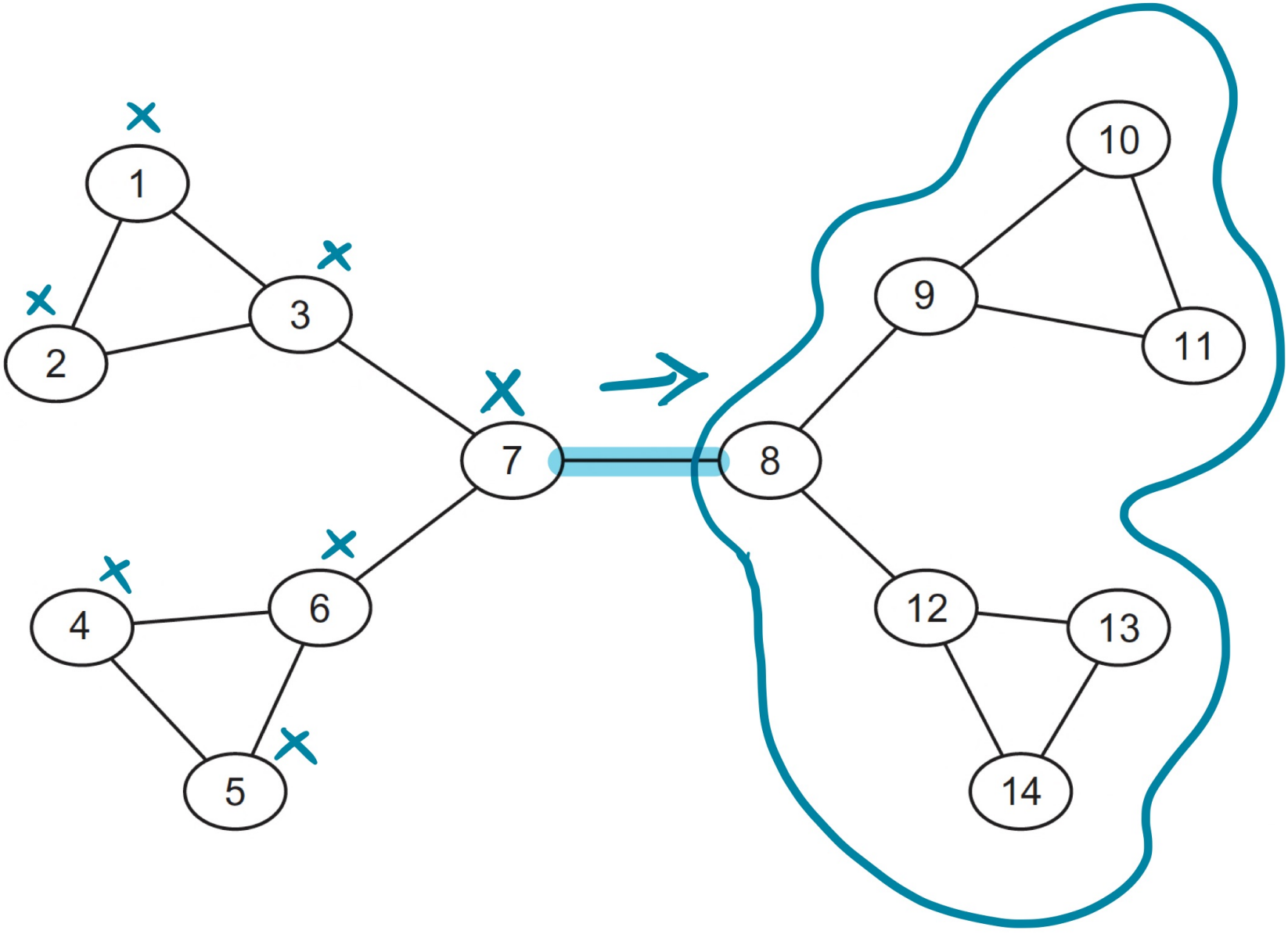
- We can define a notion of *traffic* on the network
- For every pair of nodes A and B in the graph connected by a path, we have 1 unit of flow along the edges from A to B
- This flow divides itself equally along all the possible shortest paths from A to B
- If there are k shortest paths, $\frac{1}{k}$ units of flow pass along each path

Betweenness

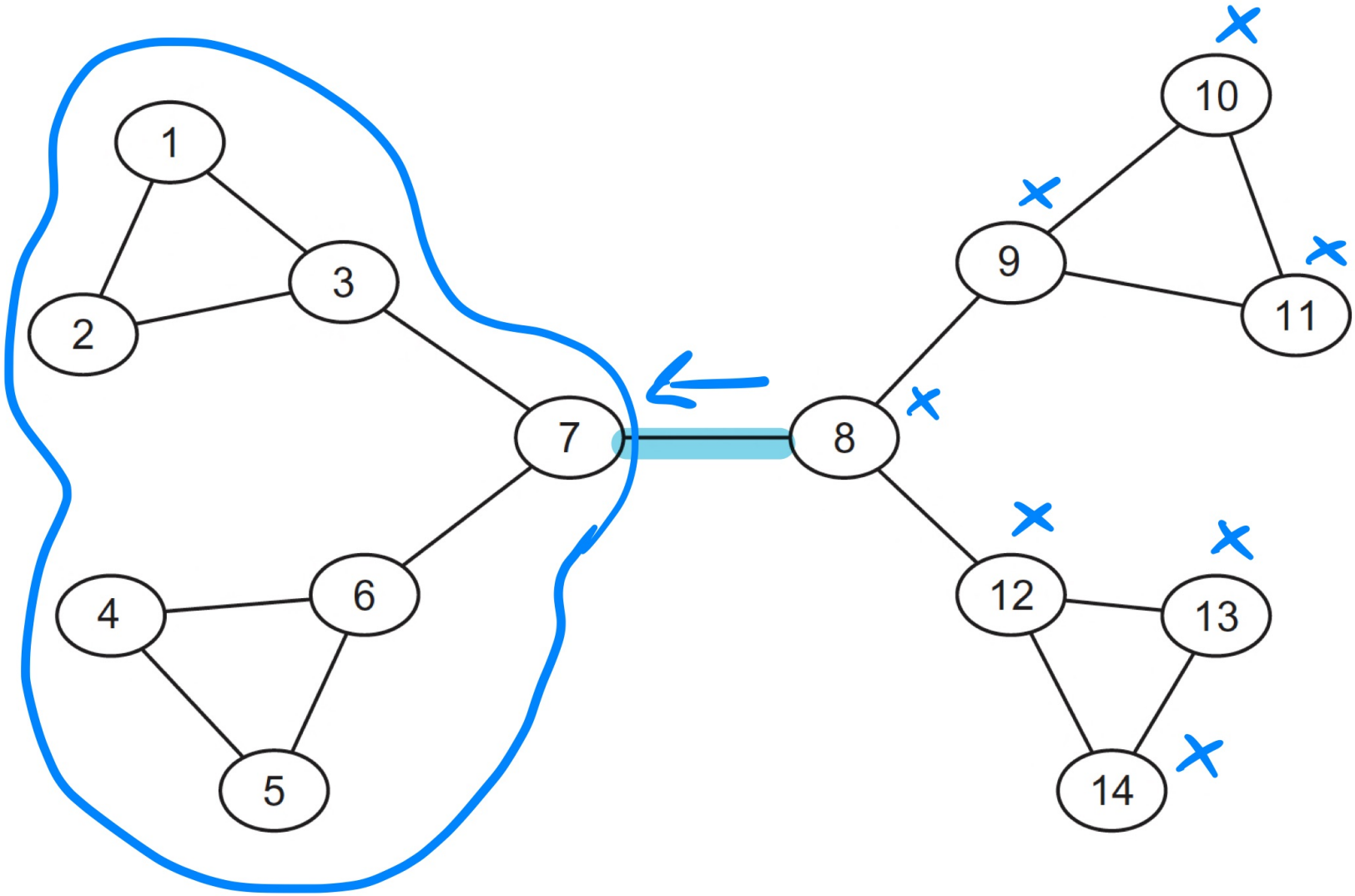
- We define the betweenness of an edge as the total amount of flow it carries
  - We take into account the flow between all pairs of nodes using the edge
- Let's see an example



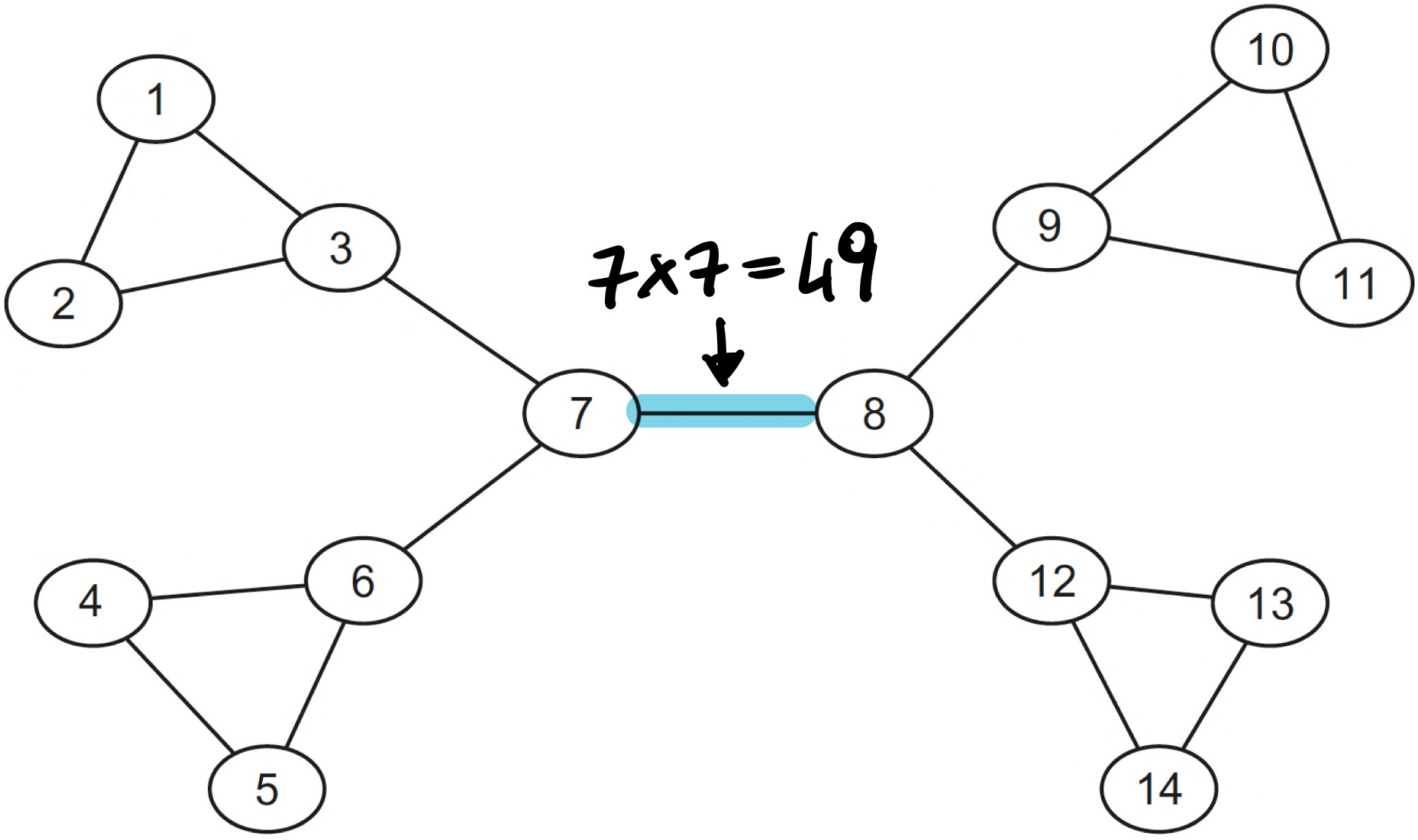
Betweenness



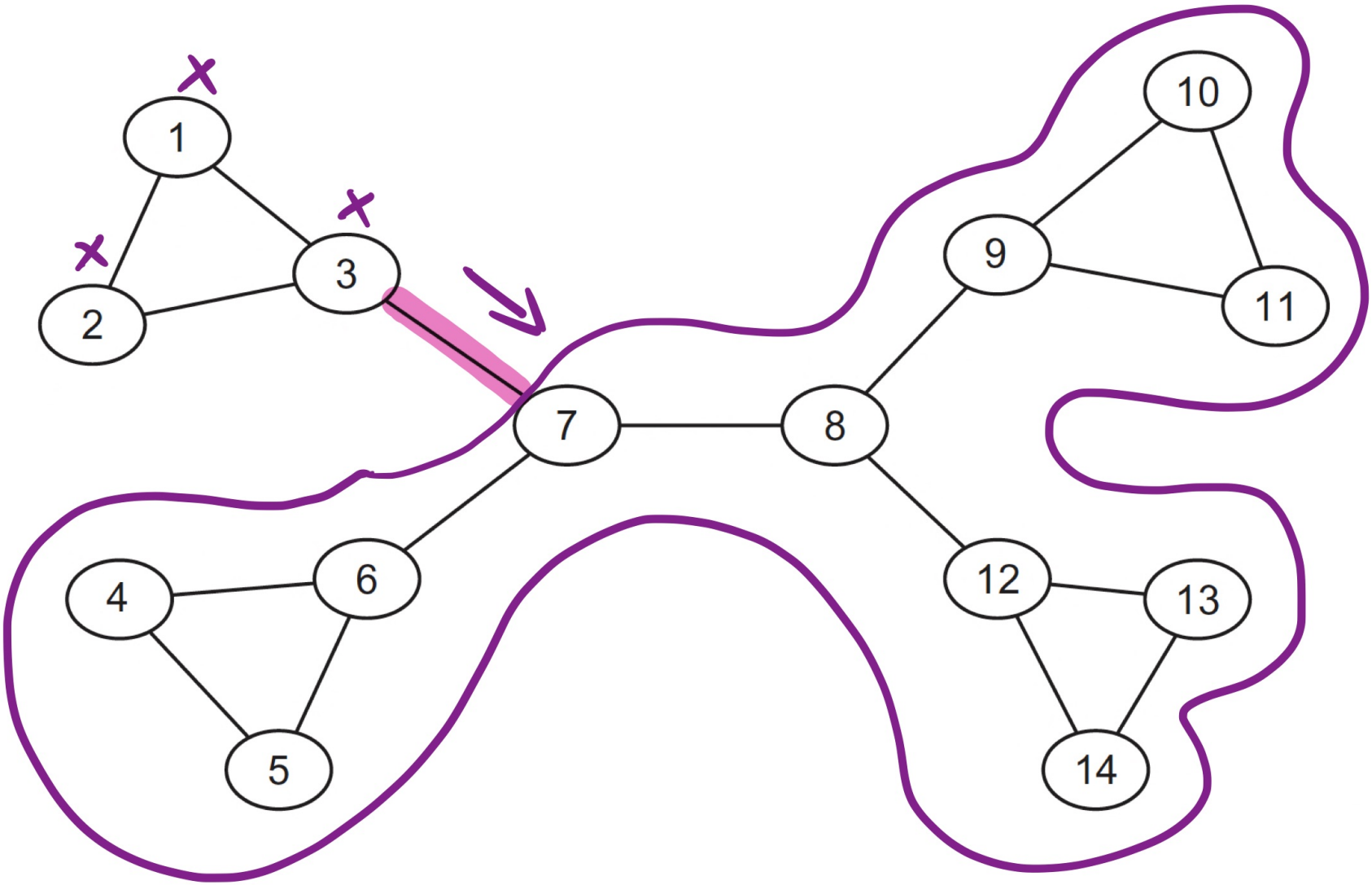
Betweenness



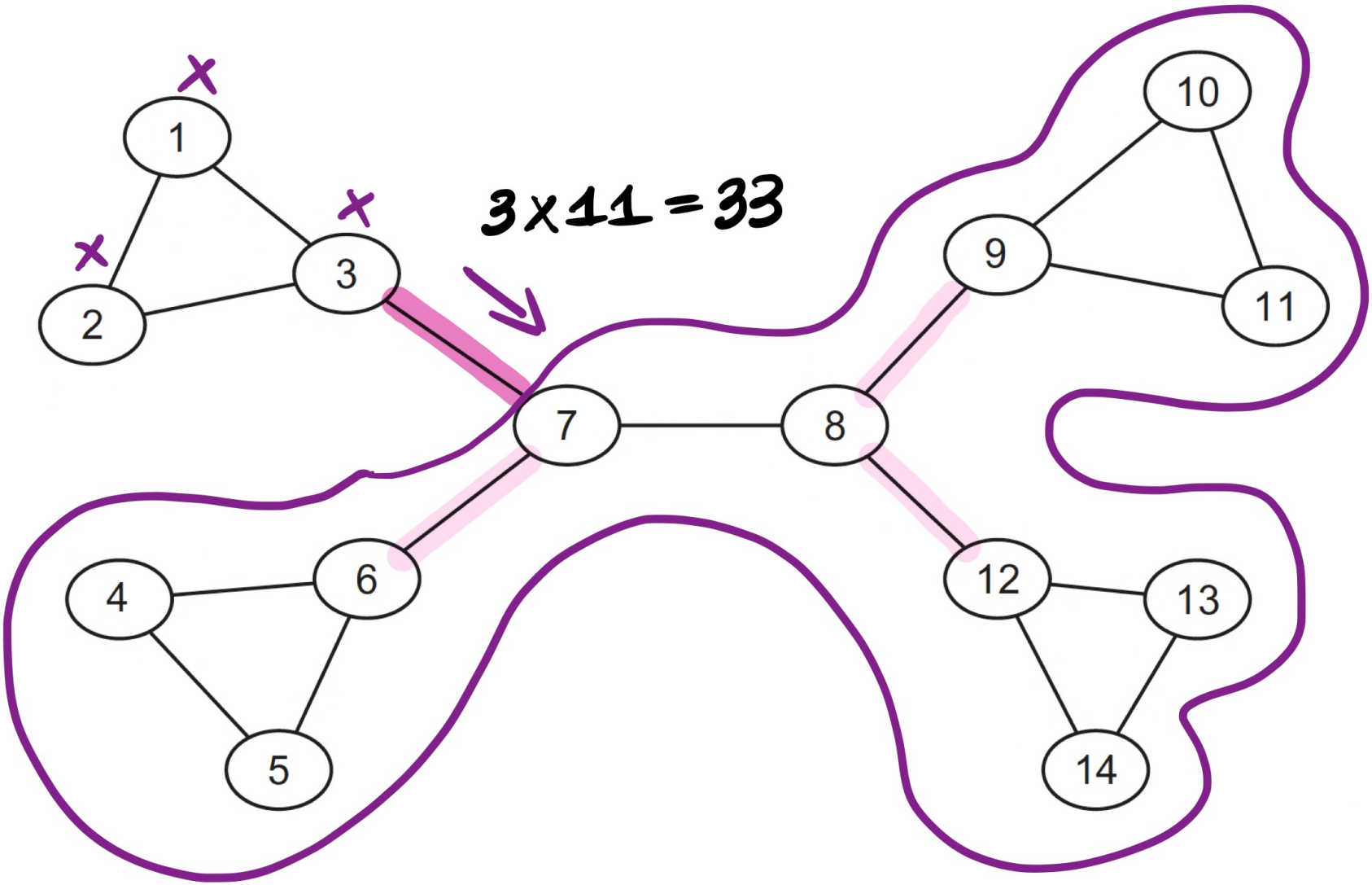
Betweenness



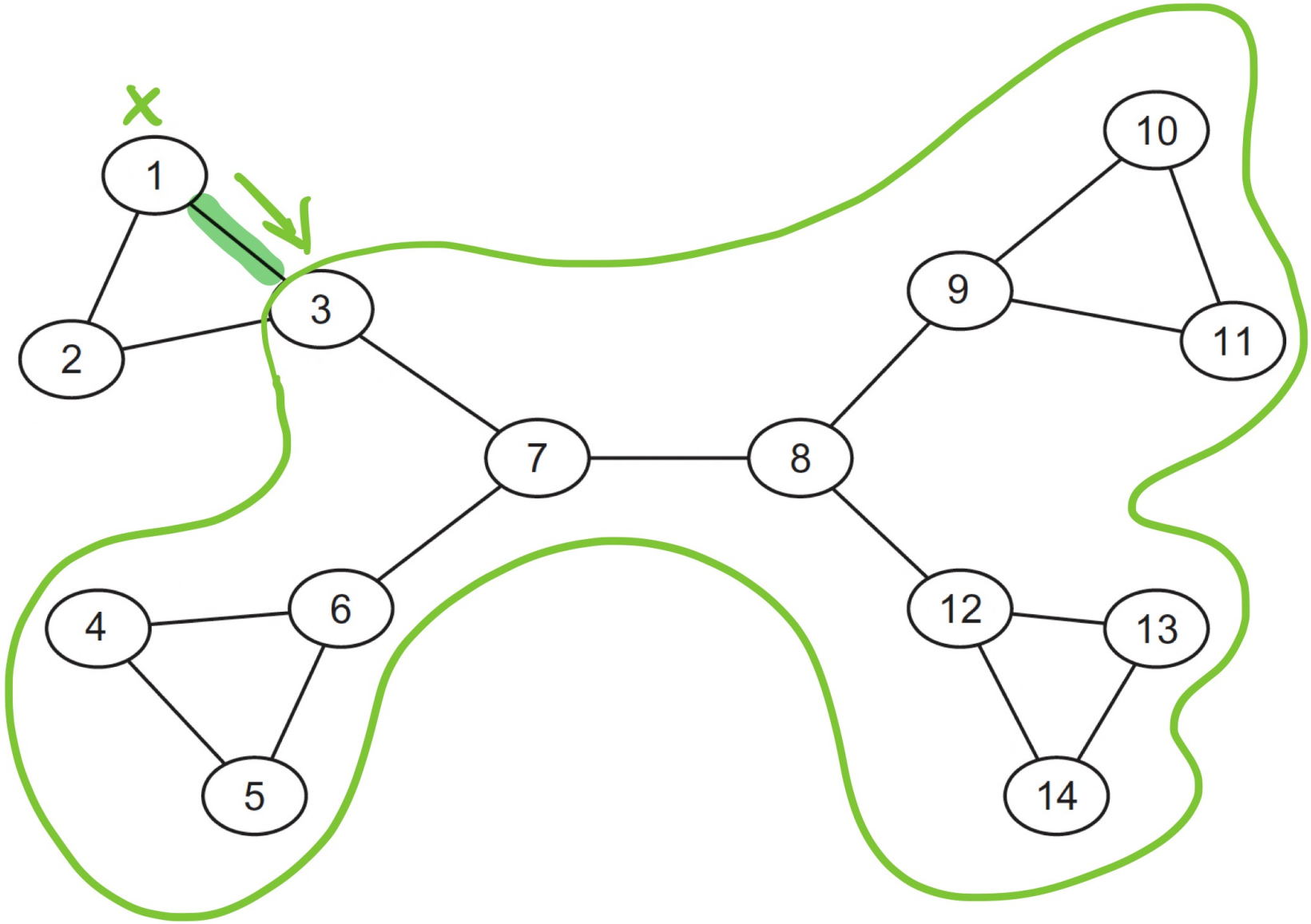
Betweenness



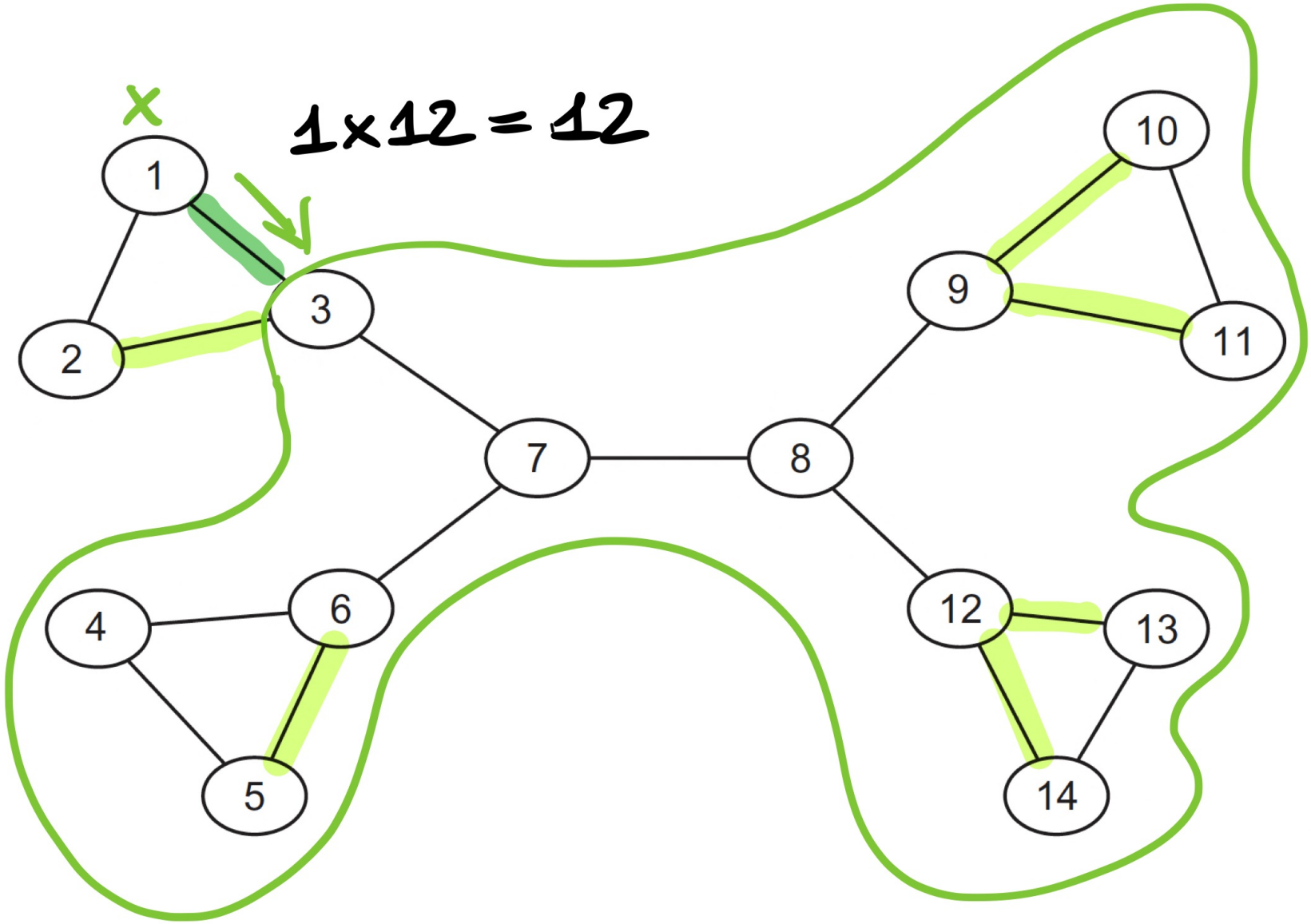
Betweenness



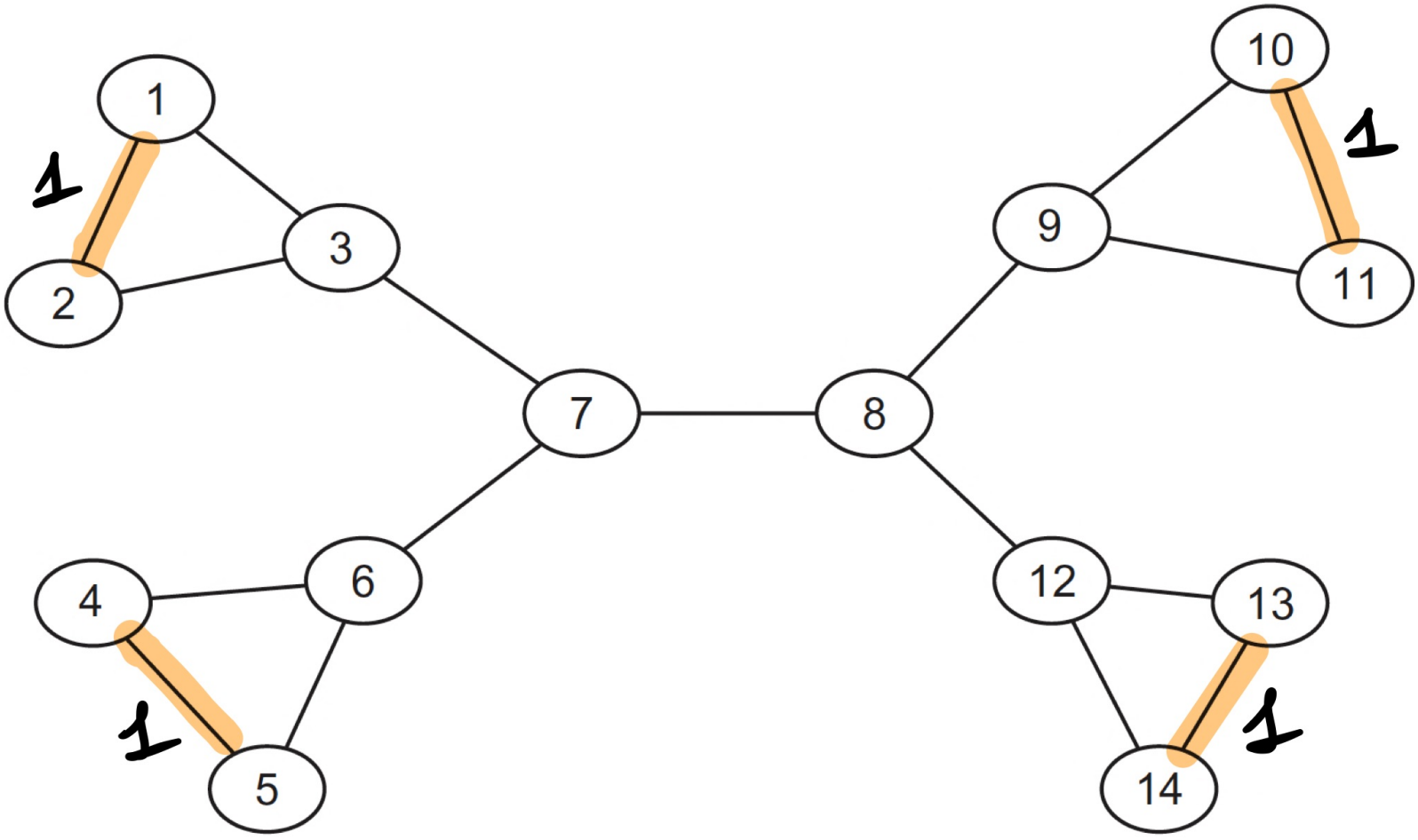
Betweenness



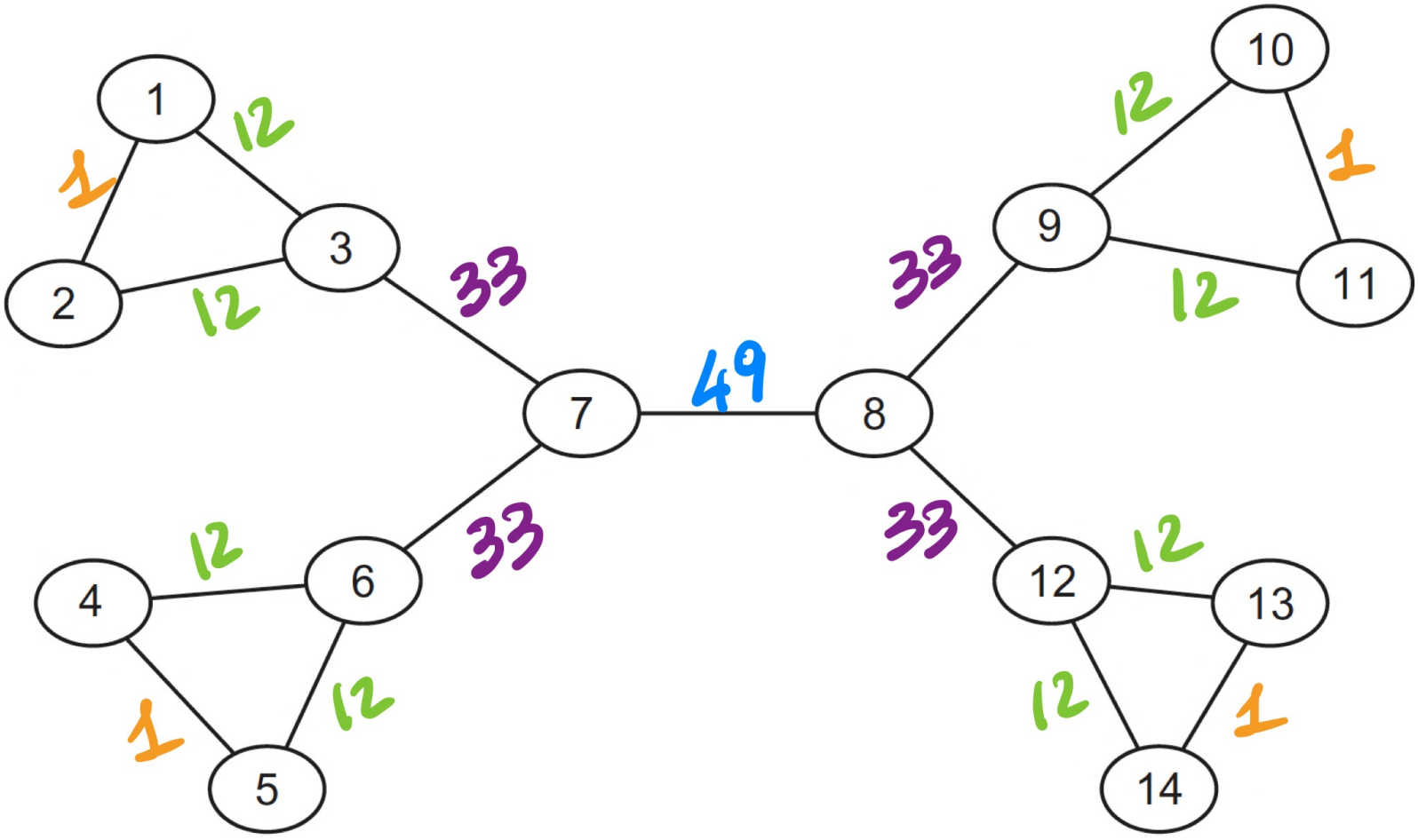
Betweenness



Betweenness

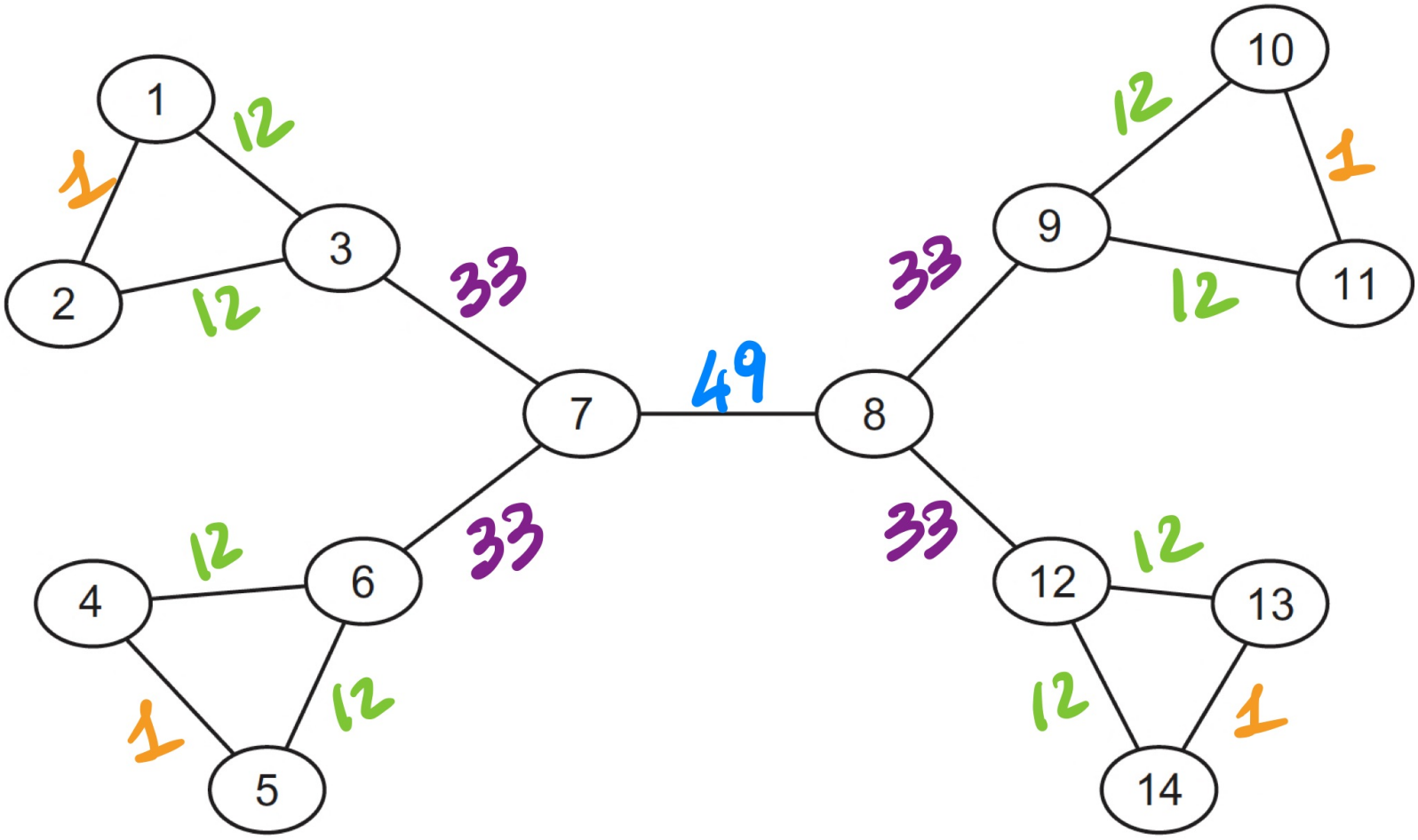


Betweenness



Betweenness

- Thus, edge 7-8 carries the most traffic



Betweenness

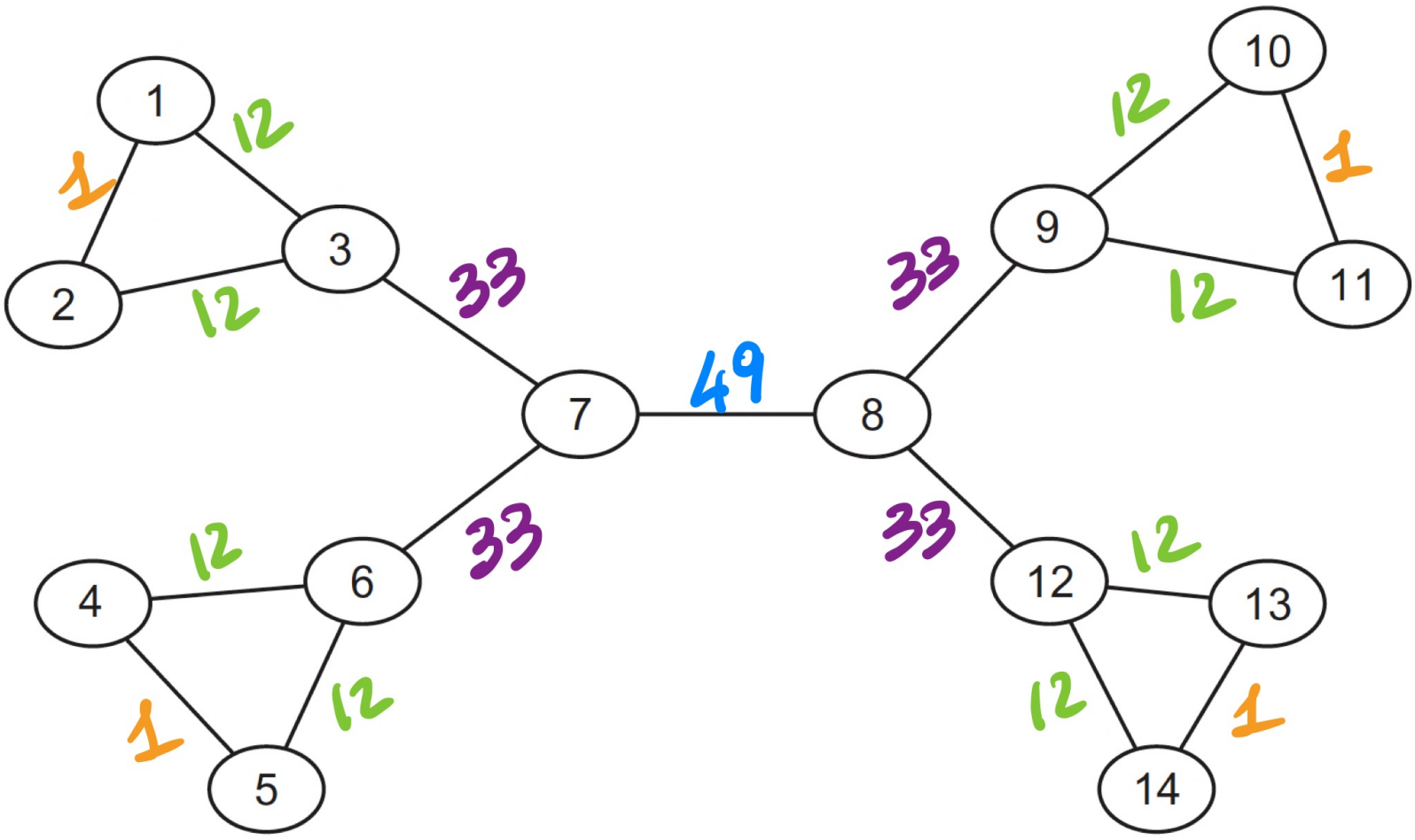
- The betweenness can also be computed at nodes' levels
- The betweenness of a node is the total amount of flow that it carries
- Like edges of high betweenness, nodes of high betweenness occupy critical roles in the network
  - Nodes that span structural holes

The Girvan-Newman Method

1. Find the edge(s) of highest betweenness and remove them
2. Recalculate all the betwennessees and remove the edge(s) of highest betweenness (nested)
3. Proceed until edges remain in the graph

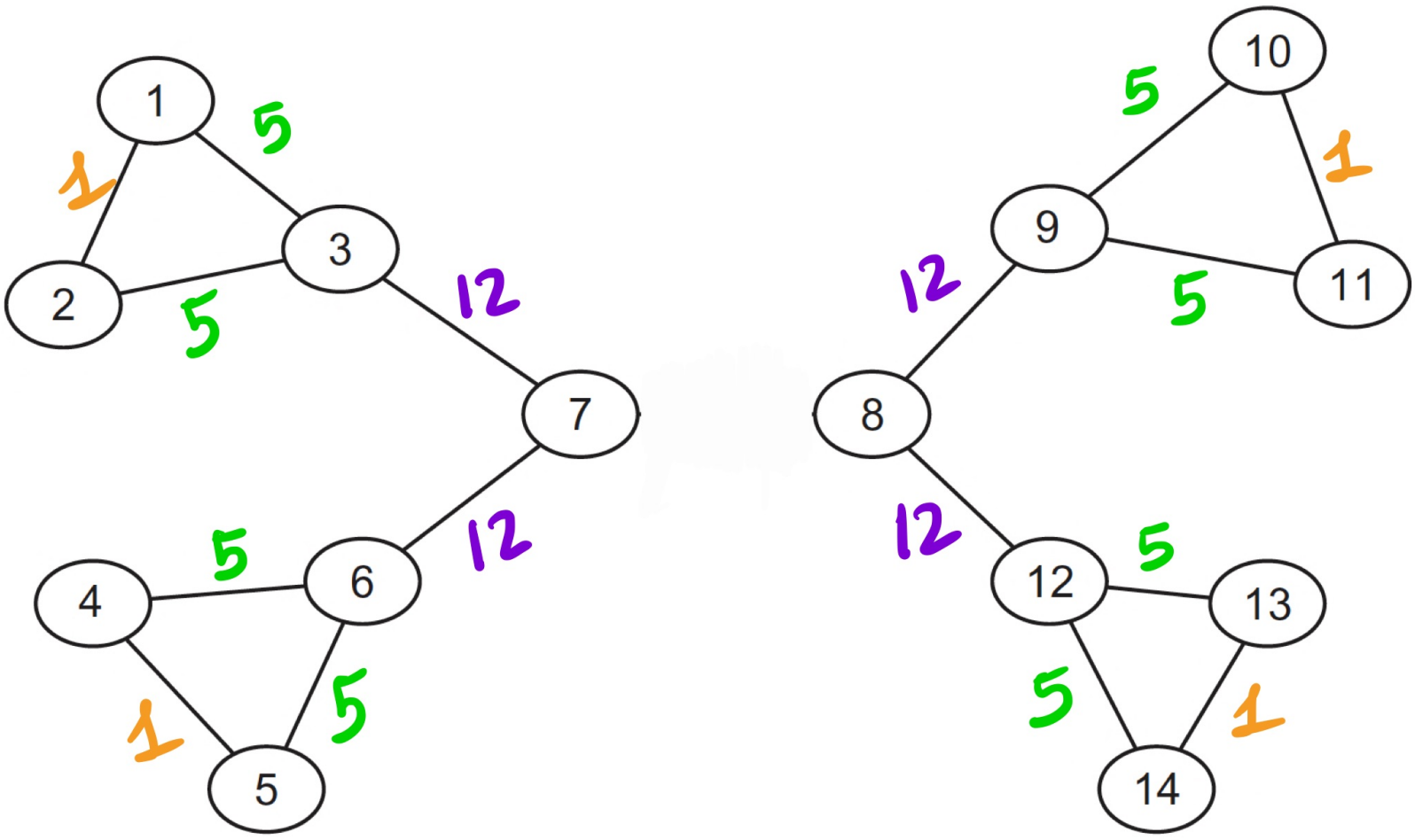
The Girvan-Newman Method

Example #1



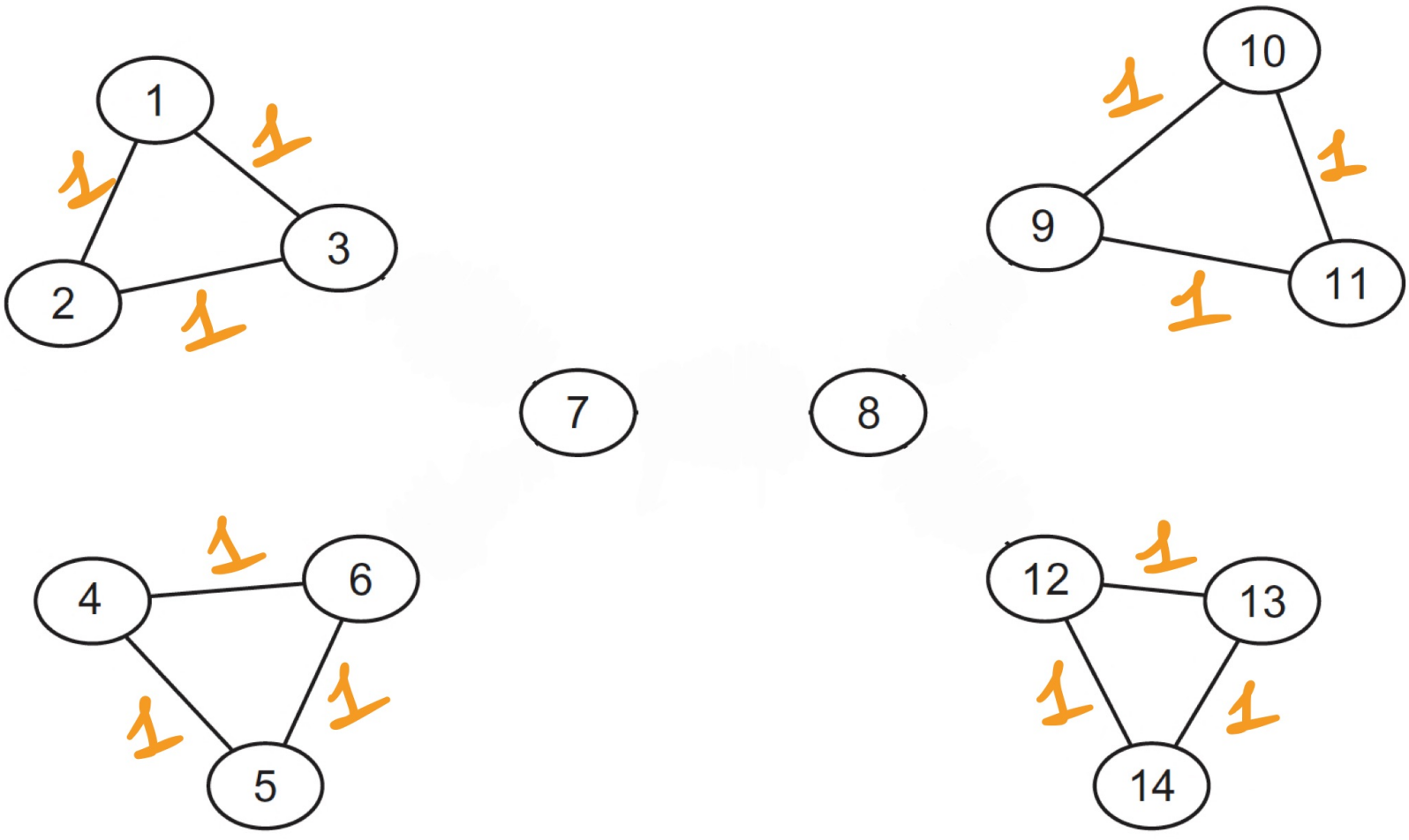
The Girvan-Newman Method

Example #1



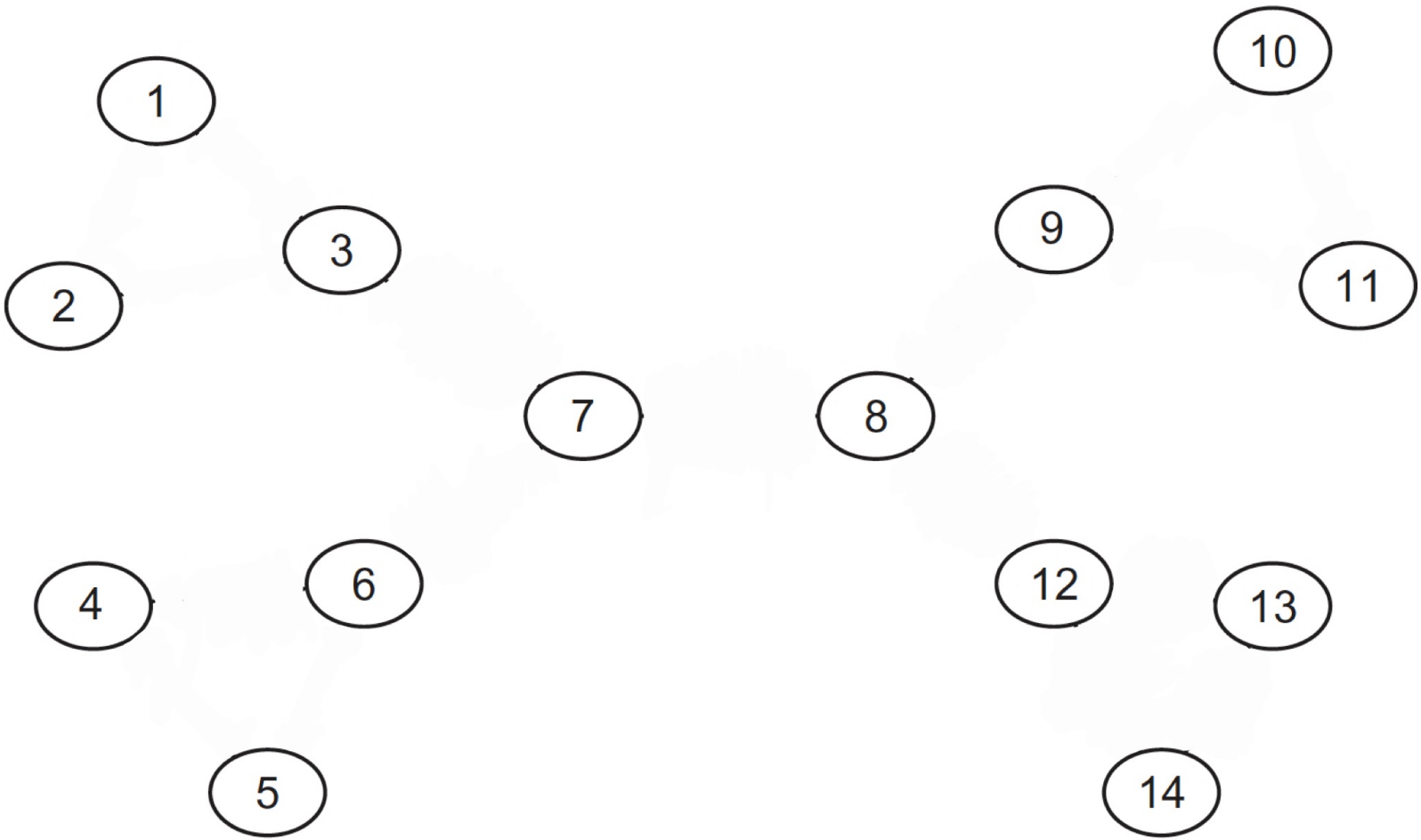
The Girvan-Newman Method

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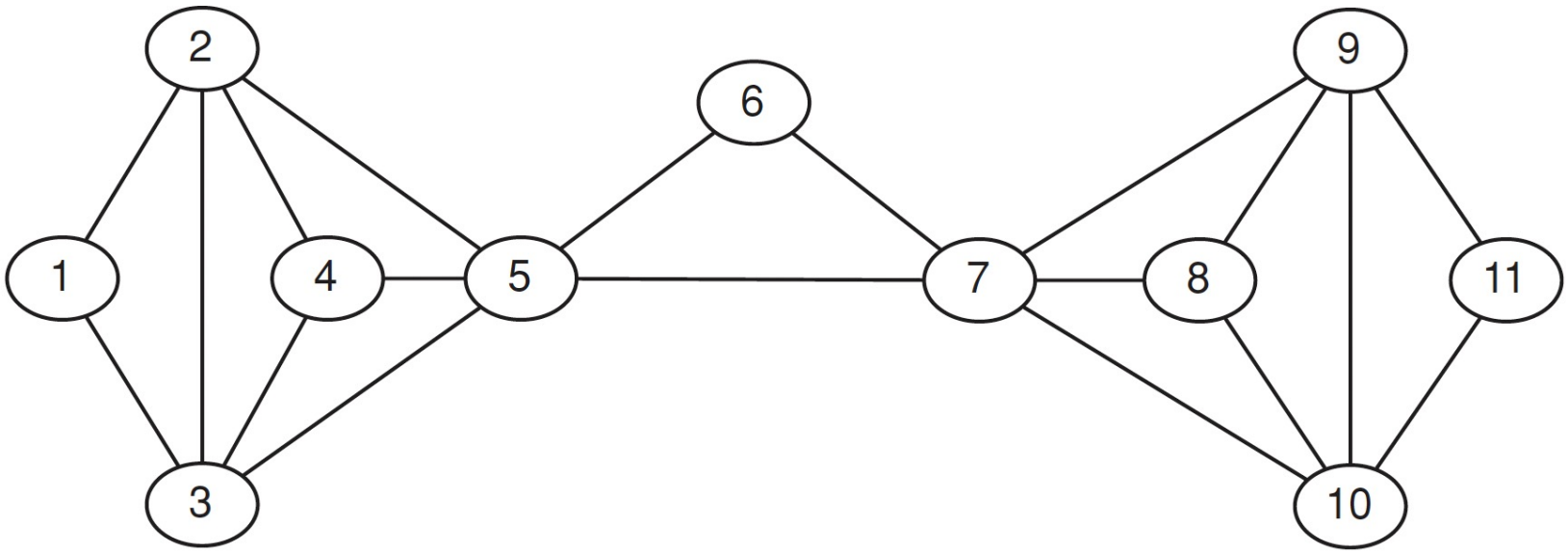
The Girvan-Newman Method

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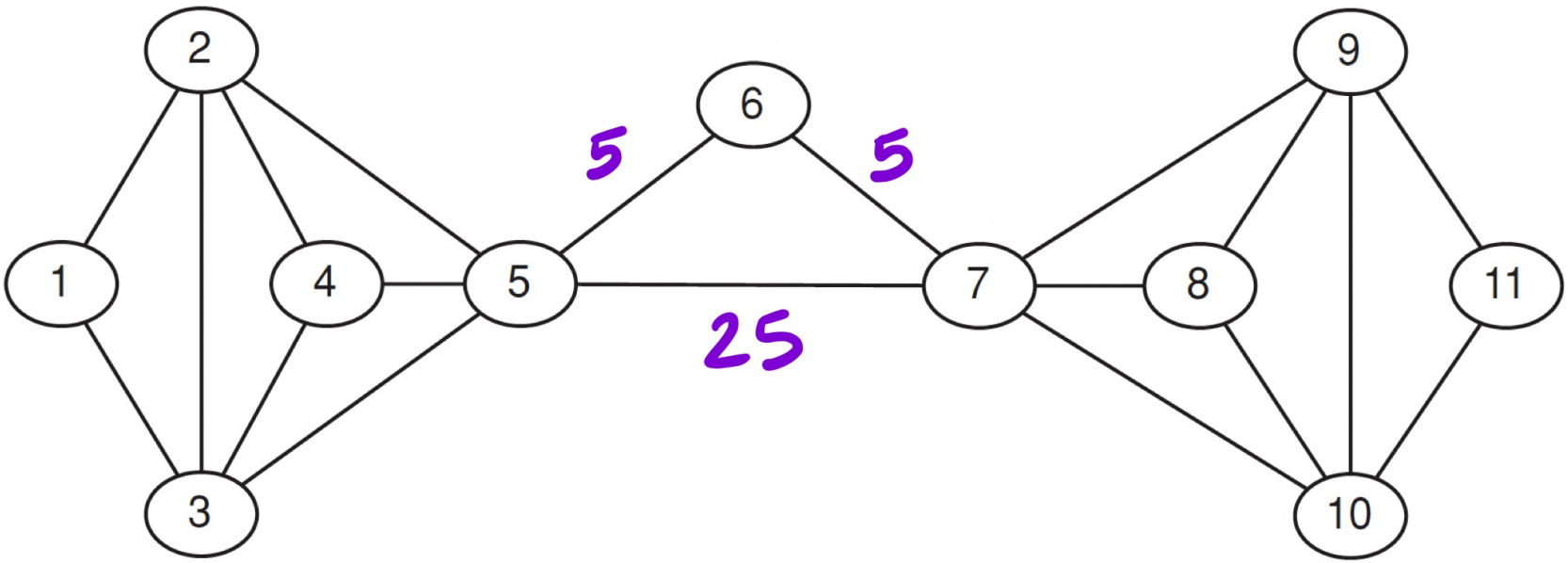
The Girvan-Newman Method

Example #2



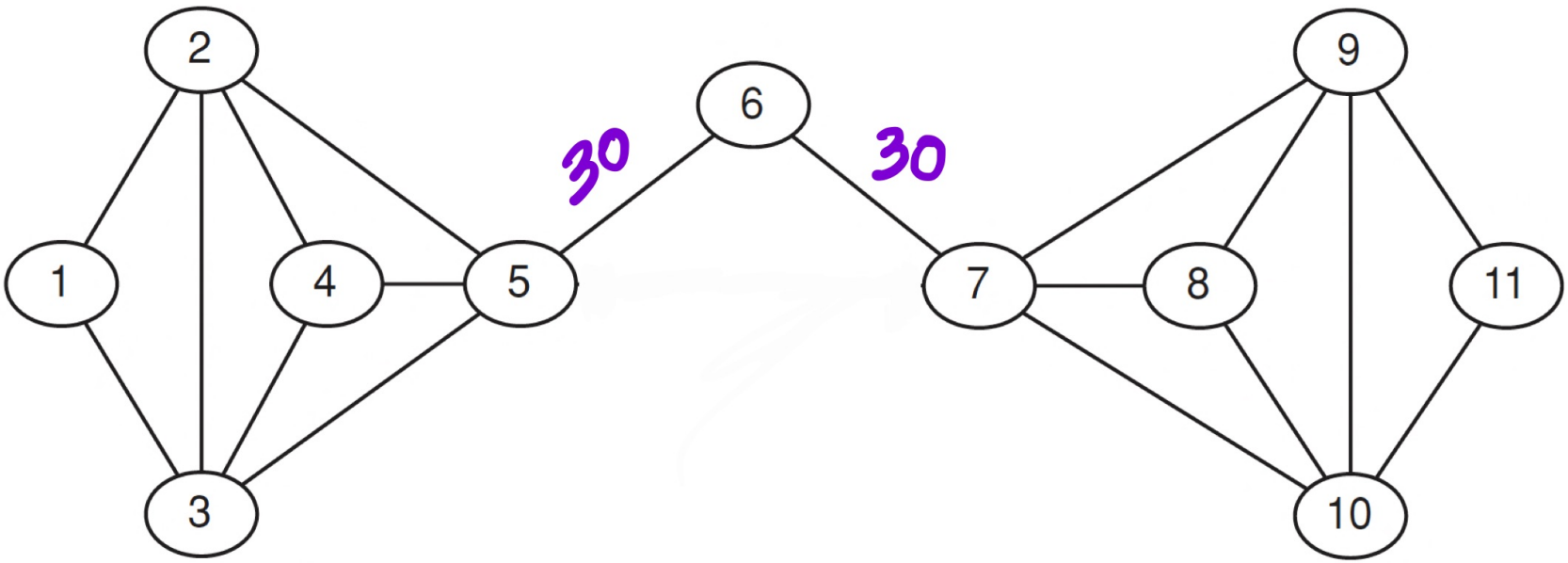
The Girvan-Newman Method

Example #2



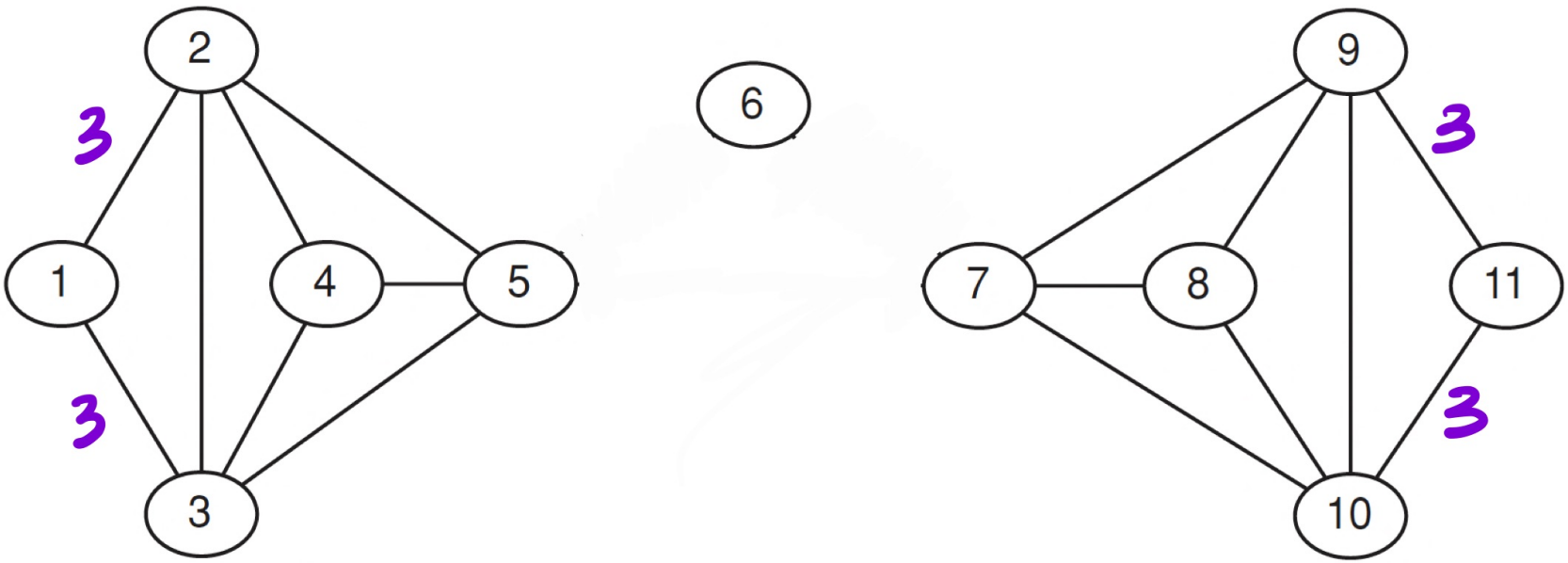
The Girvan-Newman Method

Example #2



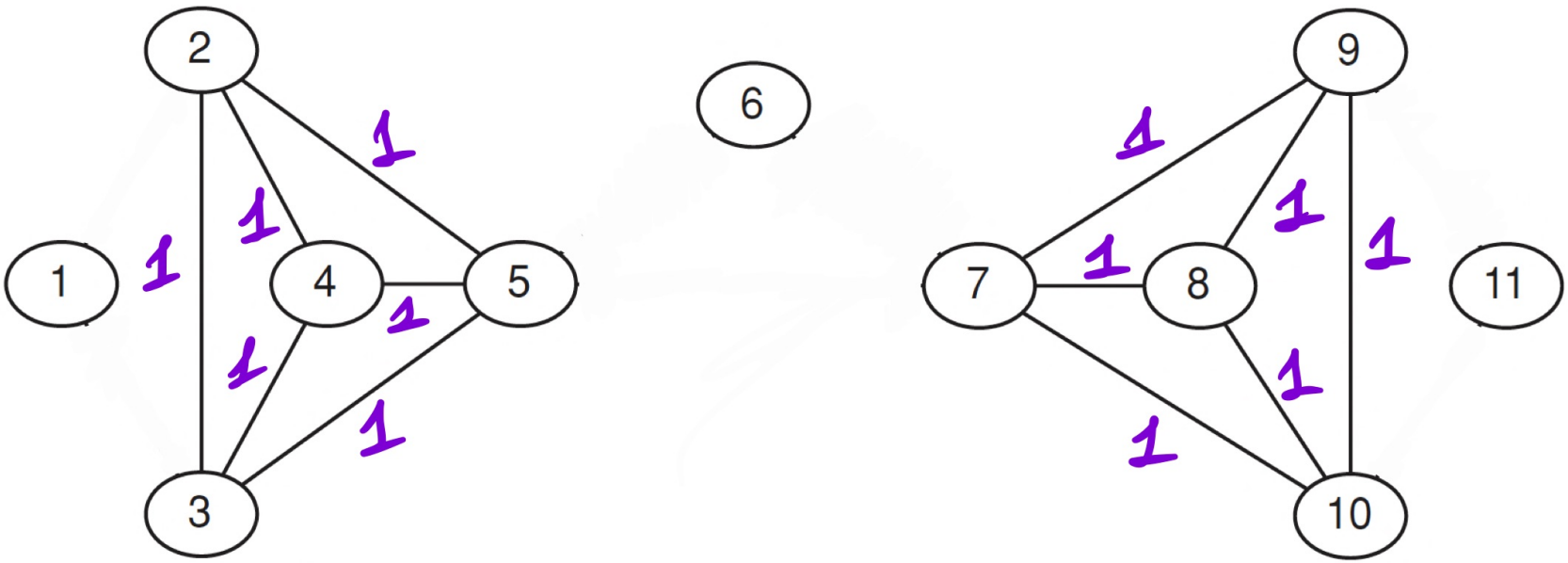
The Girvan-Newman Method

Example #2



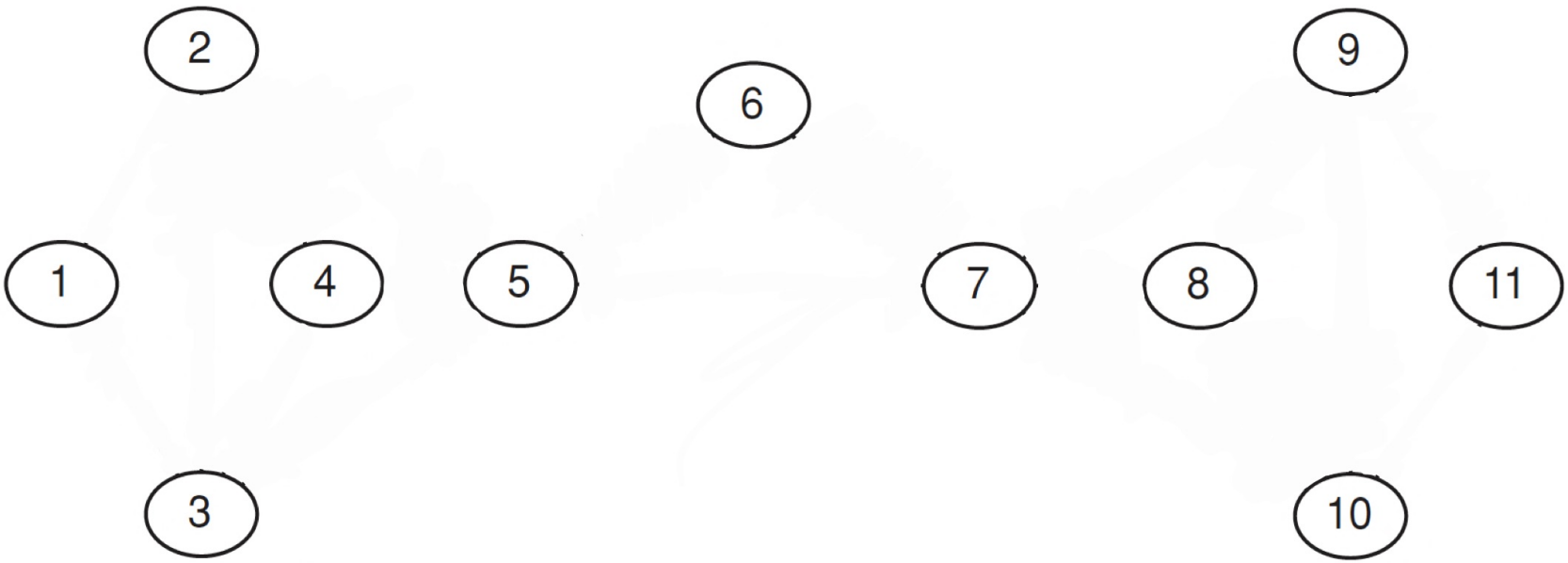
The Girvan-Newman Method

Example #2



The Girvan-Newman Method

Example #2



The Girvan-Newman Method

- Girvan and Newman showed the effect of their method on real-world datasets
- Zachary's Karate Club
- The method works well for networks of moderate size (few thousands nodes)
- For larger networks, computing the betweenness at each step is expensive
  - Approximating the betweenness
  - Agglomerative methods

Notes

This presentation is based on:

- Chapter 2 of Albert Laszlo Barabasi, *Network Science*, Cambridge University Press, 2016
- Chapter 3 of * David Easley and Jon Kleinberg, *Networks, crowds, and markets*, Cambridge University Press, 2012